

Model Analysis

Table of contents

1	Data analysis /Dataframe import and filters	3
1.1	DataFrame import and filters	3
1.2	Metadata	5
2	Model(s) analysis	6
2.1	Explanation legend tables	6
2.2	Repartition of train and test	7
2.2.1	External Split: X_train vs X_test	7
2.2.2	Internal CV Folds	7
2.3	Model : Linear	9
2.3.1	Gridsearch — Group: 3	9
2.3.2	Scatter	11
2.3.3	Scatter residuals	12
2.3.4	Learning curve — Group: 3	12
2.3.5	Gridsearch — Group: 4	13
2.3.6	Scatter	15
2.3.7	Scatter residuals	16
2.3.8	Learning curve — Group: 4	16
2.3.9	Gridsearch — Group: 5	17
2.3.10	Scatter	19
2.3.11	Scatter residuals	20
2.3.12	Learning curve — Group: 5	20
2.3.13	Gridsearch — Group: 11	21
2.3.14	Scatter	23
2.3.15	Scatter residuals	24
2.3.16	Learning curve — Group: 11	24
2.4	Model : LGBM	25
2.4.1	Gridsearch — Group: 3	25
2.4.2	Scatter	29

2.4.3	Scatter residuals	30
2.4.4	Learning curve — Group: 3	30
2.4.5	Gridsearch — Group: 4	31
2.4.6	Scatter	36
2.4.7	Scatter residuals	37
2.4.8	Learning curve — Group: 4	37
2.4.9	Gridsearch — Group: 5	38
2.4.10	Scatter	42
2.4.11	Scatter residuals	43
2.4.12	Learning curve — Group: 5	43
2.4.13	Gridsearch — Group: 11	44
2.4.14	Scatter	48
2.4.15	Scatter residuals	49
2.4.16	Learning curve — Group: 11	49
2.5	Model : CatBoost	50
2.5.1	Gridsearch — Group: 3	50
2.5.2	Scatter	56
2.5.3	Scatter residuals	57
2.5.4	Learning curve — Group: 3	57
2.5.5	Gridsearch — Group: 4	58
2.5.6	Scatter	64
2.5.7	Scatter residuals	65
2.5.8	Learning curve — Group: 4	65
2.5.9	Gridsearch — Group: 5	66
2.5.10	Scatter	70
2.5.11	Scatter residuals	71
2.5.12	Learning curve — Group: 5	71
2.5.13	Gridsearch — Group: 11	72
2.5.14	Scatter	78
2.5.15	Scatter residuals	79
2.5.16	Learning curve — Group: 11	79
2.6	Sumup-table	80
3	Plot	81
3.1	Standalone plots	81
3.1.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	81
3.1.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	84
3.1.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	87
3.1.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	90
3.1.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	93
3.1.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	96
3.1.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	99
3.1.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	102

3.2	Compare plots	105
3.2.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	105
3.2.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	106
3.2.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	108
3.2.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	109
3.2.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	110
3.2.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	112
3.2.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	113
3.2.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	114

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                             feats_to_balance + feats_grouping +
                             feats_reference+ feats_postproc))

    df = df[cols]
    #df['campaign'] = "ALL"

    return df

def import_data_IR():
    if target_col[0] == "i_NH3":
```

```

        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

df = pd.read_csv(file_path, sep=";")
df["datetime"] = pd.to_datetime(df["datetime"])
df.sort_values("datetime").reset_index(drop=True)

if target_col[0] == "i_NH3":
    df["campaign"] = df["datetime"].apply(assign_campaign)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

```

```

# if camp == "2025_Jun":
#     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
# elif camp == "2025_Sep":
#     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
if camp == "2025_Apr":
    #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
    #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
    mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
elif camp == "2025_Jun":
    mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
    #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
elif camp == "2025_Sep":
    mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
    #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
elif camp == "2026_Gen":
    #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
    #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
    mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
else:
    mapping = {}

return mapping.get(rid, "no")

```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : id_channel

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : id_channel

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

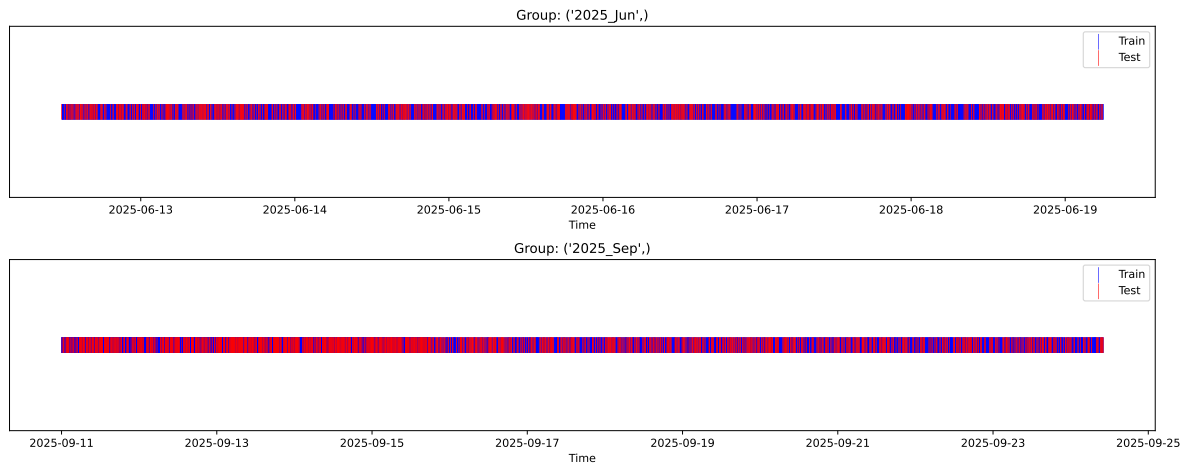
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

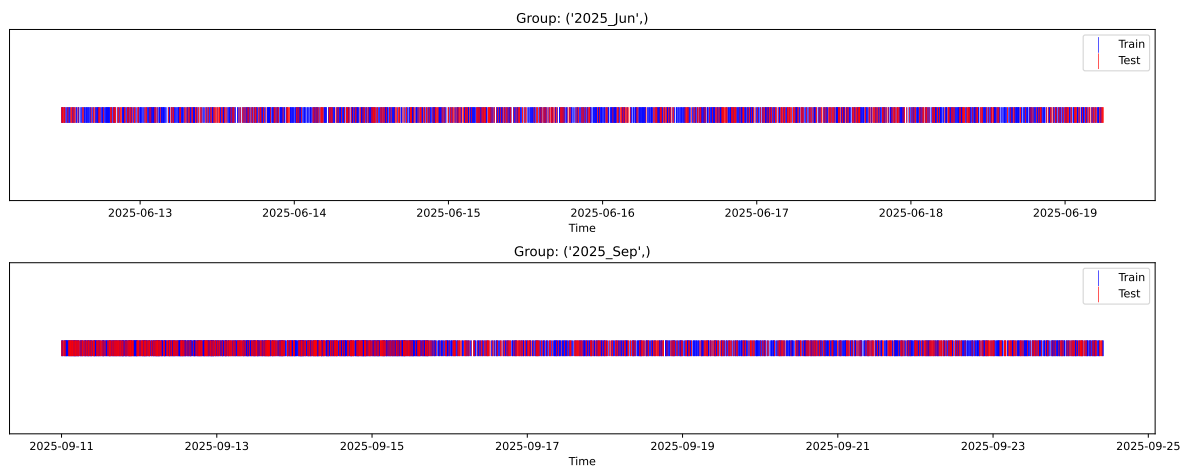
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

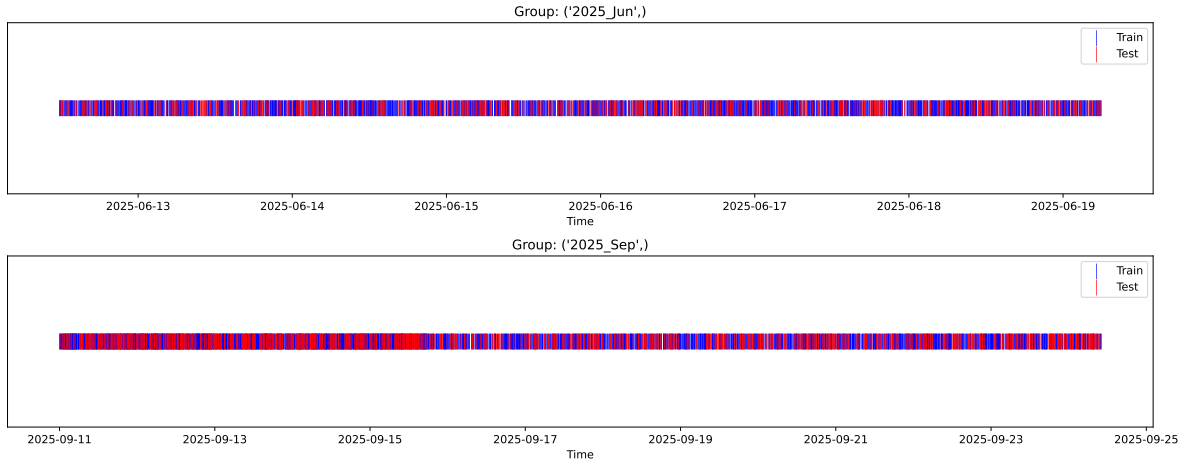
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



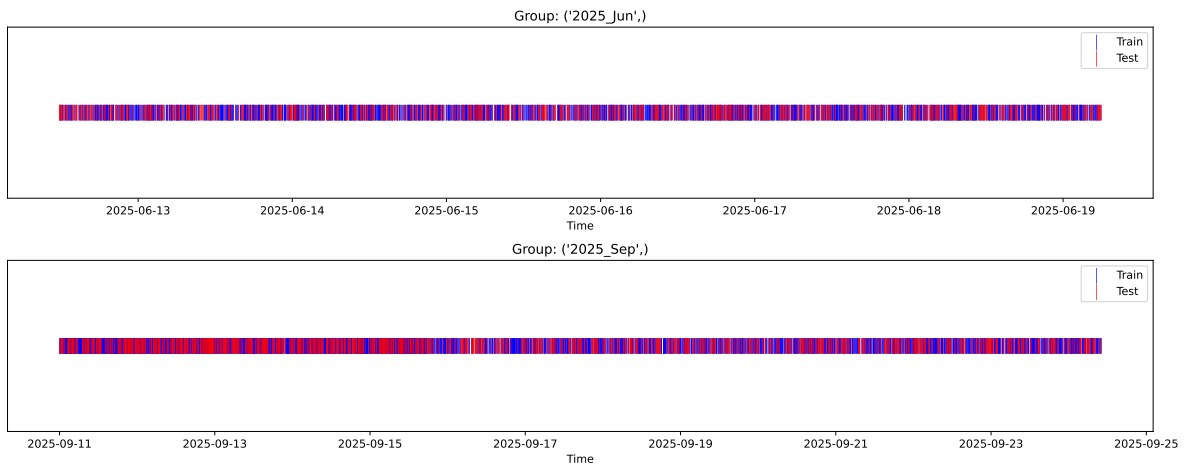
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



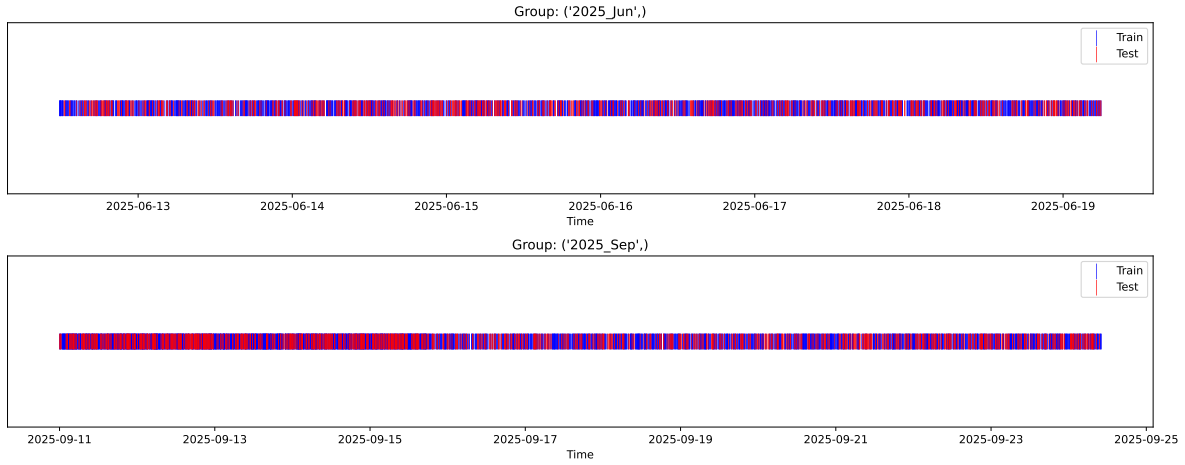
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



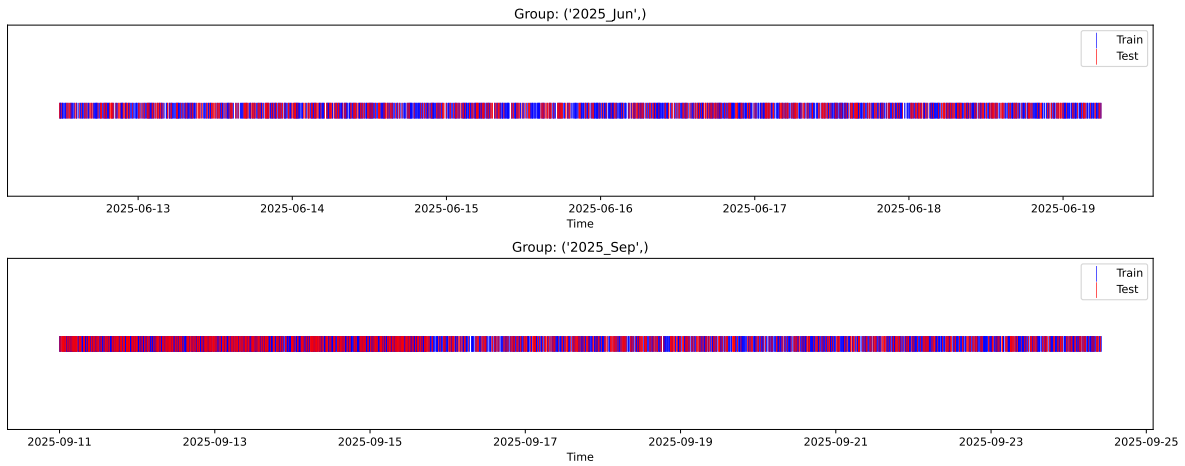
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch — Group: 3

Len(group) : 1356, Len(training) : 950, Len(test) : 406

Distribution y_train vs y_test: y_train: mean=2.317, std=0.708, min=1.041, max=6.086
y_test: mean=2.318, std=0.740, min=1.068, max=5.447

Fold 0 — y_test: mean=2.39, std=0.76, y_train: mean=2.29, std=0.68

Fold 1 — y_test: mean=2.34, std=0.73, y_train: mean=2.31, std=0.70

Fold 2 — y_test: mean=2.33, std=0.71, y_train: mean=2.31, std=0.71

Fold 3 — y_test: mean=2.33, std=0.69, y_train: mean=2.31, std=0.71

Fold 4 — y_test: mean=2.30, std=0.73, y_train: mean=2.32, std=0.70

train scores CV: [0.62268767 0.60265136 0.58040391 0.56906438 0.60353391]

test scores CV: [0.49521601 0.53276144 0.58141377 0.60879453 0.53165288]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5957	0.55	0.0402	0.6403	0.6408	0.0908	1.1651	1.0831

Coefficients:

coef_r_CO2 = -0.0897

coef_r_NH3 = 0.0392

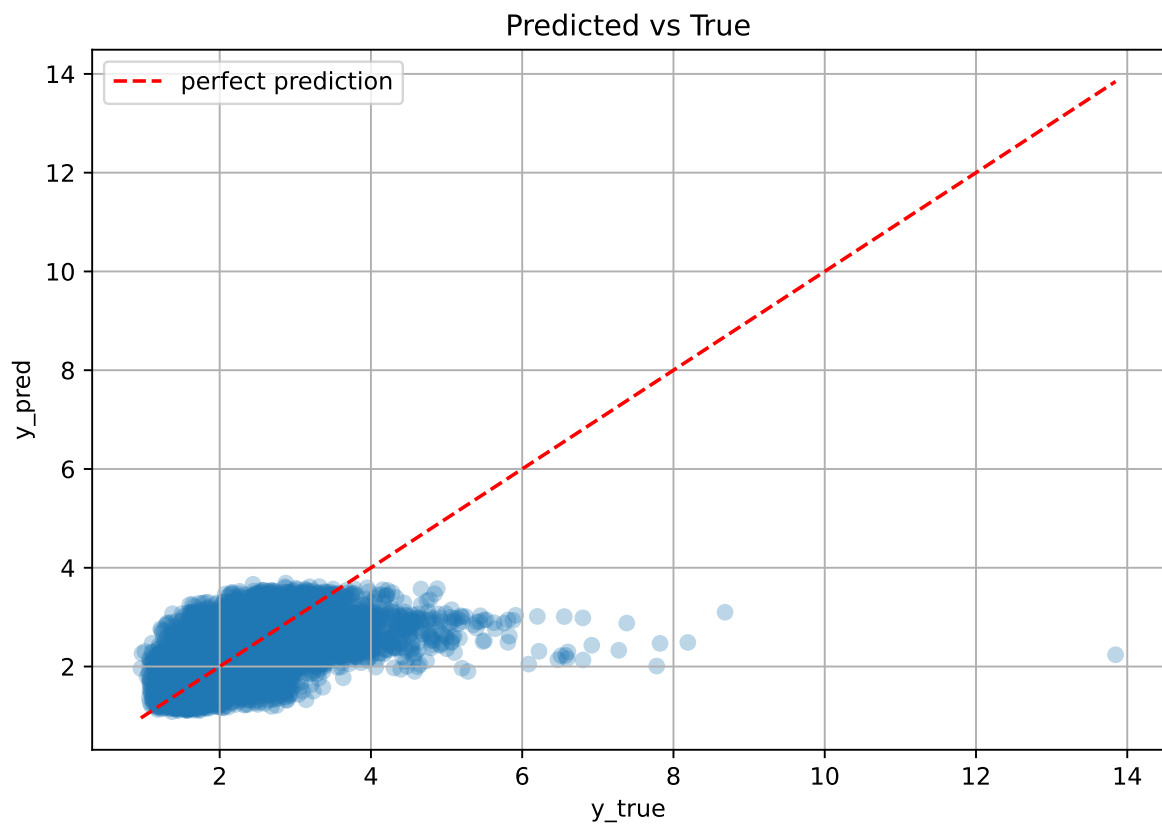
coef_r_THI = 0.4472

coef_r_temperature = -0.3735

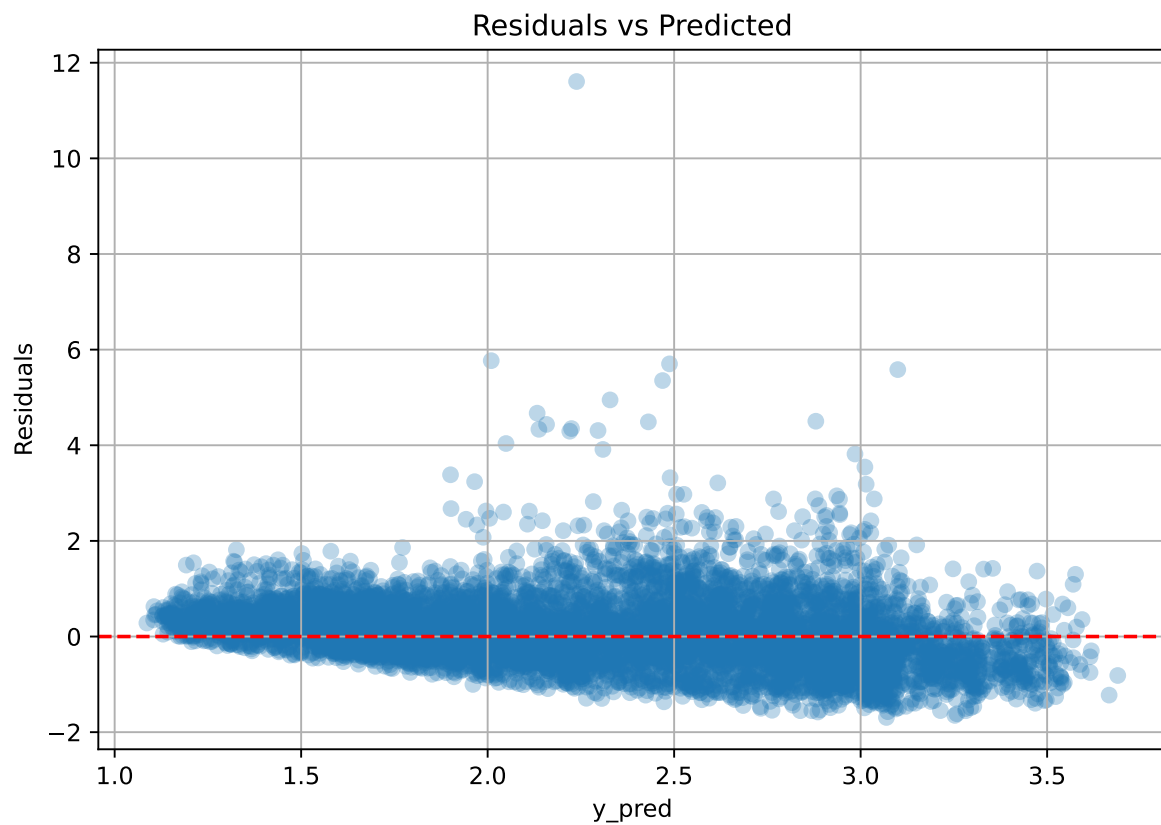
coef_r_umidity = -0.5516

intercept = 2.3165

2.3.2 Scatter



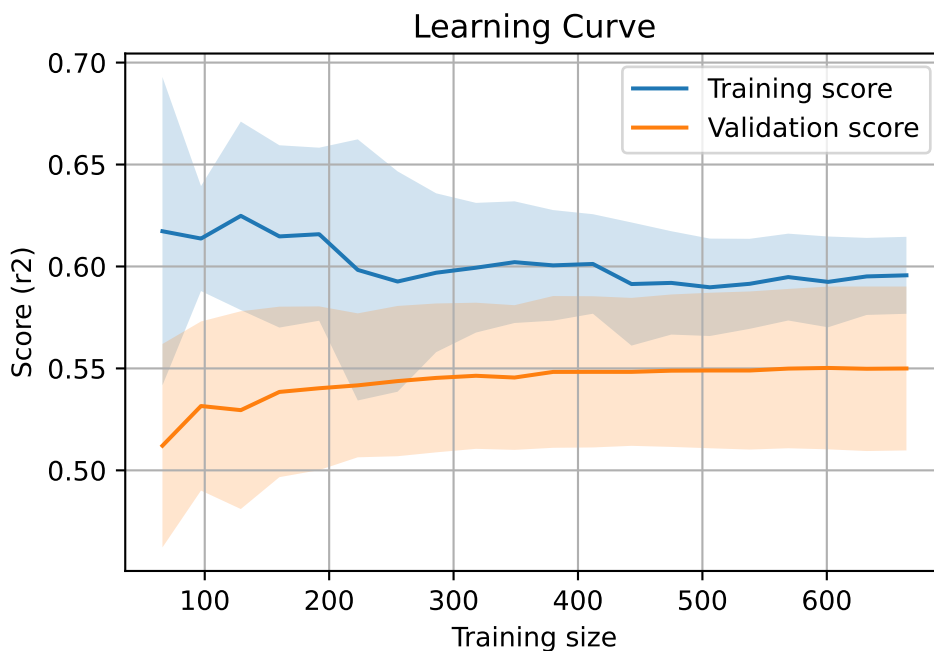
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 3

`Len(group)` : 1356, `Len(training)` : 950,

`Len(training_real)` : 665, `Len(test_of_training)` : 285, `Len(test)` : 406



2.3.5 Gridsearch — Group: 4

Len(group) : 1360, Len(training) : 952, Len(test) : 408

Distribution y_train vs y_test: y_train: mean=2.239, std=0.831, min=0.959, max=7.383
y_test: mean=2.202, std=0.791, min=1.062, max=7.822

Fold 0 — y_test: mean=2.22, std=0.80, y_train: mean=2.25, std=0.84

Fold 1 — y_test: mean=2.22, std=0.70, y_train: mean=2.25, std=0.88

Fold 2 — y_test: mean=2.29, std=0.86, y_train: mean=2.22, std=0.81

Fold 3 — y_test: mean=2.31, std=0.86, y_train: mean=2.21, std=0.82

Fold 4 — y_test: mean=2.22, std=0.83, y_train: mean=2.25, std=0.83

train scores CV: [0.31407089 0.27350601 0.3045029 0.31978092 0.29846023]

test scores CV: [0.23746916 0.36893865 0.26706608 0.23176705 0.28402149]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3021	0.2779	0.0494	0.2765	0.2771	-0.0008	0.9973	1.0871

Coefficients:

$\text{coef_r_CO2} = -0.1021$

$\text{coef_r_NH3} = 0.1428$

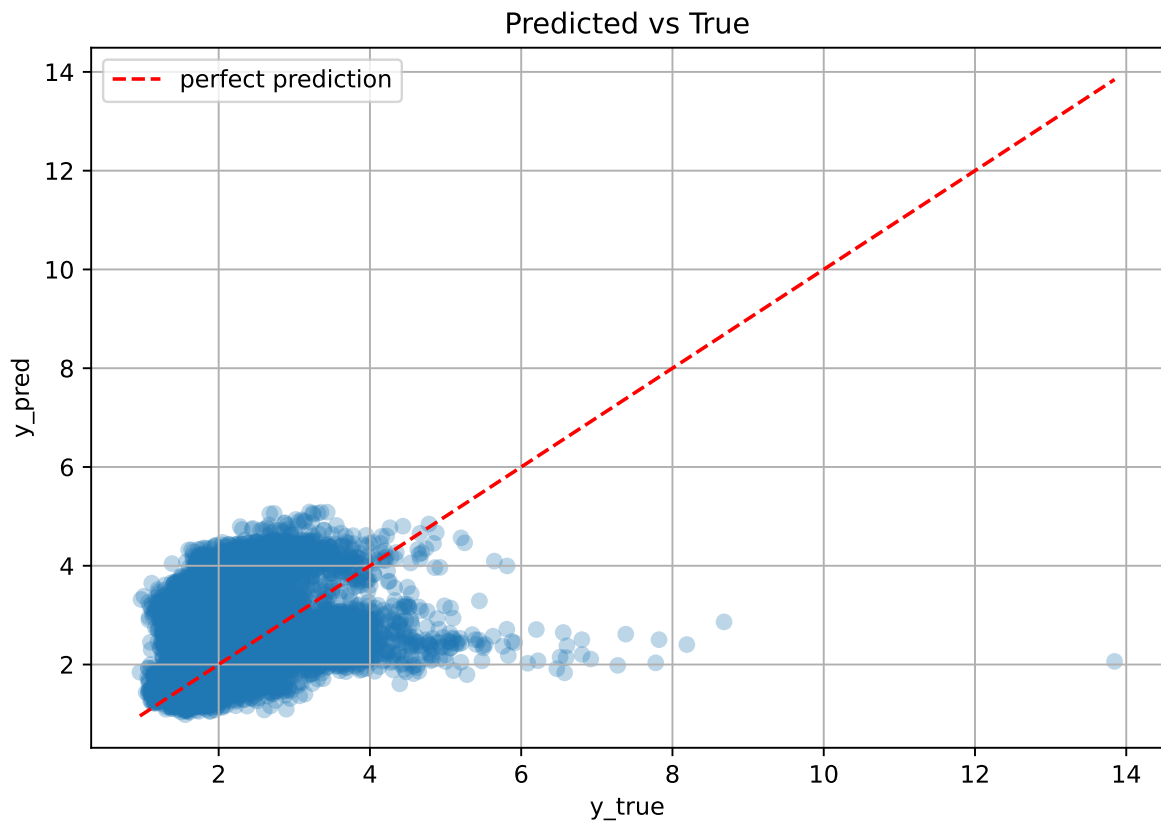
$\text{coef_r_THI} = 1.0882$

$\text{coef_r_temperature} = -1.7734$

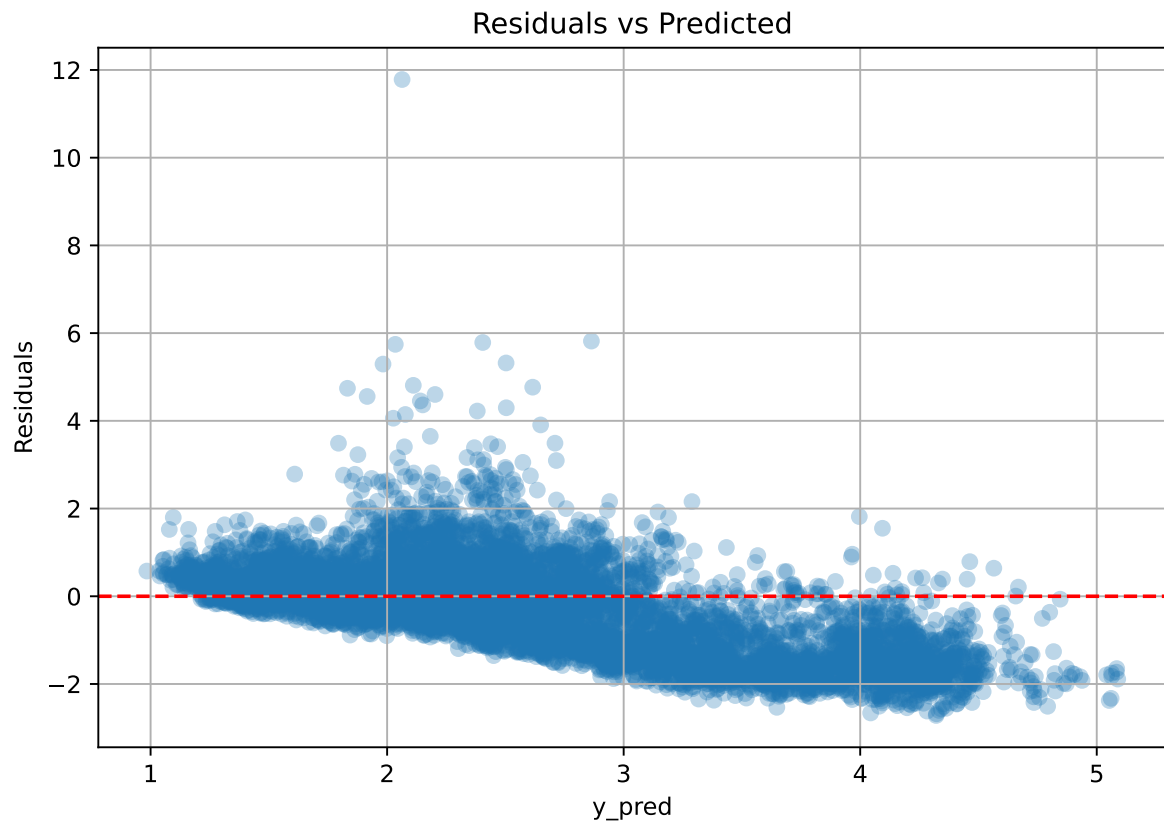
$\text{coef_r_umidity} = -1.1164$

$\text{intercept} = 2.2387$

2.3.6 Scatter



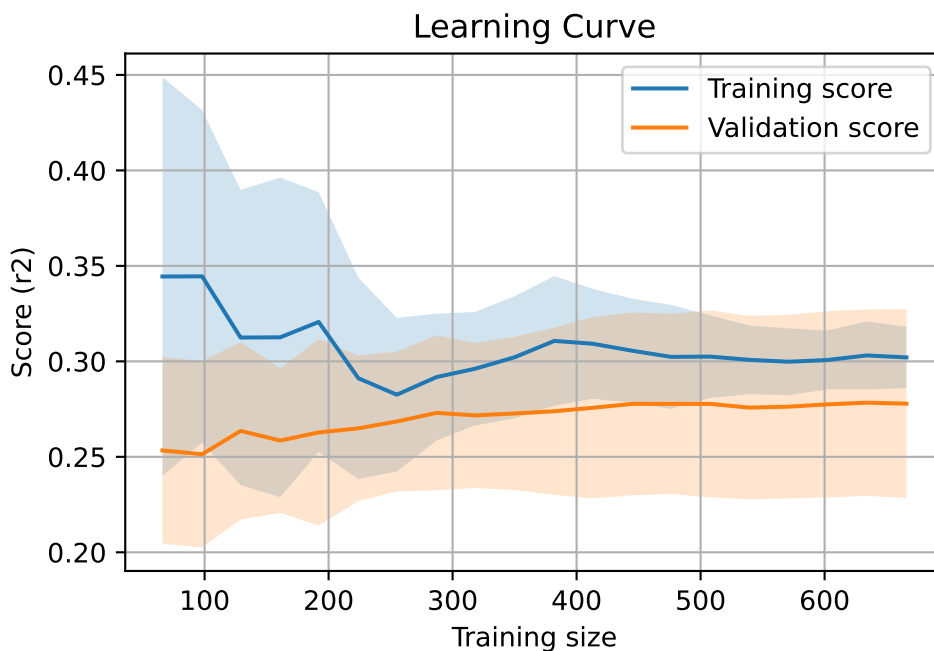
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 4

`Len(group)` : 1360, `Len(training)` : 952,

`Len(training_real)` : 667, `Len(test_of_training)` : 285, `Len(test)` : 408



2.3.9 Gridsearch — Group: 5

Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

Fold 0 — y_test: mean=2.27, std=0.68, y_train: mean=2.22, std=0.68

Fold 1 — y_test: mean=2.25, std=0.71, y_train: mean=2.23, std=0.67

Fold 2 — y_test: mean=2.20, std=0.65, y_train: mean=2.25, std=0.69

Fold 3 — y_test: mean=2.21, std=0.65, y_train: mean=2.25, std=0.70

Fold 4 — y_test: mean=2.22, std=0.63, y_train: mean=2.24, std=0.70

train scores CV: [0.4277706 0.44021497 0.41172833 0.43714743 0.44005866]

test scores CV: [0.41299685 0.39069188 0.45040747 0.3878951 0.37549212]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4314	0.4035	0.0264	0.4566	0.4572	0.0537	1.1332	1.0691

Coefficients:

$\text{coef_r_CO2} = -0.0952$

$\text{coef_r_NH3} = -0.1254$

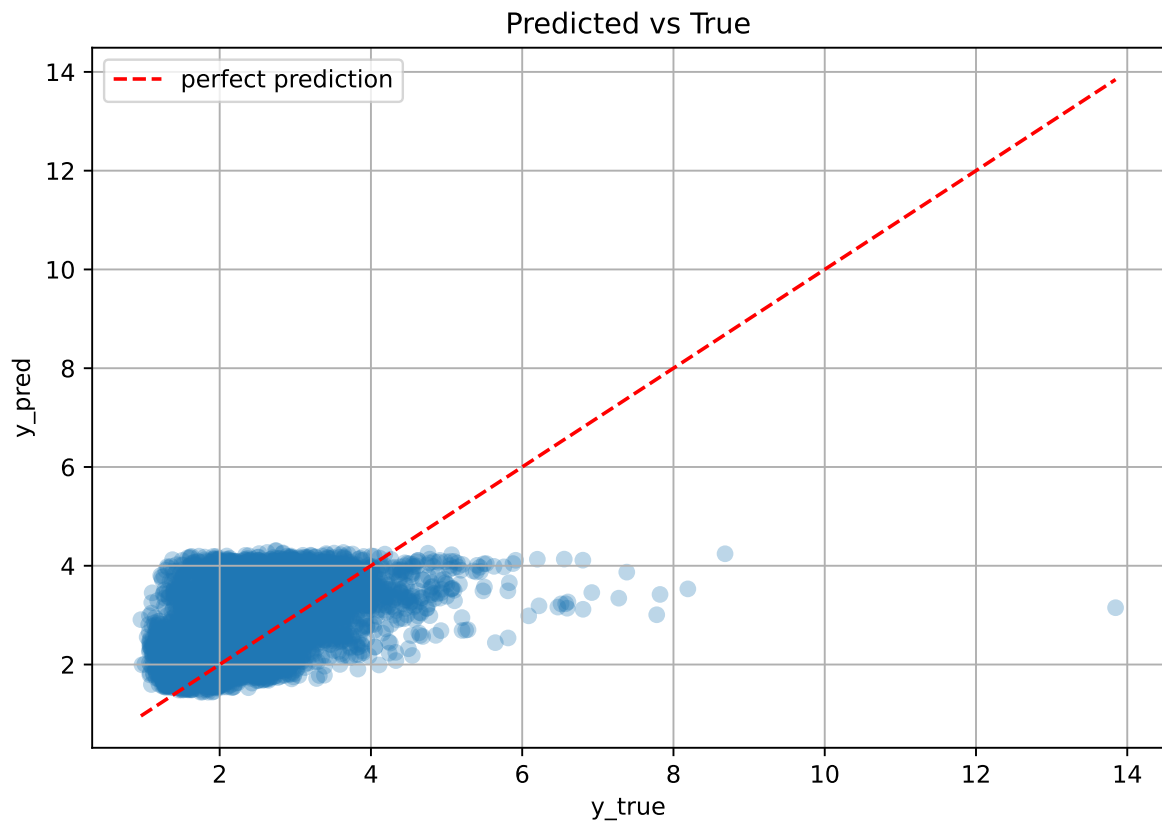
$\text{coef_r_THI} = 0.0994$

$\text{coef_r_temperature} = 0.0362$

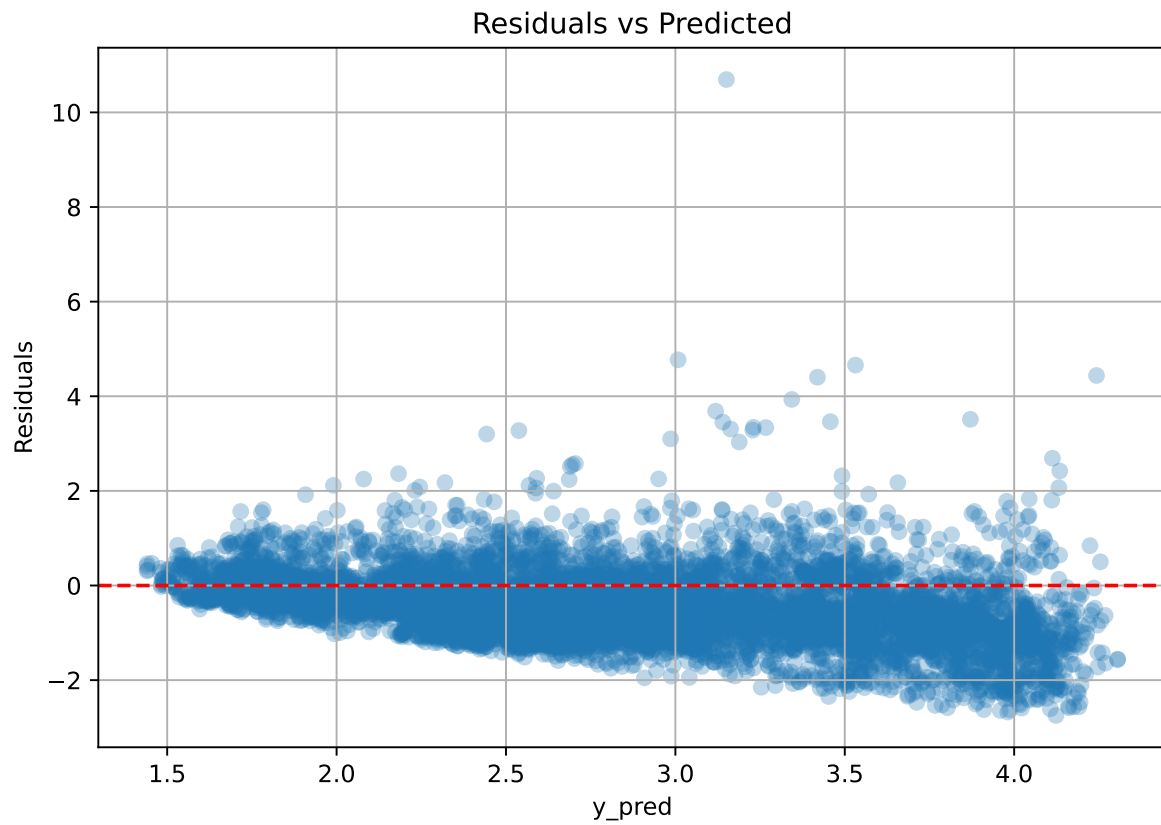
$\text{coef_r_umidity} = -0.4502$

$\text{intercept} = 2.2379$

2.3.10 Scatter



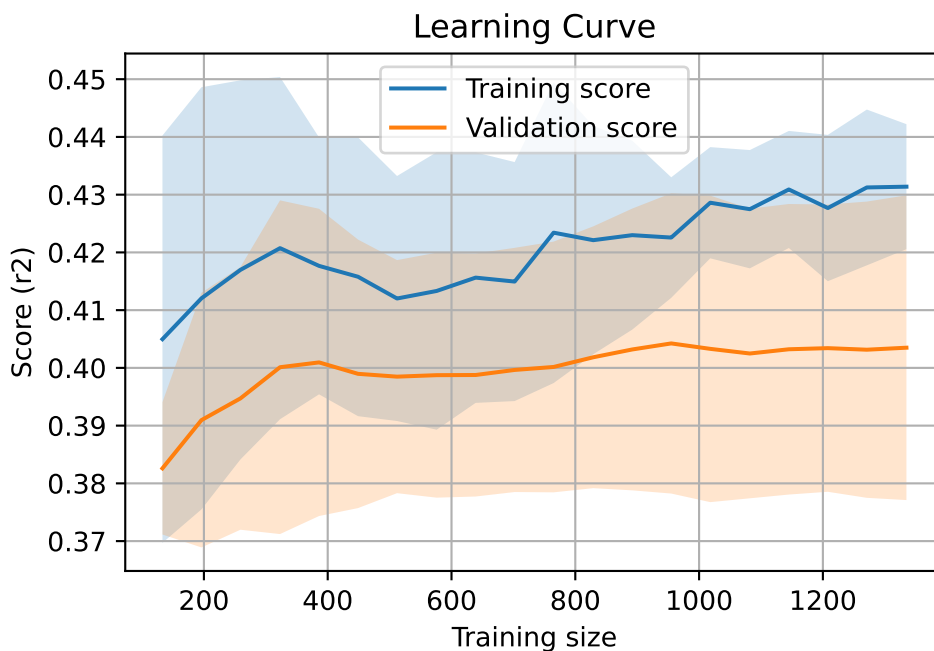
2.3.11 Scatter residuals



2.3.12 Learning curve — Group: 5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.3.13 Gridsearch — Group: 11

Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

Fold 0 — y_test: mean=2.58, std=0.86, y_train: mean=2.52, std=0.86

Fold 1 — y_test: mean=2.56, std=0.84, y_train: mean=2.53, std=0.87

Fold 2 — y_test: mean=2.50, std=0.84, y_train: mean=2.55, std=0.87

Fold 3 — y_test: mean=2.53, std=0.87, y_train: mean=2.55, std=0.85

Fold 4 — y_test: mean=2.53, std=0.82, y_train: mean=2.54, std=0.87

train scores CV: [0.46497694 0.46580479 0.46495948 0.46844345 0.45091902]

test scores CV: [0.45757013 0.45504757 0.45666765 0.44978339 0.49647352]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.463	0.4631	0.0169	0.3942	0.3945	-0.0686	0.8519	0.9998

Coefficients:

coef_r_CO2 = 0.0009

coef_r_NH3 = 0.0485

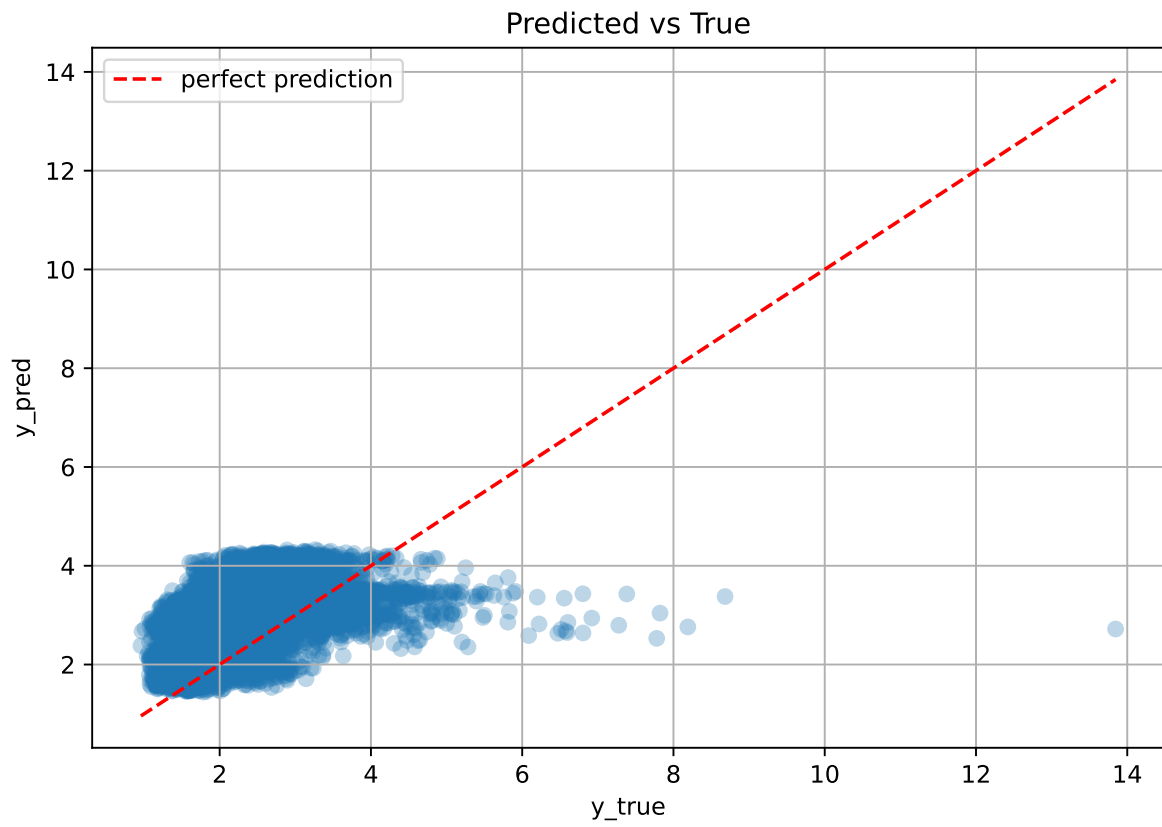
coef_r_THI = 0.8323

coef_r_temperature = -0.9731

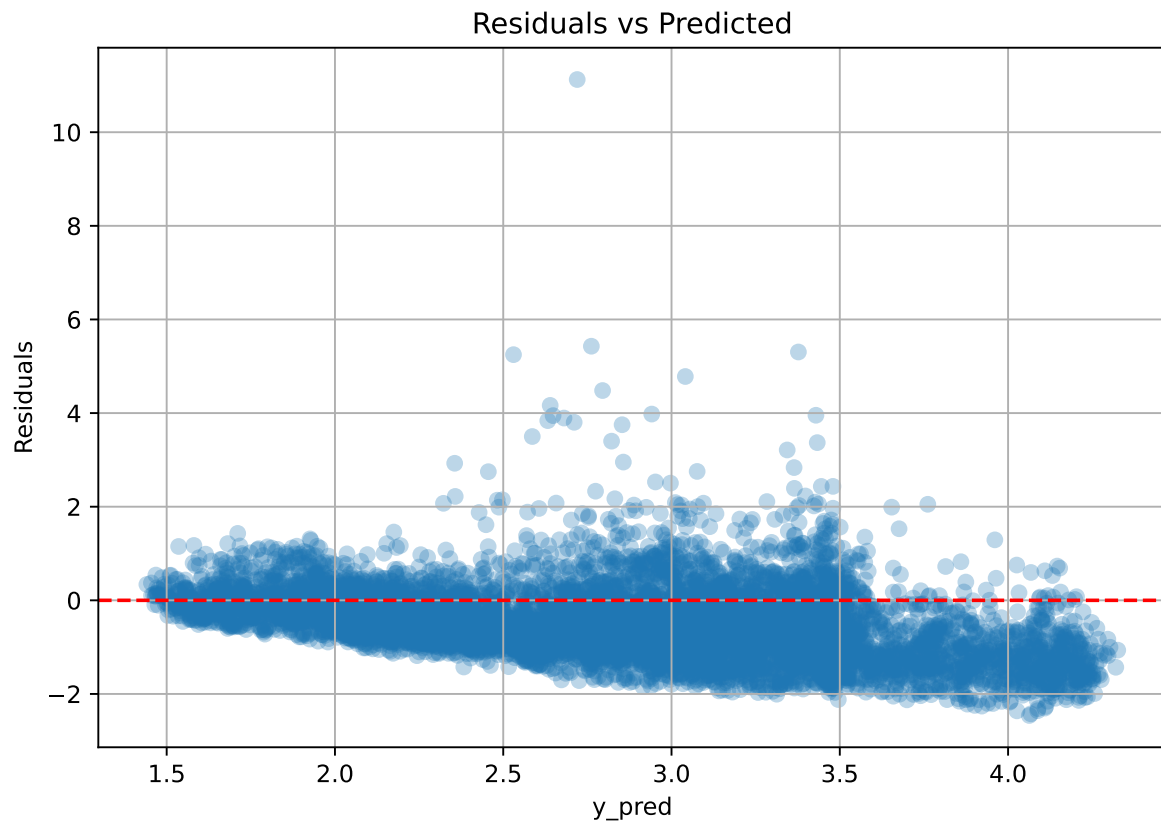
coef_r_umidity = -0.7852

intercept = 2.5398

2.3.14 Scatter



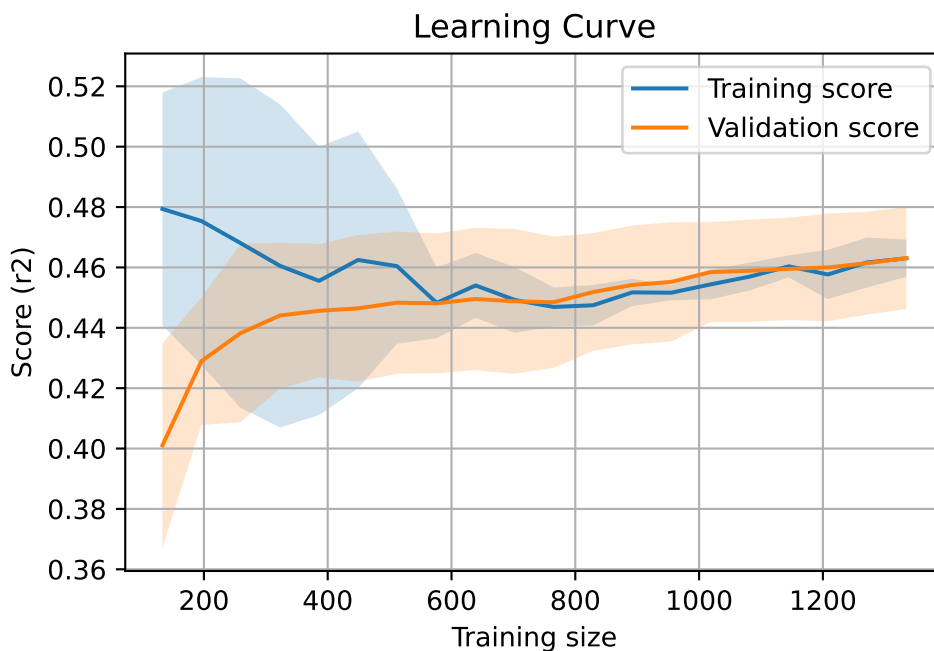
2.3.15 Scatter residuals



2.3.16 Learning curve — Group: 11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



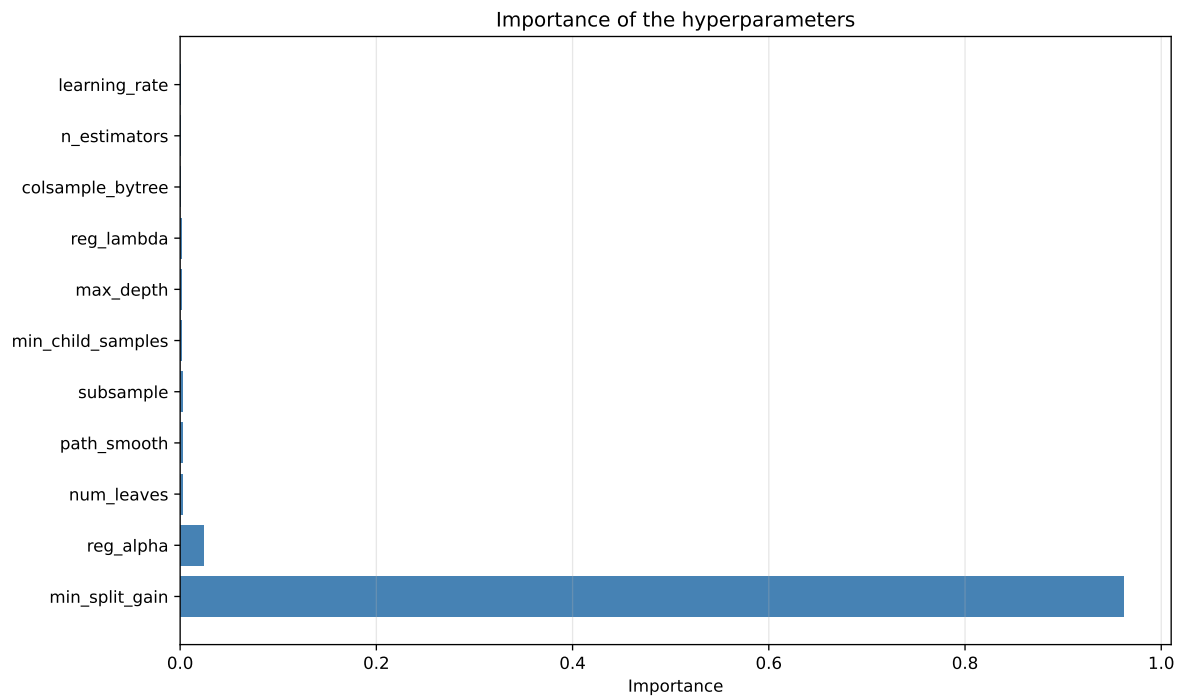
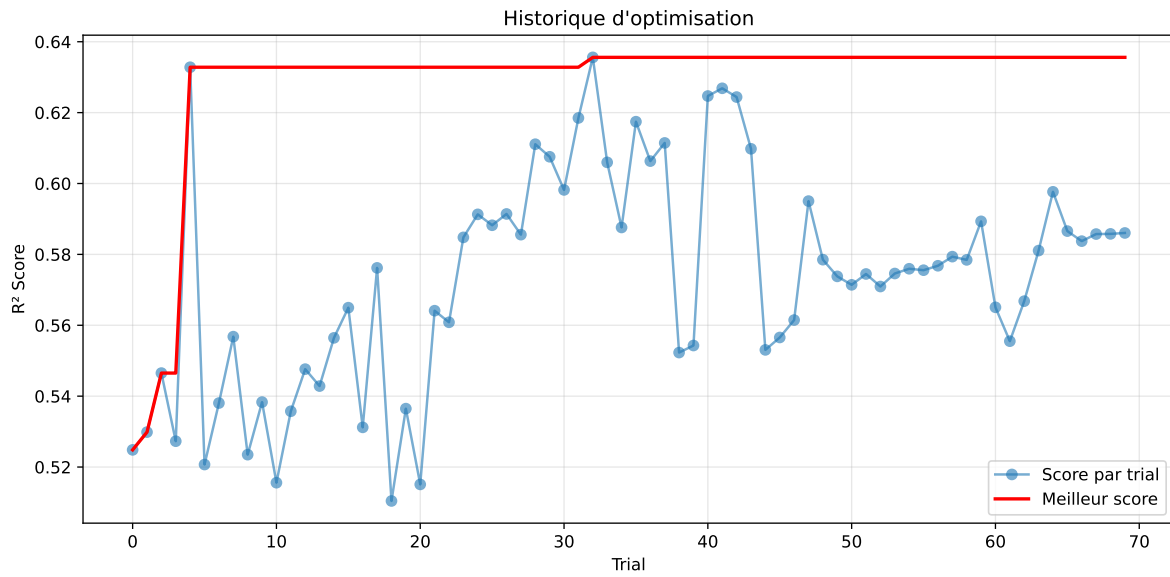
2.4 Model : LGBM

2.4.1 Gridsearch — Group: 3

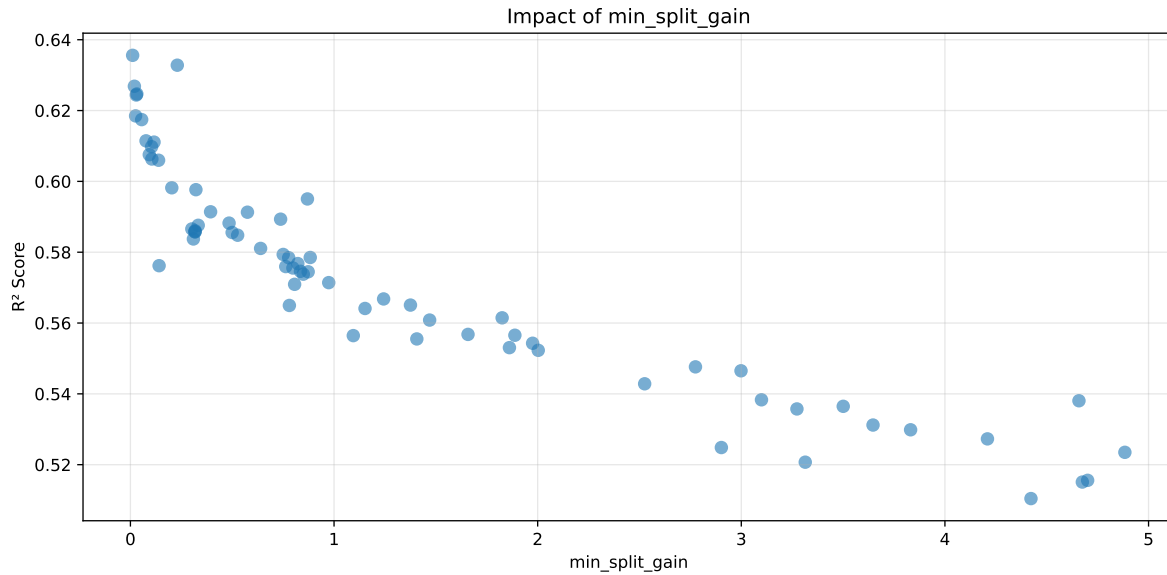
Len(group) : 1356, Len(training) : 950, Len(test) : 406

Distribution y_train vs y_test: y_train: mean=2.317, std=0.708, min=1.041, max=6.086
y_test: mean=2.318, std=0.740, min=1.068, max=5.447

===== TOP Importance FEATURES =====
feature importance r_umidity 72 r_NH3 42 r_temperature 27 r_THI 24 r_CO2 4



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6438	0.5714	0.6557	0.0381	0.3541	0.6692	0.0977	1.1711	1.1267
2	0.6499	0.5755	0.6607	0.0372	0.3525	0.6737	0.0981	1.1705	1.1292
3	0.6487	0.5746	0.6598	0.0369	0.3524	0.6728	0.0982	1.1708	1.1289
4	0.6491	0.5745	0.66	0.0365	0.3507	0.6725	0.098	1.1706	1.1298
5	0.6483	0.5738	0.659	0.0367	0.3504	0.6712	0.0974	1.1698	1.1298
6	0.6514	0.5759	0.6623	0.0375	0.3495	0.675	0.099	1.1719	1.131
7	0.6322	0.5615	0.6454	0.0387	0.3492	0.6566	0.0951	1.1694	1.126
8	0.6551	0.5785	0.6634	0.0361	0.3487	0.6744	0.0959	1.1658	1.1324
9	0.6265	0.5566	0.6412	0.0421	0.3467	0.6564	0.0998	1.1793	1.1257
10	0.6532	0.5762	0.6641	0.0378	0.3451	0.6851	0.1089	1.1889	1.1337
---	---	---	---	---	---	---	---	---	---
70	0.7686	0.6356	0.7283	0.0321	0.2366	0.7388	0.1032	1.1624	1.2092

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.06439845828749653, 'reg__num_leaves': 39, 'reg__subsample': 0.6458795907800848, 'reg__colsample_bytree': 0.8094101327740247, 'reg__min_child_samples': 16, 'reg__reg_alpha': 4.227379152005538, 'reg__reg_lambda': 37.54902264320484, 'reg__min_split_gain': 0.9738364606064964, 'reg__path_smooth': 4.358127236283715}

Rank 2: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.06455982724720963, 'reg__num_leaves': 42, 'reg__subsample': 0.6700746683672629,

'reg__colsample_bytree': 0.8167135038009465, 'reg__min_child_samples': 15, 'reg__reg_alpha': 4.384636494665206, 'reg__reg_lambda': 37.463953746740344, 'reg__min_split_gain': 0.7978904533557163, 'reg__path_smooth': 4.656997916314734}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.06371170941800725, 'reg__num_leaves': 39, 'reg__subsample': 0.6699953458557266, 'reg__colsample_bytree': 0.7207505780402307, 'reg__min_child_samples': 15, 'reg__reg_alpha': 4.3765565777632425, 'reg__reg_lambda': 37.390620130734824, 'reg__min_split_gain': 0.8350798035433348, 'reg__path_smooth': 4.535663114274643}

Rank 4: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.030984568436428925, 'reg__num_leaves': 42, 'reg__subsample': 0.6518050901752701, 'reg__colsample_bytree': 0.8181855639775194, 'reg__min_child_samples': 15, 'reg__reg_alpha': 4.332032070011099, 'reg__reg_lambda': 27.82282323766205, 'reg__min_split_gain': 0.872905814481186, 'reg__path_smooth': 4.221200756315664}

Rank 5: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.03835226139140415, 'reg__num_leaves': 42, 'reg__subsample': 0.6581174509595373, 'reg__colsample_bytree': 0.71331672830569, 'reg__min_child_samples': 16, 'reg__reg_alpha': 4.285251181160183, 'reg__reg_lambda': 37.02299623689498, 'reg__min_split_gain': 0.8488719644178531, 'reg__path_smooth': 4.213693909140222}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.06406000412484027, 'reg__num_leaves': 44, 'reg__subsample': 0.67465492589569, 'reg__colsample_bytree': 0.810915202736558, 'reg__min_child_samples': 15, 'reg__reg_alpha': 4.264260649460563, 'reg__reg_lambda': 37.57342720748057, 'reg__min_split_gain': 0.7627570571765452, 'reg__path_smooth': 4.58627209978228}

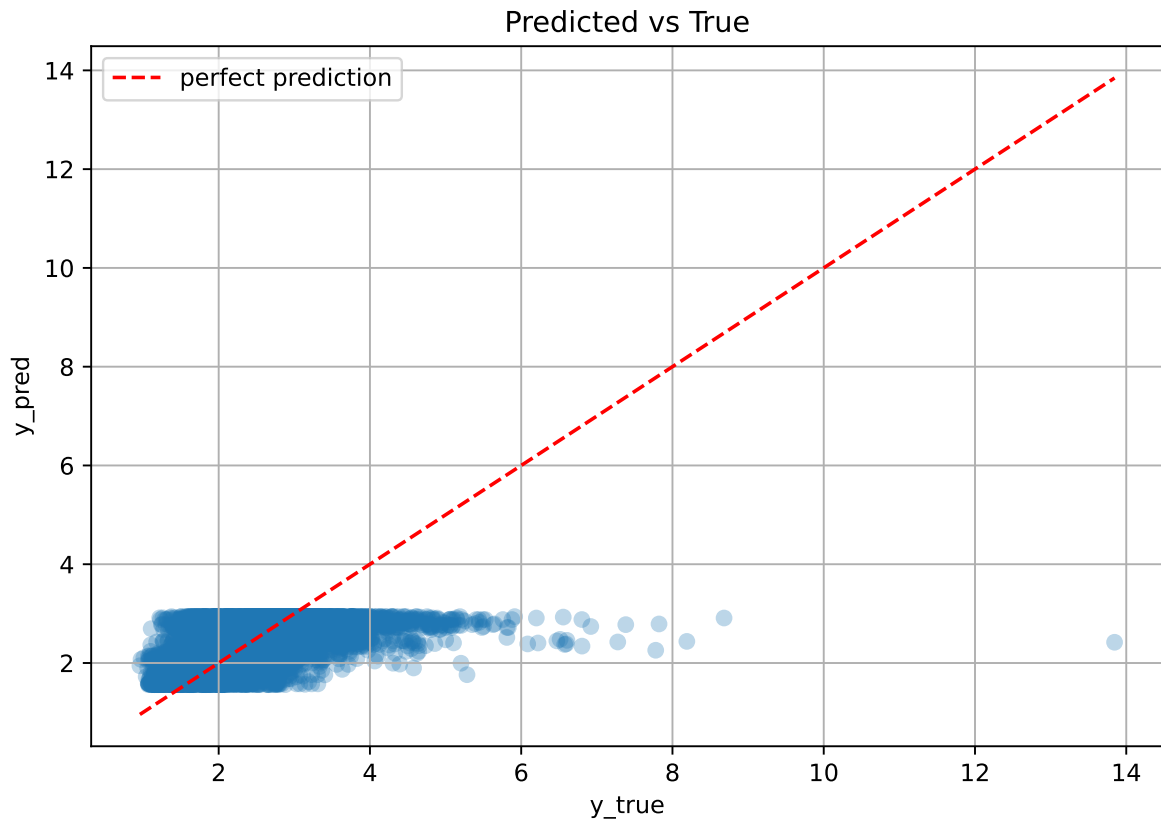
Rank 7: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.036041918117023, 'reg__num_leaves': 56, 'reg__subsample': 0.6496398821663146, 'reg__colsample_bytree': 0.7188818411565281, 'reg__min_child_samples': 21, 'reg__reg_alpha': 4.219882680021165, 'reg__reg_lambda': 10.462933026110893, 'reg__min_split_gain': 1.8255123735552214, 'reg__path_smooth': 4.43045529390465}

Rank 8: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.03850291812679062, 'reg__num_leaves': 43, 'reg__subsample': 0.6634953520179327, 'reg__colsample_bytree': 0.7081319405675965, 'reg__min_child_samples': 21, 'reg__reg_alpha': 4.247024913853151, 'reg__reg_lambda': 9.712430239335337, 'reg__min_split_gain': 0.8830833442397785, 'reg__path_smooth': 4.423724309796411}

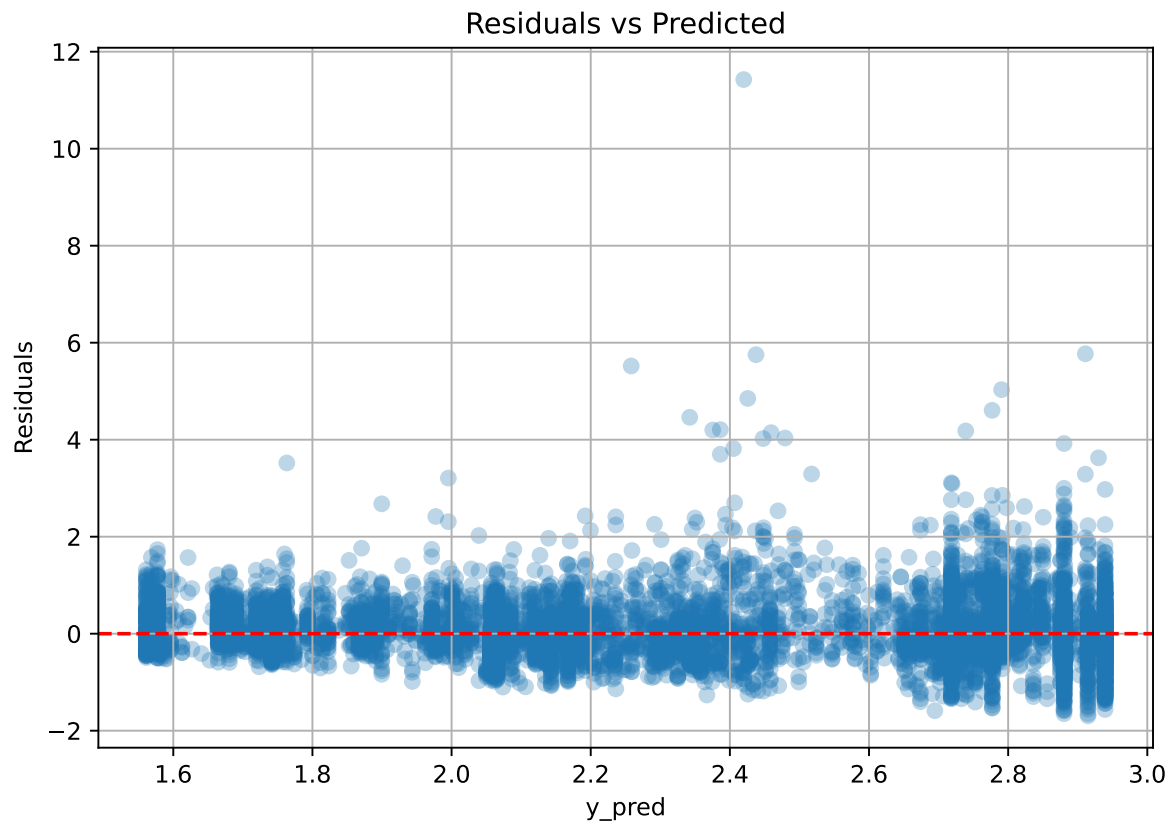
Rank 9: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.09991214495586141, 'reg__num_leaves': 57, 'reg__subsample': 0.6493418376796486, 'reg__colsample_bytree': 0.8103465923323926, 'reg__min_child_samples': 21, 'reg__reg_alpha': 4.074890413687864, 'reg__reg_lambda': 29.661953389608158, 'reg__min_split_gain': 1.8880410217228385, 'reg__path_smooth': 4.171558010532843}

Rank 10: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.014892676269611832, 'reg__num_leaves': 34, 'reg__subsample': 0.5980164861507173, 'reg__colsample_bytree': 0.7266093989220606, 'reg__min_child_samples': 44, 'reg__reg_alpha': 9.615034143397834, 'reg__reg_lambda': 34.69494990611248, 'reg__min_split_gain': 0.14083798420266802, 'reg__path_smooth': 3.5696136582772686}

2.4.2 Scatter



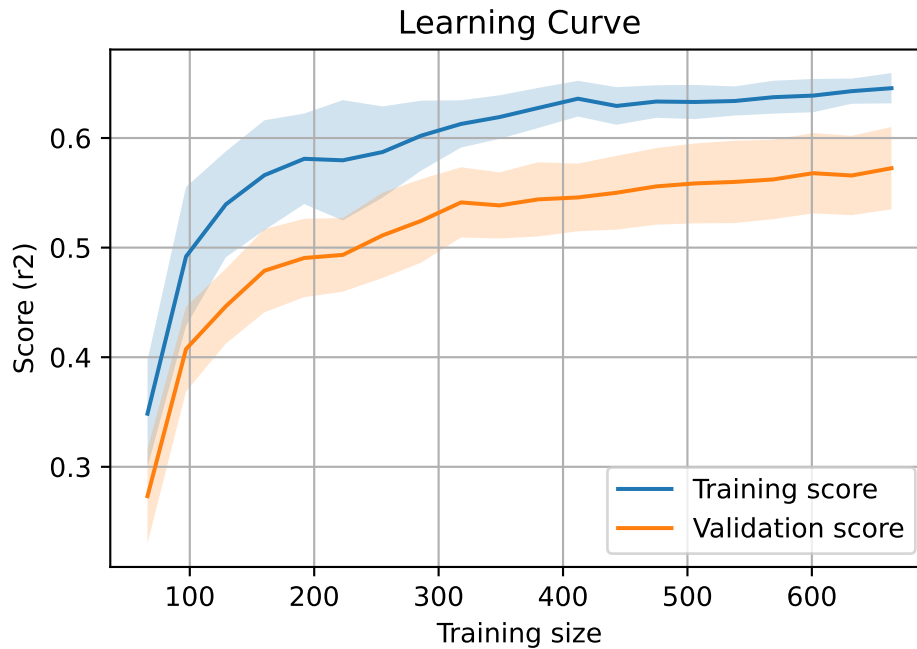
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 3

`Len(group)` : 1356, `Len(training)` : 950,

`Len(training_real)` : 665, `Len(test_of_training)` : 285, `Len(test)` : 406

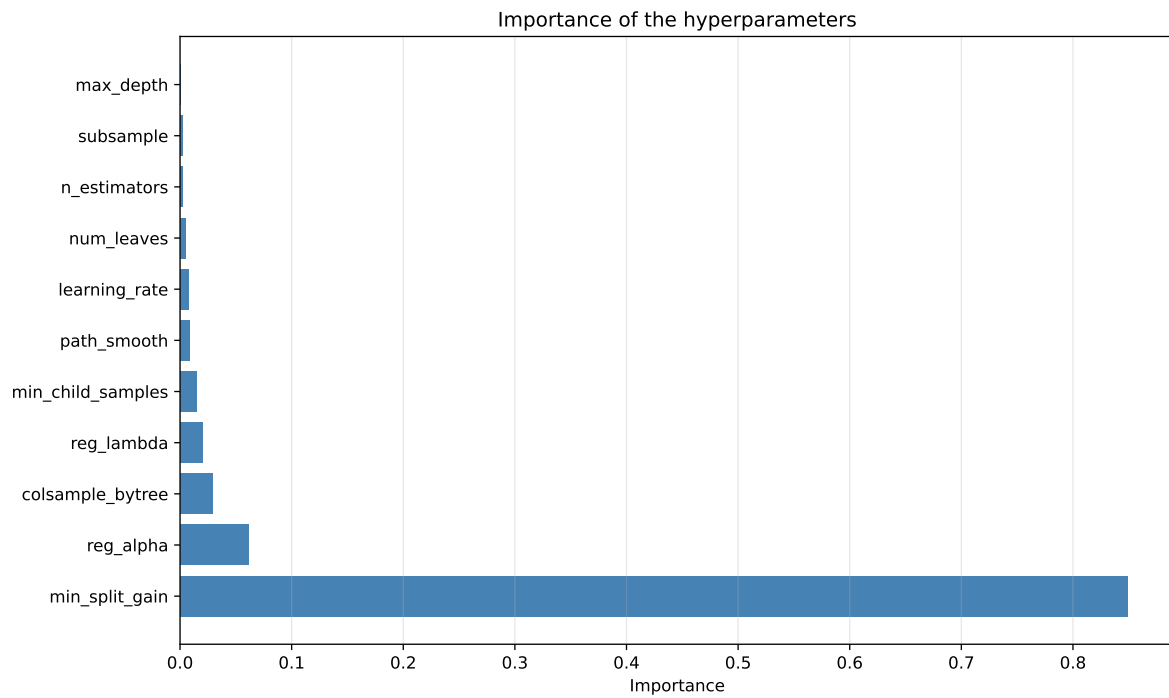
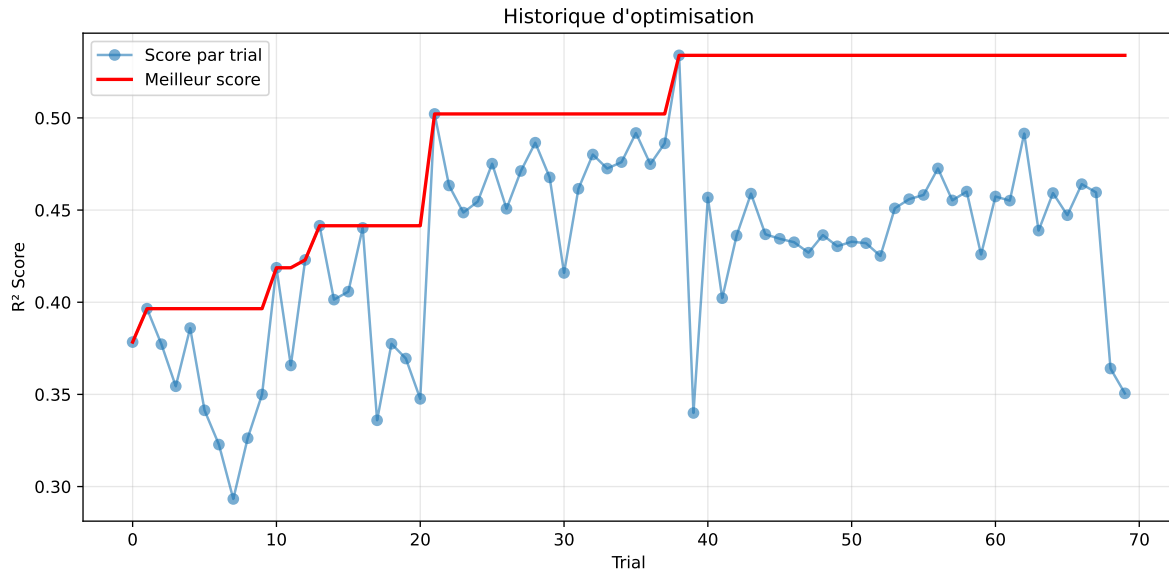


2.4.5 Gridsearch — Group: 4

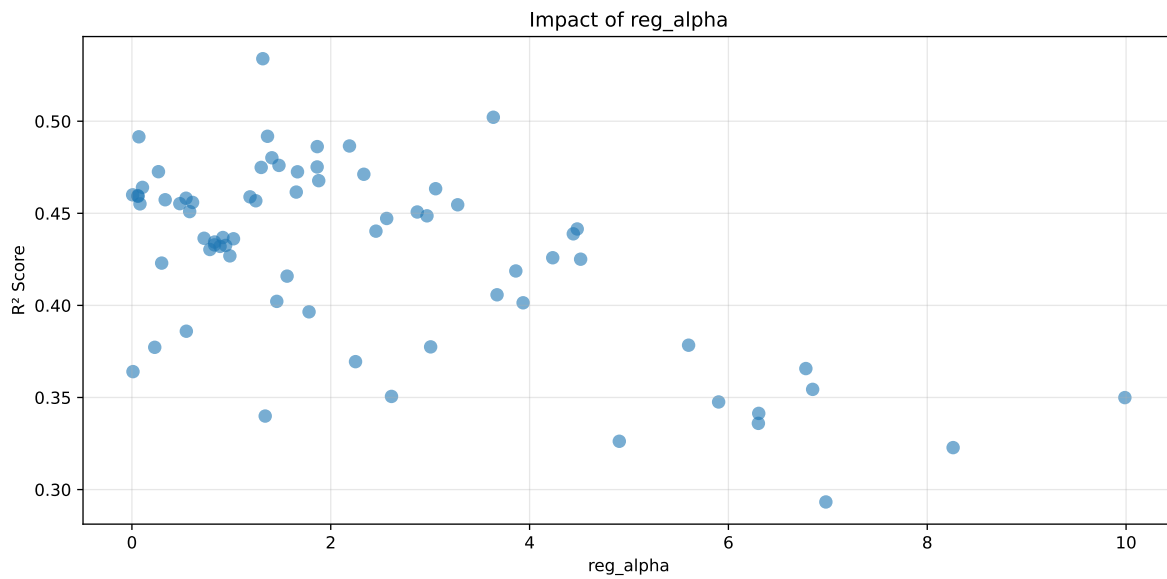
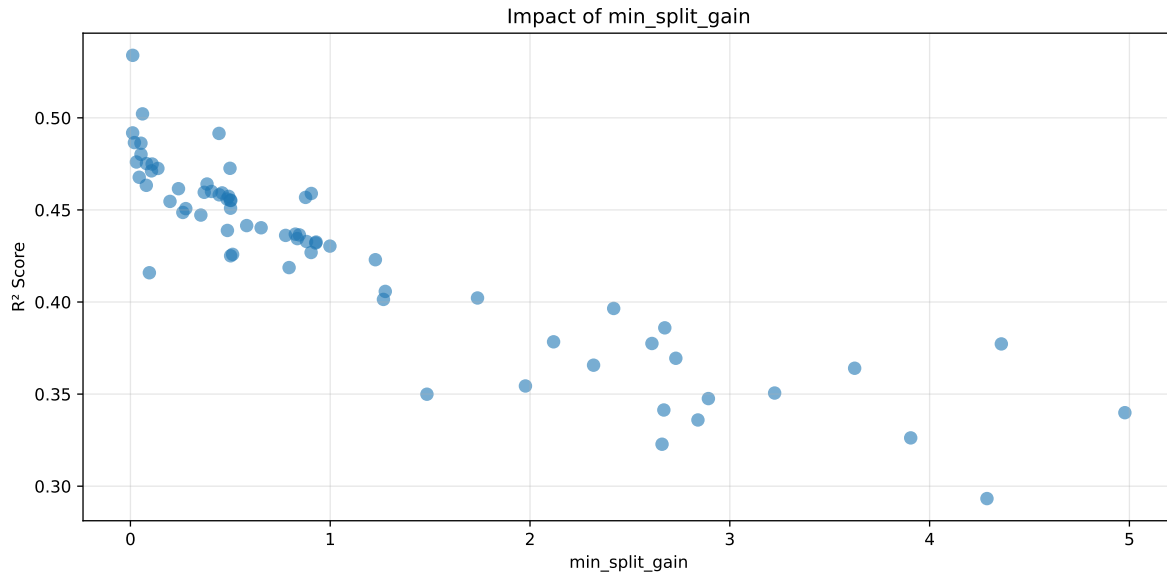
Len(group) : 1360, Len(training) : 952, Len(test) : 408

Distribution y_train vs y_test: y_train: mean=2.239, std=0.831, min=0.959, max=7.383
y_test: mean=2.202, std=0.791, min=1.062, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 242 r_THI 155 r_CO2 147 r_NH3 92 r_temperature 77

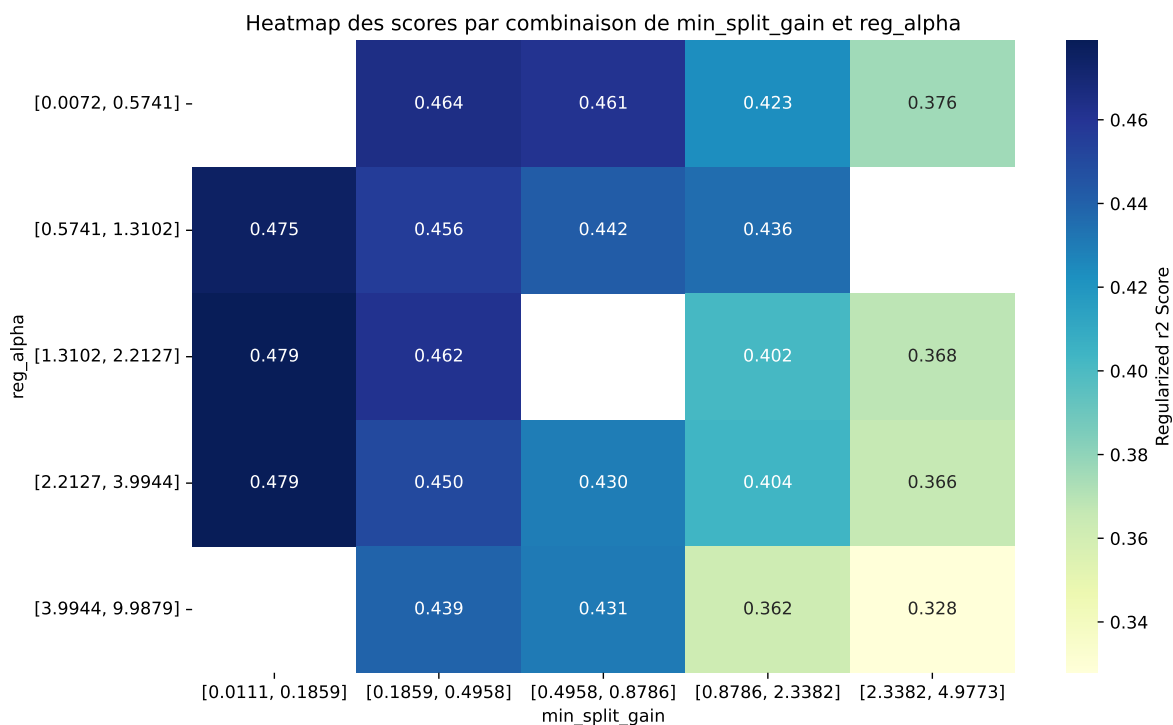


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of `observed=False` is deprecated and will change to `observed=True` in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4053	0.3414	0.3567	0.0412	0.1494	0.4055	0.0642	1.188	1.1874
2	0.3854	0.3262	0.3383	0.0427	0.1488	0.3886	0.0623	1.1911	1.1813
3	0.4562	0.3784	0.3947	0.0421	0.145	0.4377	0.0593	1.1567	1.2056
4	0.4009	0.336	0.3521	0.0433	0.1411	0.3969	0.0609	1.1814	1.1934
5	0.3441	0.2933	0.3025	0.0434	0.1408	0.3531	0.0599	1.2042	1.1733
6	0.4266	0.3544	0.3721	0.0459	0.1378	0.409	0.0546	1.154	1.2037
7	0.4176	0.3476	0.3641	0.043	0.1376	0.4037	0.0561	1.1614	1.2014
8	0.442	0.3657	0.3876	0.0366	0.1368	0.435	0.0693	1.1895	1.2086
9	0.4579	0.3775	0.3986	0.0374	0.1362	0.4359	0.0584	1.1548	1.2131
10	0.4212	0.3499	0.3729	0.0458	0.1362	0.4178	0.0678	1.1938	1.2036
---	---	---	---	---	---	---	---	---	---
70	0.8505	0.5339	0.5416	0.0256	-0.4157	0.5813	0.0474	1.0888	1.5928

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.014369576637020123, 'reg__num_leaves': 51, 'reg__subsample': 0.8622841075530947, 'reg__colsample_bytree': 0.8983829375648729, 'reg__min_child_samples': 43, 'reg__reg_alpha': 6.304965683690464, 'reg__reg_lambda': 37.908407970603726, 'reg__min_split_gain': 2.669794170308327, 'reg__path_smooth': 2.949132410422566}

Rank 2: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.0238000450419931, 'reg__num_leaves': 56, 'reg__subsample': 0.8679535393371401, 'reg__colsample_bytree': 0.7201578533860122, 'reg__min_child_samples': 20, 'reg__reg_alpha': 4.902493958829107, 'reg__reg_lambda': 34.757690273707134, 'reg__min_split_gain': 3.9053729156395285, 'reg__path_smooth': 1.3854902893594017}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.017674002346325563, 'reg__num_leaves': 60, 'reg__subsample': 0.8163651500826361, 'reg__colsample_bytree': 0.8522237698831805, 'reg__min_child_samples': 13, 'reg__reg_alpha': 5.599421824675166, 'reg__reg_lambda': 22.539393566381623, 'reg__min_split_gain': 2.1178999882887197, 'reg__path_smooth': 1.404353093174127}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.010995930667347697, 'reg__num_leaves': 56, 'reg__subsample': 0.516488162213834, 'reg__colsample_bytree': 0.5785907010951531, 'reg__min_child_samples': 27, 'reg__reg_alpha': 6.301452525151042, 'reg__reg_lambda': 32.494152188515145, 'reg__min_split_gain': 2.8404703514484604, 'reg__path_smooth': 1.050397243923341}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.011887128735056976, 'reg__num_leaves': 37, 'reg__subsample': 0.6251805036747234, 'reg__colsample_bytree': 0.6297440156789651, 'reg__min_child_samples': 17, 'reg__reg_alpha': 6.98025845898096, 'reg__reg_lambda': 46.02606290690619, 'reg__min_split_gain': 4.287099550580928, 'reg__path_smooth': 0.4489673942202732}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.01898890900025002, 'reg__num_leaves': 54, 'reg__subsample': 0.5373725524916763, 'reg__colsample_bytree': 0.5588973226856792, 'reg__min_child_samples': 28, 'reg__reg_alpha': 6.847231118767282, 'reg__reg_lambda': 42.4327099182998, 'reg__min_split_gain': 1.9770779981765259, 'reg__path_smooth': 2.661527300908934}

Rank 7: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.07204210204705366, 'reg__num_leaves': 38, 'reg__subsample': 0.7673873579054451, 'reg__colsample_bytree': 0.7271831789024619, 'reg__min_child_samples': 23, 'reg__reg_alpha': 5.901577273590778, 'reg__reg_lambda': 30.82923489602285, 'reg__min_split_gain': 2.8927492364395984, 'reg__path_smooth': 0.4770923634604368}

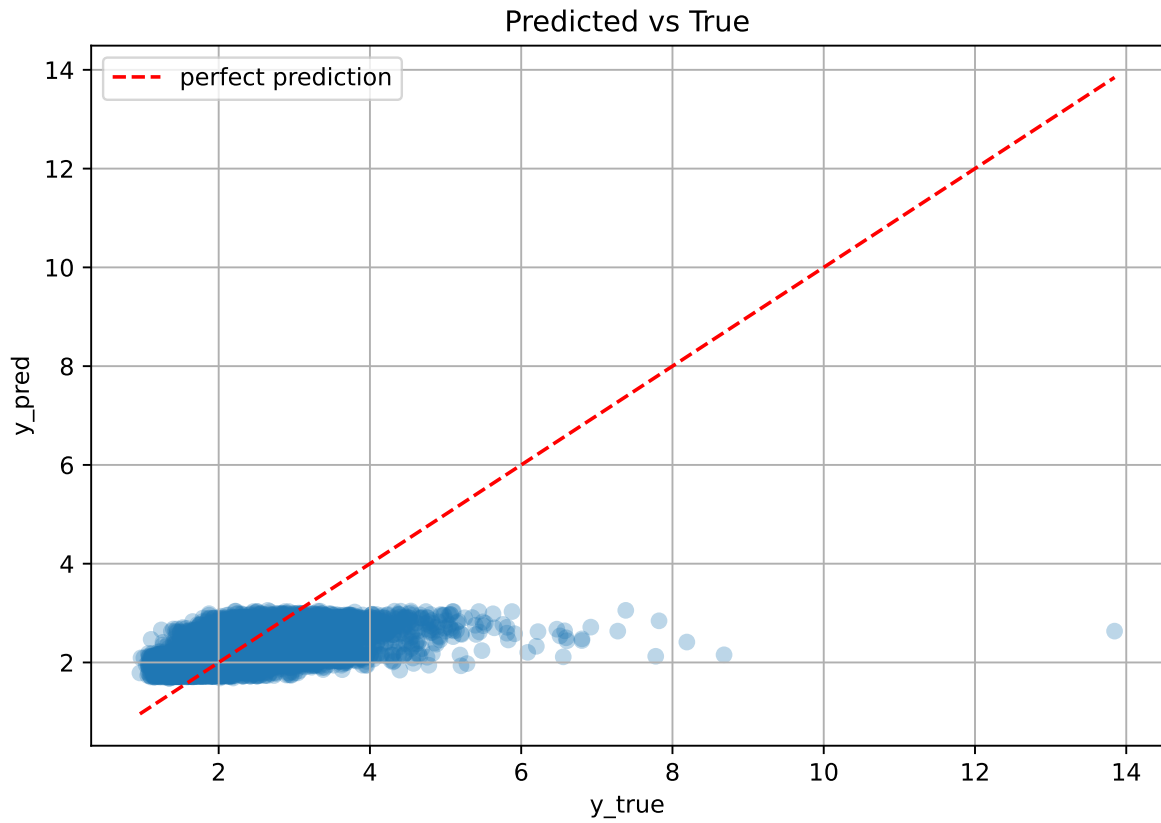
Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.013098626332747568, 'reg__num_leaves': 26, 'reg__subsample': 0.8053281793251781, 'reg__colsample_bytree': 0.7295571270730677, 'reg__min_child_samples': 24, 'reg__reg_alpha': 6.779027235720806, 'reg__reg_lambda': 11.973177420590542, 'reg__min_split_gain': 2.3180540298173242, 'reg__path_smooth': 1.5276785116366998}

Rank 9: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.02254654808039732, 'reg__num_leaves': 28, 'reg__subsample': 0.6555799411358907,

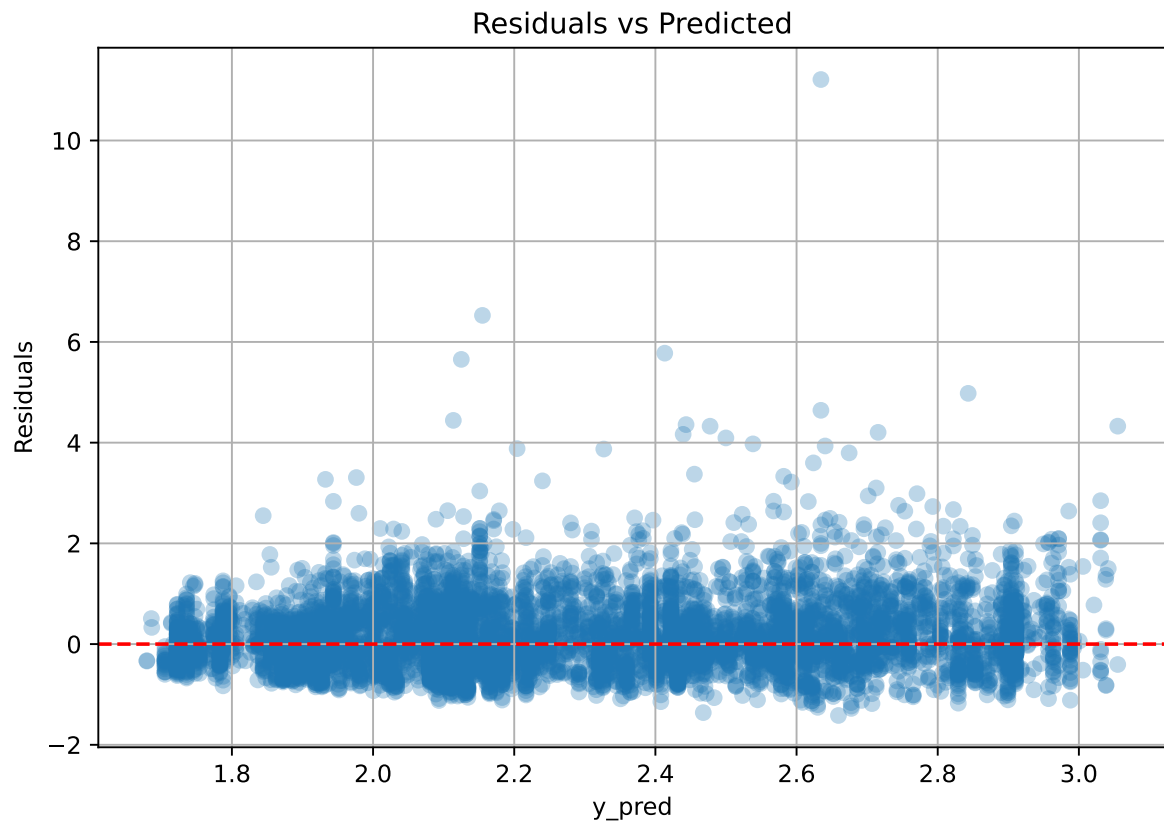
'reg__colsample_bytree': 0.8769672192819055, 'reg__min_child_samples': 25, 'reg__reg_alpha': 3.0047774586778386, 'reg__reg_lambda': 26.027950584344268, 'reg__min_split_gain': 2.610220561682908, 'reg__path_smooth': 4.437837447677641}

Rank 10: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.04953142783093542, 'reg__num_leaves': 61, 'reg__subsample': 0.5023778480595895, 'reg__colsample_bytree': 0.507839668533145, 'reg__min_child_samples': 21, 'reg__reg_alpha': 9.987919840242629, 'reg__reg_lambda': 32.68101690058394, 'reg__min_split_gain': 1.4840709064455249, 'reg__path_smooth': 0.3708852784272493}

2.4.6 Scatter



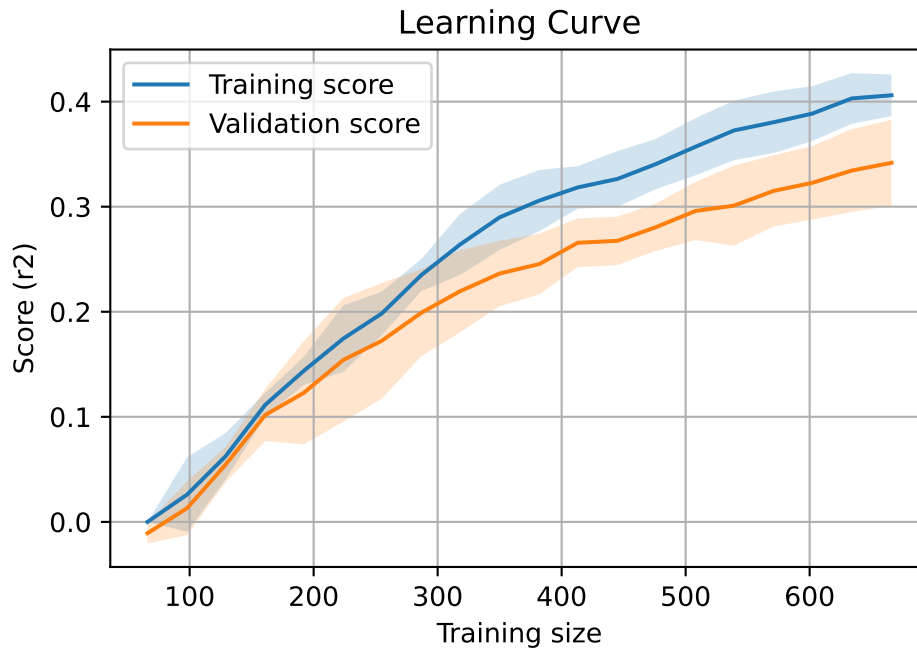
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 4

`Len(group)` : 1360, `Len(training)` : 952,

`Len(training_real)` : 667, `Len(test_of_training)` : 285, `Len(test)` : 408

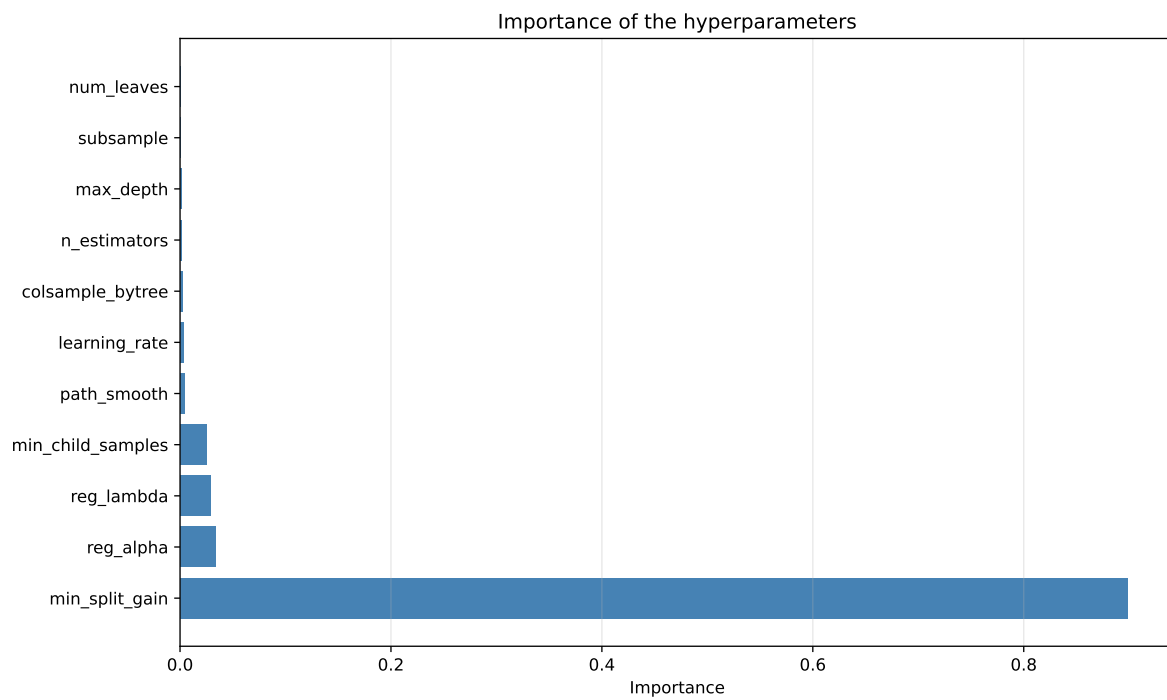
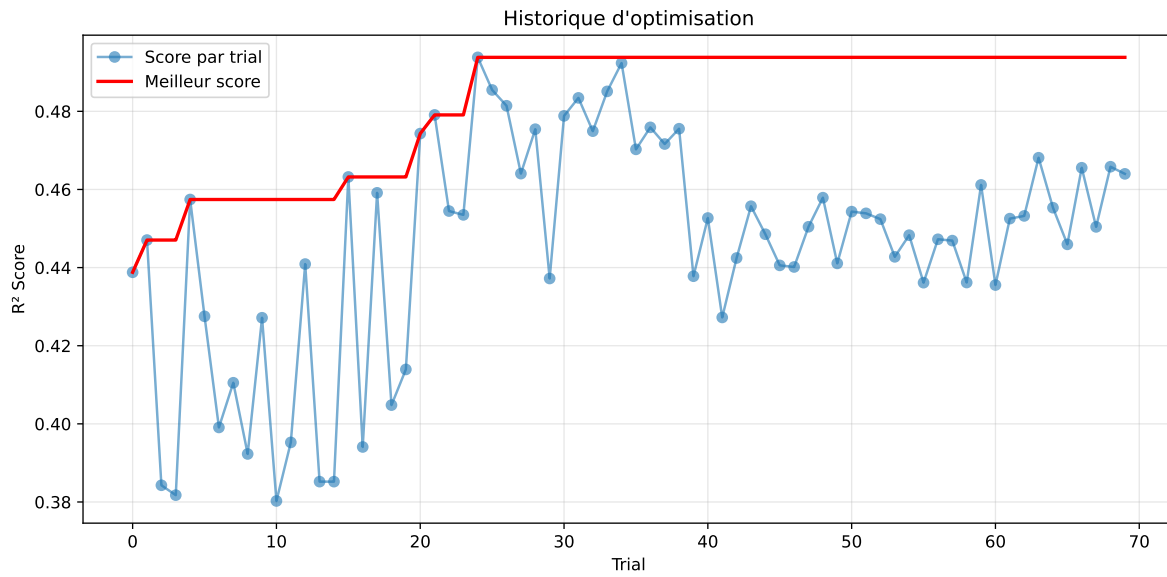


2.4.9 Gridsearch — Group: 5

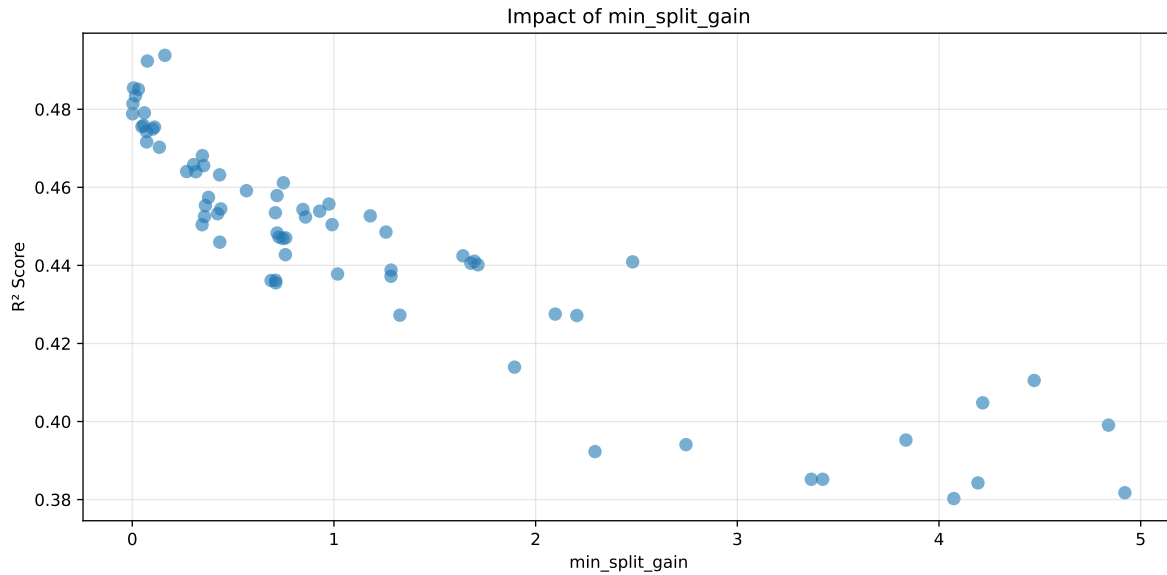
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

===== TOP Importance FEATURES =====
feature importance r_umidity 340 r_NH3 127 r_temperature 86 r_CO2 79 r_THI 10



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4241	0.3923	0.4457	0.0235	0.2968	0.4667	0.0744	1.1897	1.0812
2	0.4304	0.3953	0.4553	0.0234	0.2899	0.4772	0.082	1.2074	1.0888
3	0.4432	0.4048	0.4639	0.0245	0.2896	0.4818	0.077	1.1903	1.0948
4	0.4179	0.3852	0.4458	0.0236	0.2871	0.4664	0.0812	1.2108	1.0849
5	0.4302	0.3941	0.4562	0.0229	0.2856	0.4782	0.0841	1.2134	1.0918
6	0.4185	0.3852	0.4462	0.024	0.2855	0.4676	0.0824	1.2138	1.0863
7	0.4173	0.3843	0.4424	0.022	0.2852	0.4602	0.0759	1.1976	1.0859
8	0.4143	0.3817	0.4428	0.0227	0.2842	0.4657	0.084	1.22	1.0851
9	0.4753	0.4272	0.4864	0.0242	0.2831	0.5069	0.0797	1.1865	1.1124
10	0.4126	0.3803	0.4419	0.0231	0.2831	0.4624	0.0822	1.2161	1.0852
---	---	---	---	---	---	---	---	---	---
70	0.6542	0.4924	0.5612	0.0292	0.0069	0.585	0.0926	1.1881	1.3286

Hyperparameters:

Rank 1: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.011964579364842017, 'reg__num_leaves': 39, 'reg__subsample': 0.5750274867632249, 'reg__colsample_bytree': 0.894065566453603, 'reg__min_child_samples': 39, 'reg__reg_alpha': 9.16719336052024, 'reg__reg_lambda': 43.036415465363106, 'reg__min_split_gain': 2.2943706248354605, 'reg__path_smooth': 0.9687913937408932}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.03503158932669869, 'reg__num_leaves': 15, 'reg__subsample': 0.675139472671423,

'reg__colsample_bytree': 0.8236246082962031, 'reg__min_child_samples': 31, 'reg__reg_alpha': 3.569315254400239, 'reg__reg_lambda': 48.76919486177043, 'reg__min_split_gain': 3.8360598586349424, 'reg__path_smooth': 1.282864233764884}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.03710425832808846, 'reg__num_leaves': 49, 'reg__subsample': 0.8313512896595172, 'reg__colsample_bytree': 0.8676693931751251, 'reg__min_child_samples': 12, 'reg__reg_alpha': 0.43686255793787265, 'reg__reg_lambda': 39.59484575089608, 'reg__min_split_gain': 4.216784372709867, 'reg__path_smooth': 4.24647921637681}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.017898864590565573, 'reg__num_leaves': 22, 'reg__subsample': 0.6031500665416467, 'reg__colsample_bytree': 0.5269849861279549, 'reg__min_child_samples': 23, 'reg__reg_alpha': 8.87355397497458, 'reg__reg_lambda': 45.20201748833022, 'reg__min_split_gain': 3.3668519172087037, 'reg__path_smooth': 4.341726483497855}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.019208319259402747, 'reg__num_leaves': 51, 'reg__subsample': 0.7729538674740043, 'reg__colsample_bytree': 0.5572941434759514, 'reg__min_child_samples': 36, 'reg__reg_alpha': 9.056224856027258, 'reg__reg_lambda': 23.99212785239081, 'reg__min_split_gain': 2.7456491174071918, 'reg__path_smooth': 3.0042017847908946}

Rank 6: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.03041024298522407, 'reg__num_leaves': 33, 'reg__subsample': 0.7307709066132572, 'reg__colsample_bytree': 0.5357182761167646, 'reg__min_child_samples': 43, 'reg__reg_alpha': 9.246238912416258, 'reg__reg_lambda': 41.90259355133416, 'reg__min_split_gain': 3.423495579656075, 'reg__path_smooth': 0.07419939274442178}

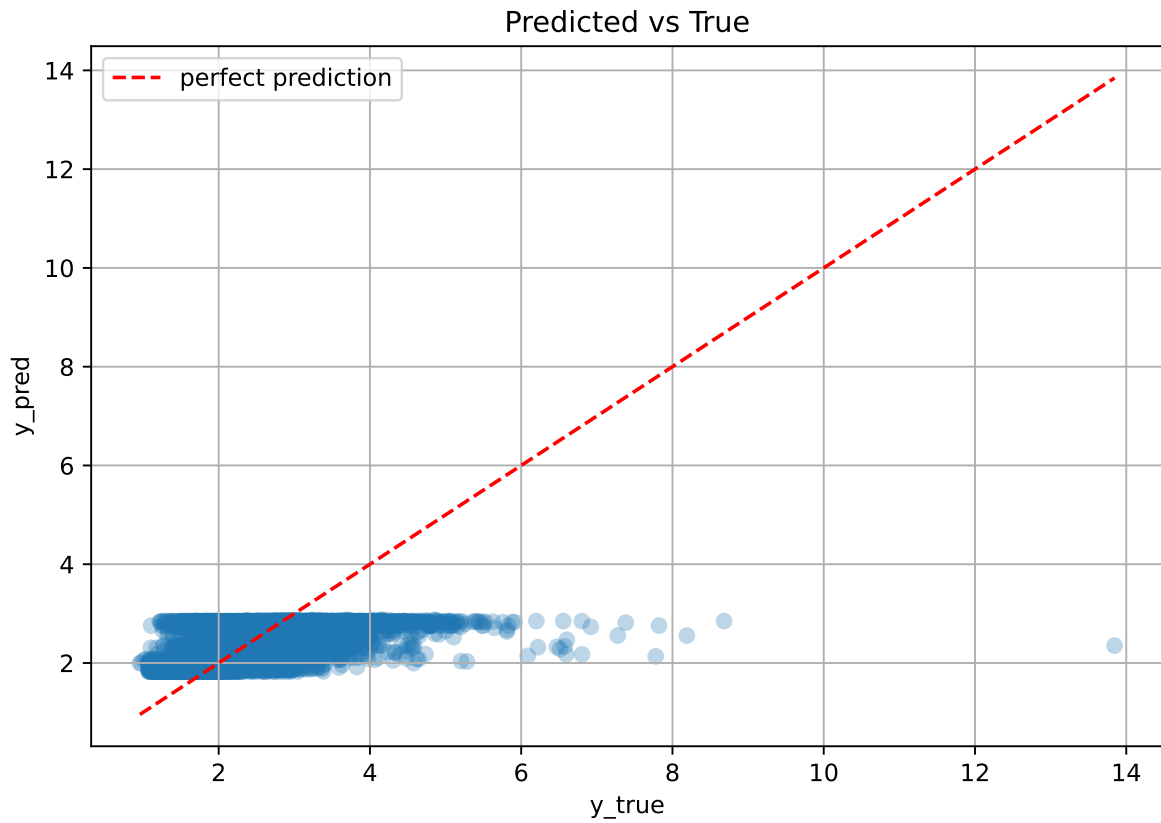
Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.01082224080897455, 'reg__num_leaves': 51, 'reg__subsample': 0.8400910266315418, 'reg__colsample_bytree': 0.674426173291734, 'reg__min_child_samples': 23, 'reg__reg_alpha': 2.8918030684040708, 'reg__reg_lambda': 31.330872077905997, 'reg__min_split_gain': 4.193186867147995, 'reg__path_smooth': 4.442914641524449}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.02383652100791048, 'reg__num_leaves': 42, 'reg__subsample': 0.7730084790214327, 'reg__colsample_bytree': 0.6204213377214353, 'reg__min_child_samples': 45, 'reg__reg_alpha': 7.021811415179754, 'reg__reg_lambda': 18.9104366259155, 'reg__min_split_gain': 4.921694850453035, 'reg__path_smooth': 4.300809253800763}

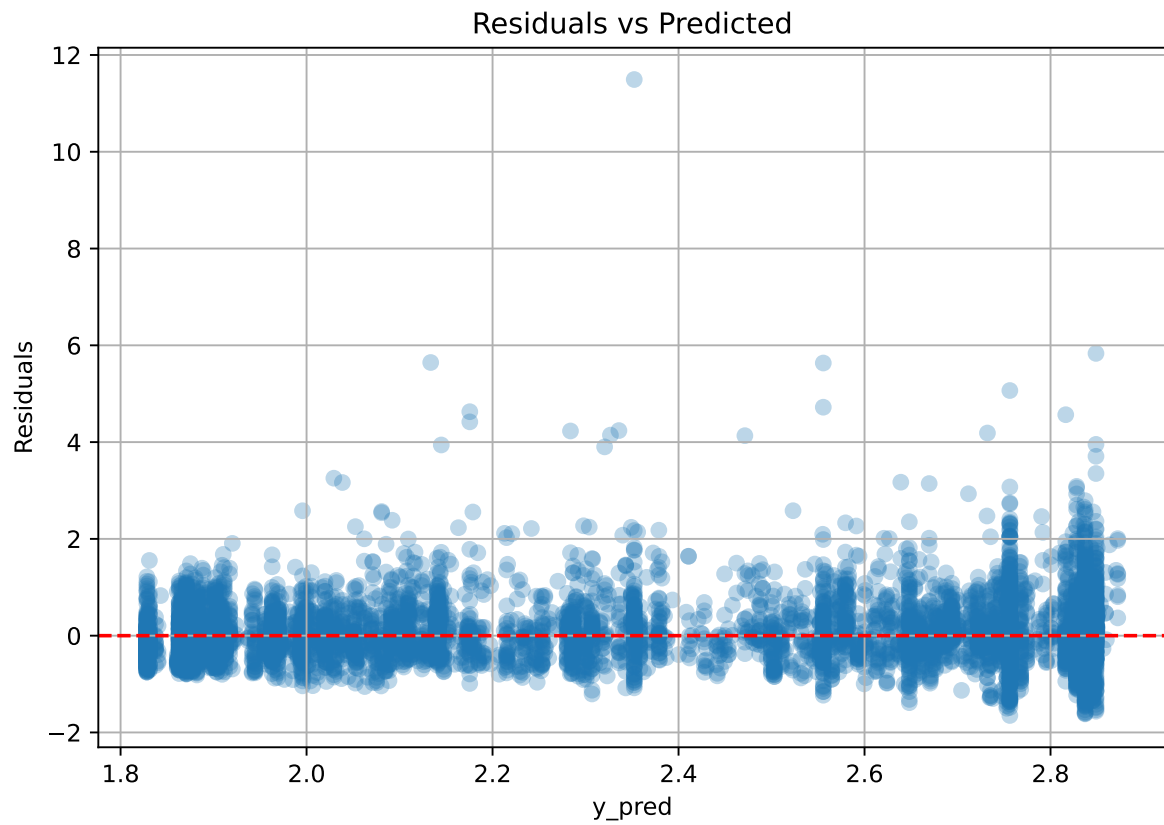
Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.07629747677651541, 'reg__num_leaves': 26, 'reg__subsample': 0.553784349624588, 'reg__colsample_bytree': 0.71297987023854, 'reg__min_child_samples': 18, 'reg__reg_alpha': 6.904048688387435, 'reg__reg_lambda': 32.40208966003114, 'reg__min_split_gain': 1.3270339087106162, 'reg__path_smooth': 1.8619309736094196}

Rank 10: {'reg__n_estimators': 200, 'reg__max_depth': 7, 'reg__learning_rate': 0.05190360819367957, 'reg__num_leaves': 21, 'reg__subsample': 0.6917794361962776, 'reg__colsample_bytree': 0.6704932251525614, 'reg__min_child_samples': 49, 'reg__reg_alpha': 7.968139805899338, 'reg__reg_lambda': 49.201968611877625, 'reg__min_split_gain': 4.0734986025358975, 'reg__path_smooth': 4.700741220257157}

2.4.10 Scatter



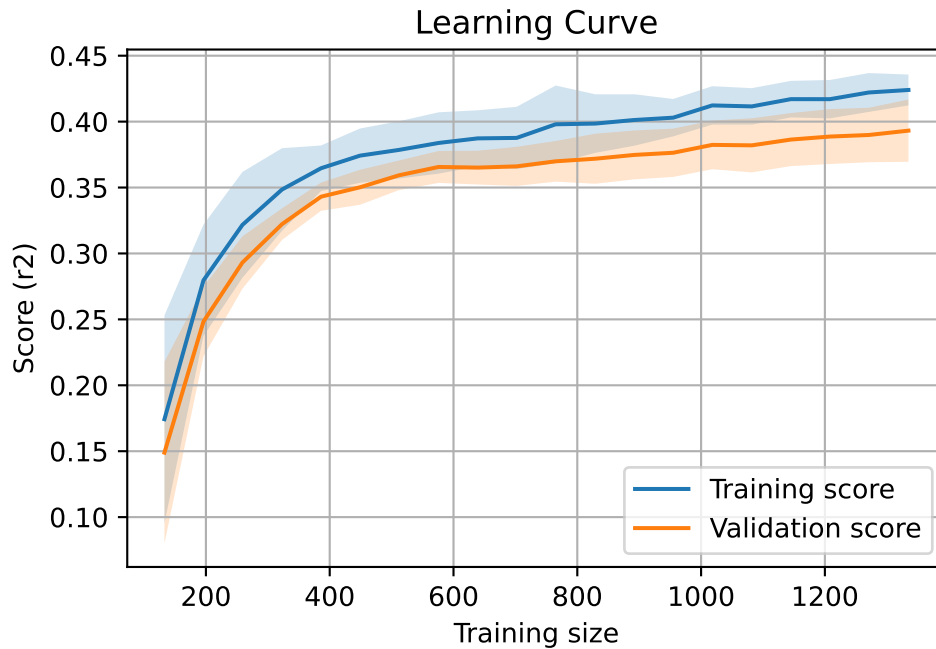
2.4.11 Scatter residuals



2.4.12 Learning curve — Group: 5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

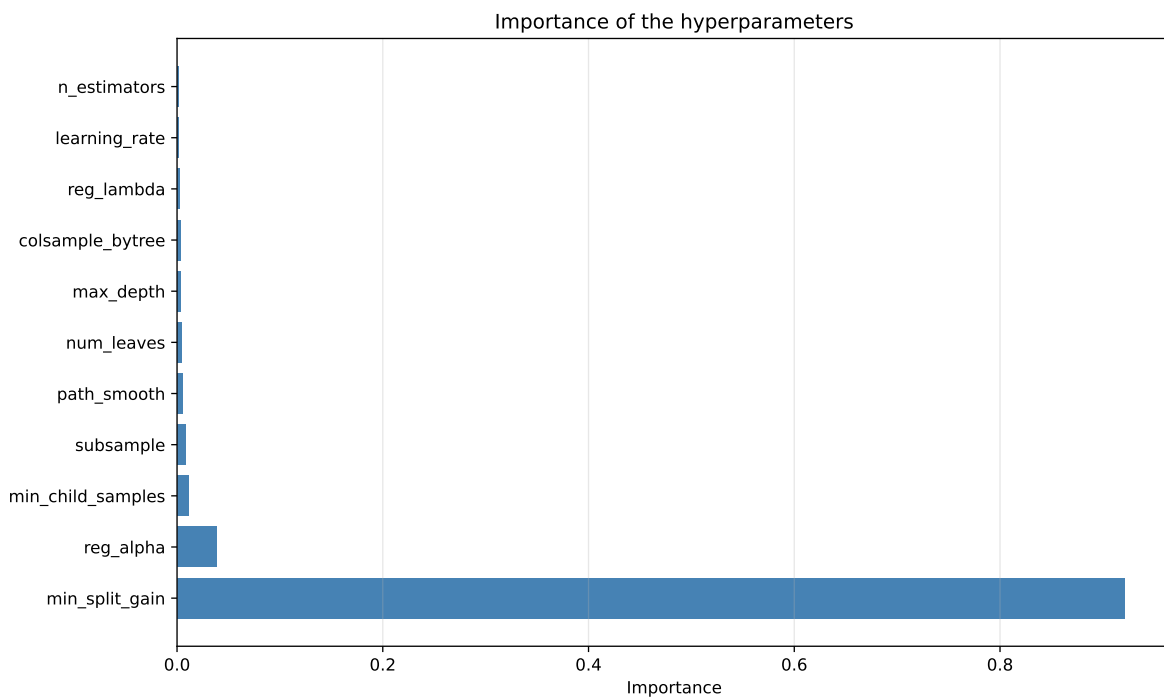
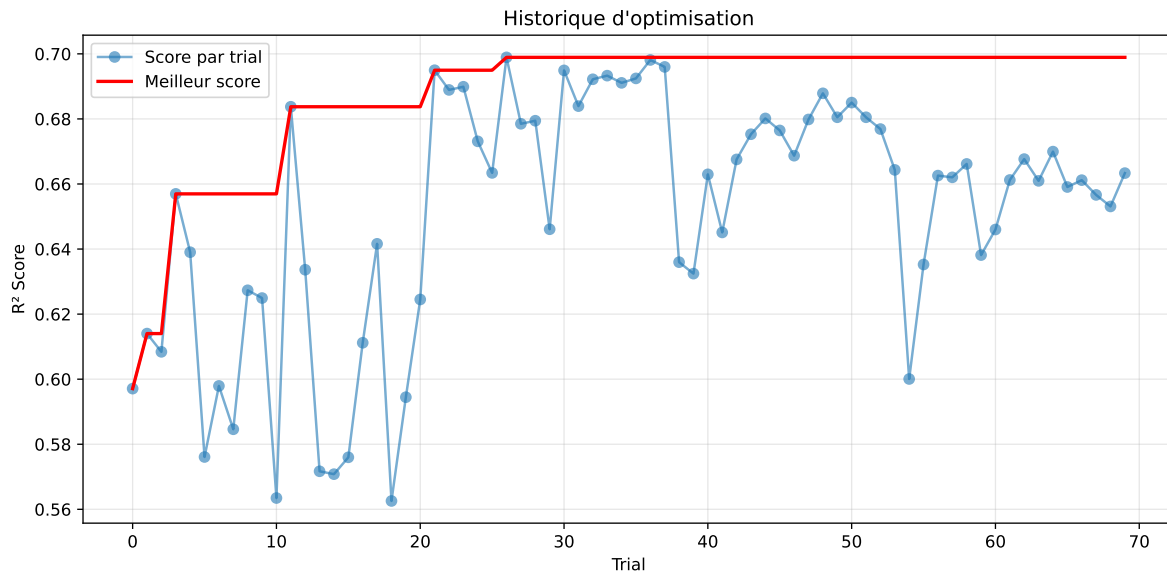


2.4.13 Gridsearch — Group: 11

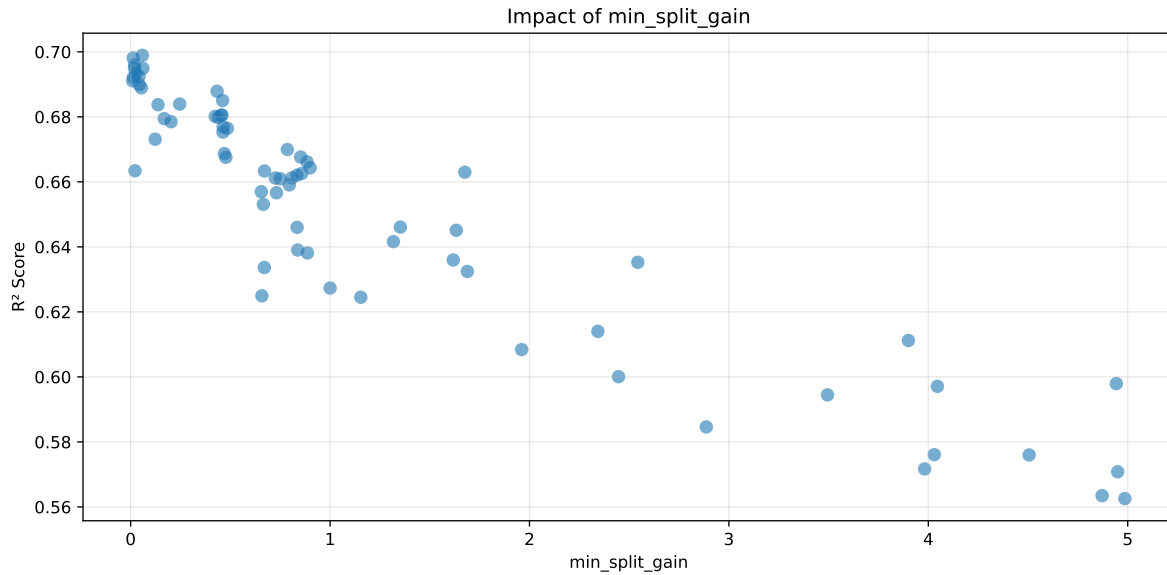
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

===== TOP Importance FEATURES =====
feature importance r_umidity 208 r_NH3 199 r_CO2 174 r_THI 110 r_temperature 81



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6847	0.6531	0.5517	0.0139	0.6531	0.5719	-0.0812	0.8756	1.0484
2	0.671	0.6461	0.5454	0.0139	0.6461	0.5589	-0.0872	0.865	1.0386
3	0.6767	0.646	0.5526	0.0136	0.646	0.57	-0.076	0.8823	1.0474
4	0.6738	0.6451	0.5478	0.0129	0.6451	0.5657	-0.0794	0.877	1.0445
5	0.6665	0.6416	0.5453	0.0129	0.6416	0.5585	-0.0831	0.8705	1.0388
6	0.662	0.639	0.5452	0.0136	0.639	0.5613	-0.0777	0.8784	1.0359
7	0.6658	0.6381	0.5421	0.0134	0.6381	0.5613	-0.0768	0.8796	1.0434
8	0.659	0.636	0.5415	0.013	0.636	0.5585	-0.0774	0.8782	1.0362
9	0.6594	0.6353	0.5376	0.0144	0.6353	0.5621	-0.0731	0.8849	1.038
10	0.6609	0.6336	0.5439	0.0105	0.6336	0.5644	-0.0693	0.8907	1.043
---	---	---	---	---	---	---	---	---	---
70	0.8586	0.695	0.5752	0.0109	0.204	0.5923	-0.1027	0.8522	1.2355

Hyperparameters:

Rank 1: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.052998039314212105, 'reg__num_leaves': 53, 'reg__subsample': 0.8315636087775528, 'reg__colsample_bytree': 0.6781309776861574, 'reg__min_child_samples': 44, 'reg__reg_alpha': 3.0904236150918423, 'reg__reg_lambda': 8.681130106112636, 'reg__min_split_gain': 0.6653868600472284, 'reg__path_smooth': 3.68345903139847}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.06039851042645481, 'reg__num_leaves': 34, 'reg__subsample': 0.8908104443554214,

'reg__colsample_bytree': 0.6929352211095419, 'reg__min_child_samples': 39, 'reg__reg_alpha': 2.6997735695737486, 'reg__reg_lambda': 5.626227198992428, 'reg__min_split_gain': 1.3524856427188796, 'reg__path_smooth': 4.6185172858422305}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.04609383304042299, 'reg__num_leaves': 50, 'reg__subsample': 0.8134461229819201, 'reg__colsample_bytree': 0.6438024681008568, 'reg__min_child_samples': 19, 'reg__reg_alpha': 4.355909096731496, 'reg__reg_lambda': 24.31309890644416, 'reg__min_split_gain': 0.8348638447181126, 'reg__path_smooth': 0.005880064745843232}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.03768701805917666, 'reg__num_leaves': 45, 'reg__subsample': 0.7883714814689133, 'reg__colsample_bytree': 0.7503252878040632, 'reg__min_child_samples': 24, 'reg__reg_alpha': 3.701189712162107, 'reg__reg_lambda': 14.134881925493426, 'reg__min_split_gain': 1.6328314365031327, 'reg__path_smooth': 4.920294377135372}

Rank 5: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.010647855119989297, 'reg__num_leaves': 28, 'reg__subsample': 0.5021560867917966, 'reg__colsample_bytree': 0.797522153764816, 'reg__min_child_samples': 45, 'reg__reg_alpha': 2.807575531397747, 'reg__reg_lambda': 33.49582685127519, 'reg__min_split_gain': 1.3174551370241638, 'reg__path_smooth': 1.712036967117219}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.010786797903629922, 'reg__num_leaves': 41, 'reg__subsample': 0.5841463272193548, 'reg__colsample_bytree': 0.531640037175902, 'reg__min_child_samples': 29, 'reg__reg_alpha': 7.053786545859488, 'reg__reg_lambda': 0.8873899026113519, 'reg__min_split_gain': 0.8379024894734727, 'reg__path_smooth': 0.5263592632612352}

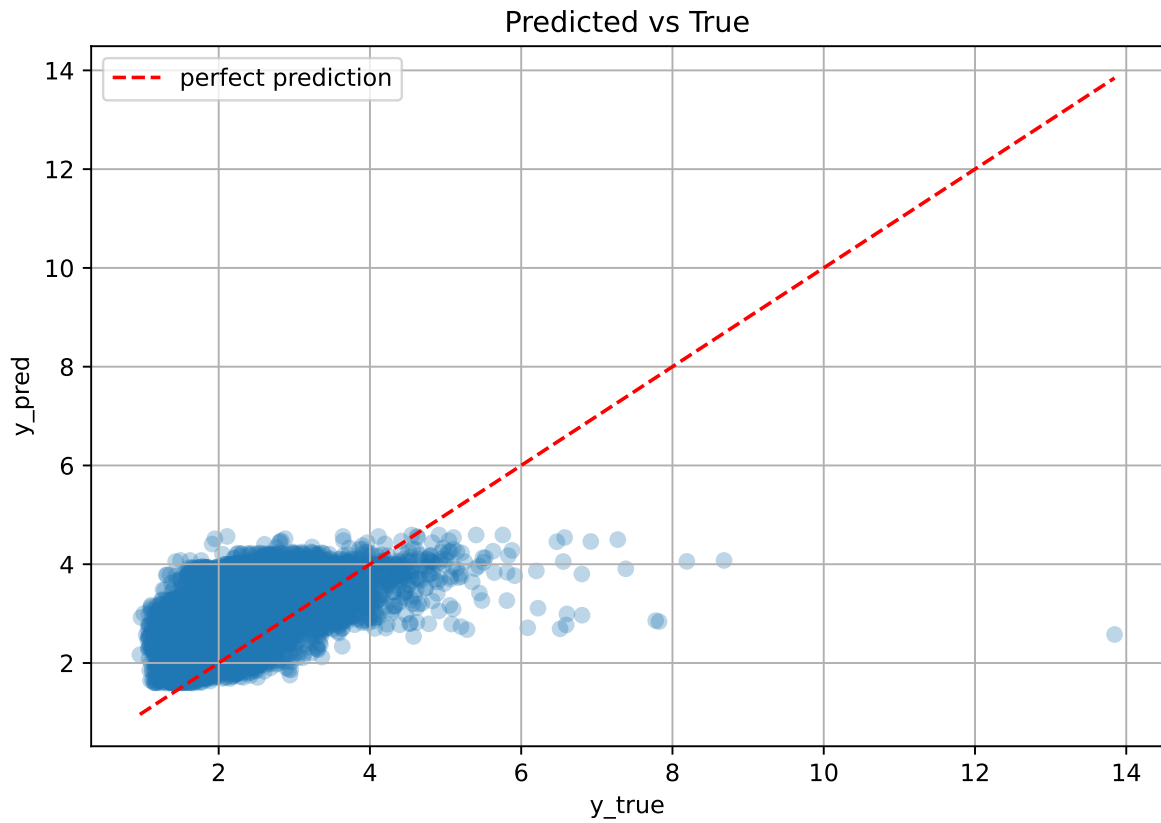
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.049637620943394675, 'reg__num_leaves': 50, 'reg__subsample': 0.8129738430767937, 'reg__colsample_bytree': 0.7176251685169835, 'reg__min_child_samples': 27, 'reg__reg_alpha': 9.769251331450711, 'reg__reg_lambda': 2.2011028061760127, 'reg__min_split_gain': 0.8867626299489658, 'reg__path_smooth': 0.023416038876329814}

Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.07080929746694824, 'reg__num_leaves': 47, 'reg__subsample': 0.7754774132925121, 'reg__colsample_bytree': 0.6616615902362429, 'reg__min_child_samples': 23, 'reg__reg_alpha': 3.7928880825530156, 'reg__reg_lambda': 11.153767627997457, 'reg__min_split_gain': 1.6177500156593778, 'reg__path_smooth': 3.059818951659014}

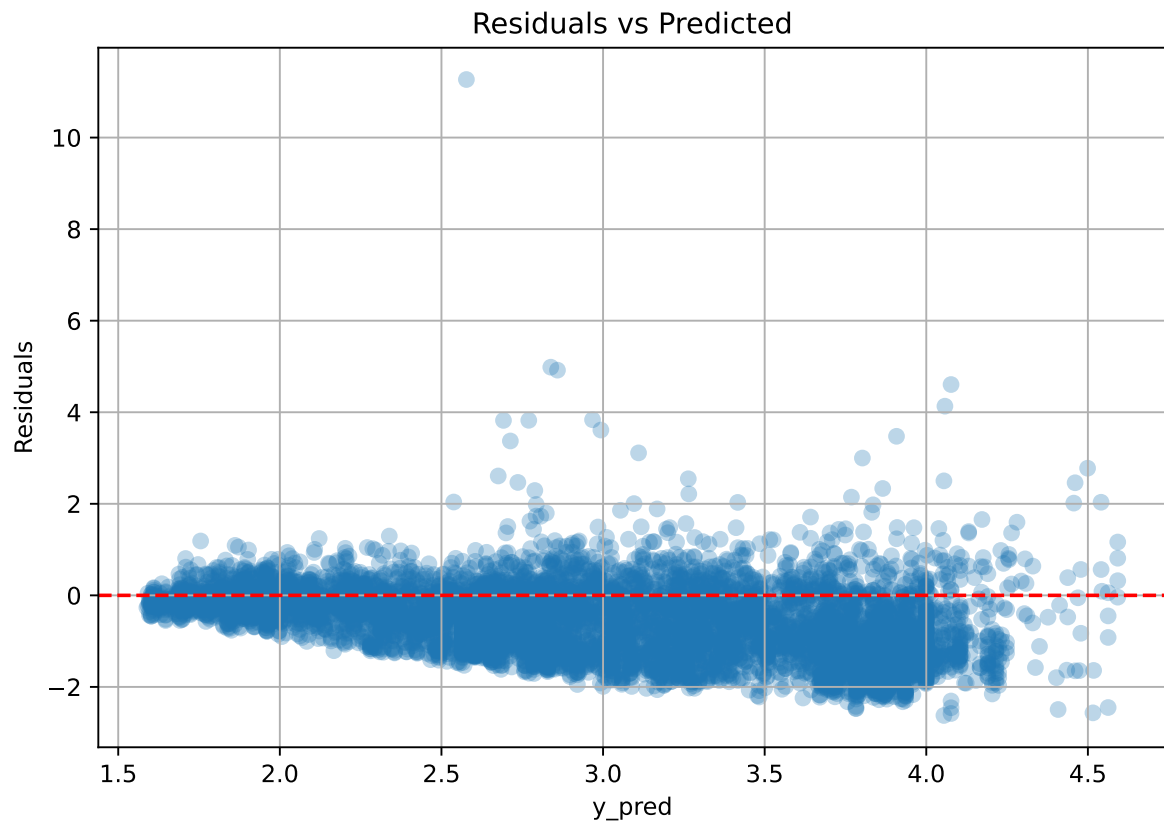
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.0459295067949791, 'reg__num_leaves': 49, 'reg__subsample': 0.8082116929696167, 'reg__colsample_bytree': 0.7205403711008868, 'reg__min_child_samples': 19, 'reg__reg_alpha': 4.74312254499941, 'reg__reg_lambda': 2.42286563889059, 'reg__min_split_gain': 2.5432034779759105, 'reg__path_smooth': 4.1238293303744085}

Rank 10: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.0209046188160892, 'reg__num_leaves': 24, 'reg__subsample': 0.5244537776507711, 'reg__colsample_bytree': 0.5076998252277924, 'reg__min_child_samples': 12, 'reg__reg_alpha': 7.298866564975745, 'reg__reg_lambda': 34.492316051267494, 'reg__min_split_gain': 0.6702344455437187, 'reg__path_smooth': 0.9748185813101379}

2.4.14 Scatter



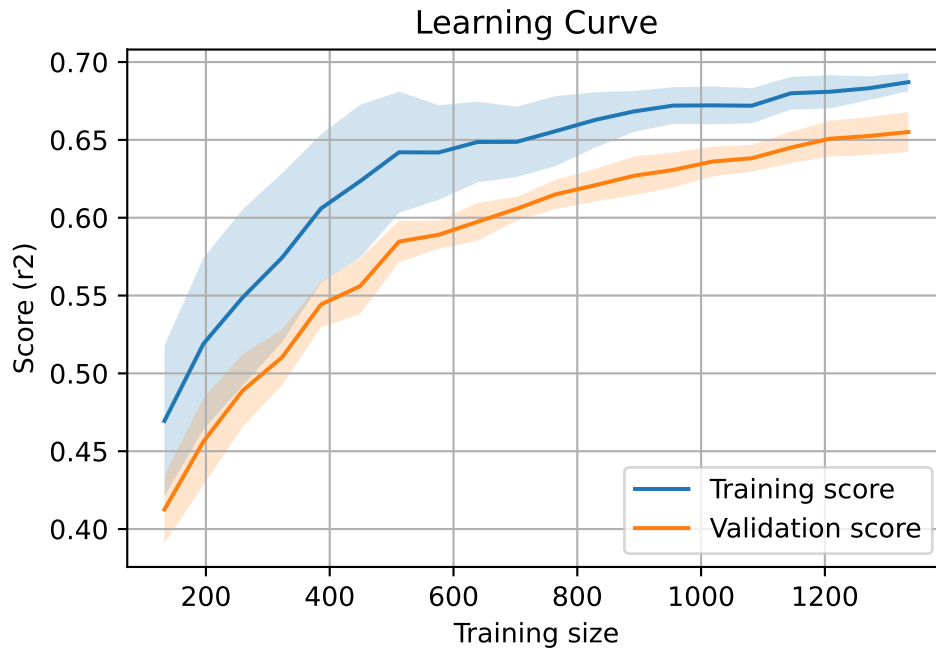
2.4.15 Scatter residuals



2.4.16 Learning curve — Group: 11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



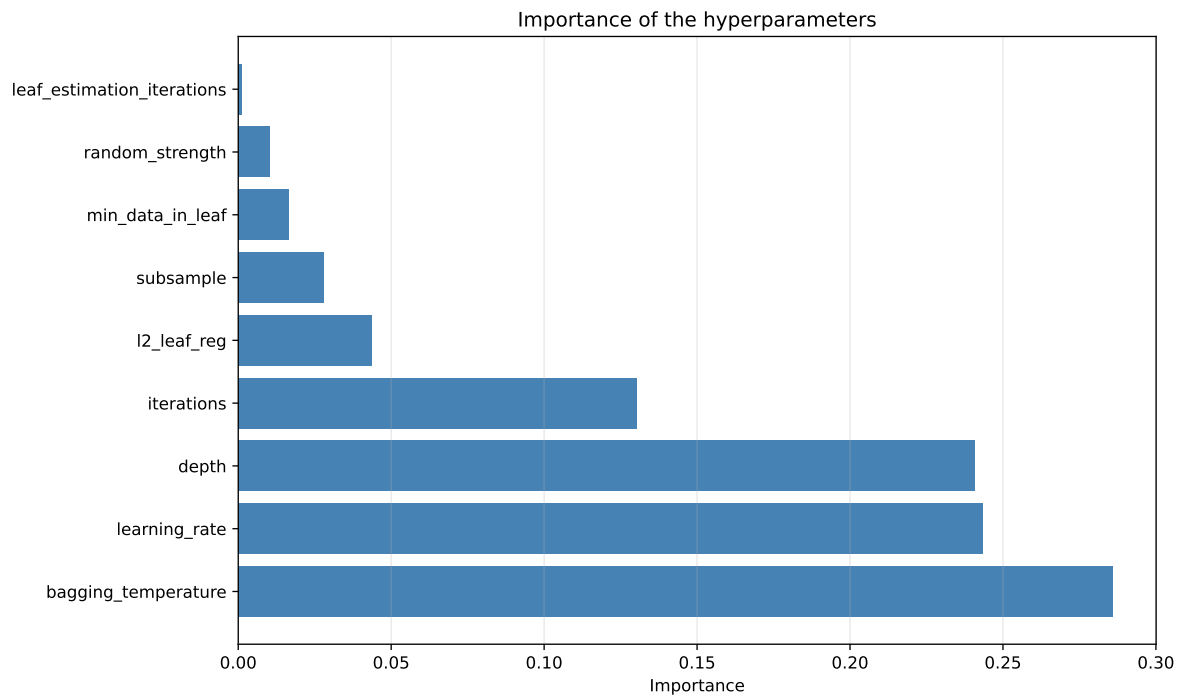
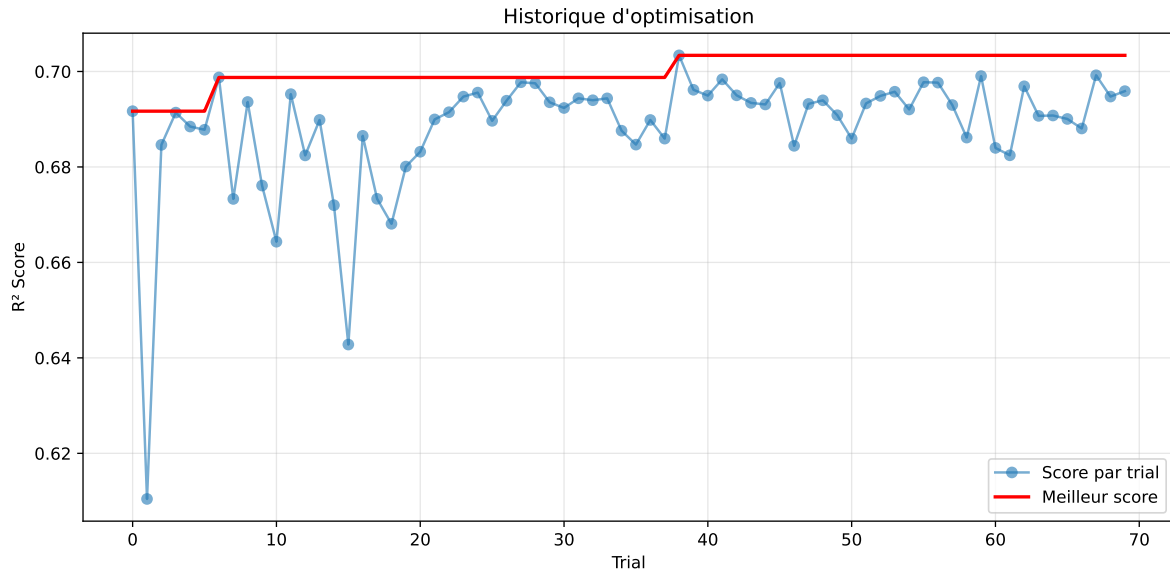
2.5 Model : CatBoost

2.5.1 Gridsearch — Group: 3

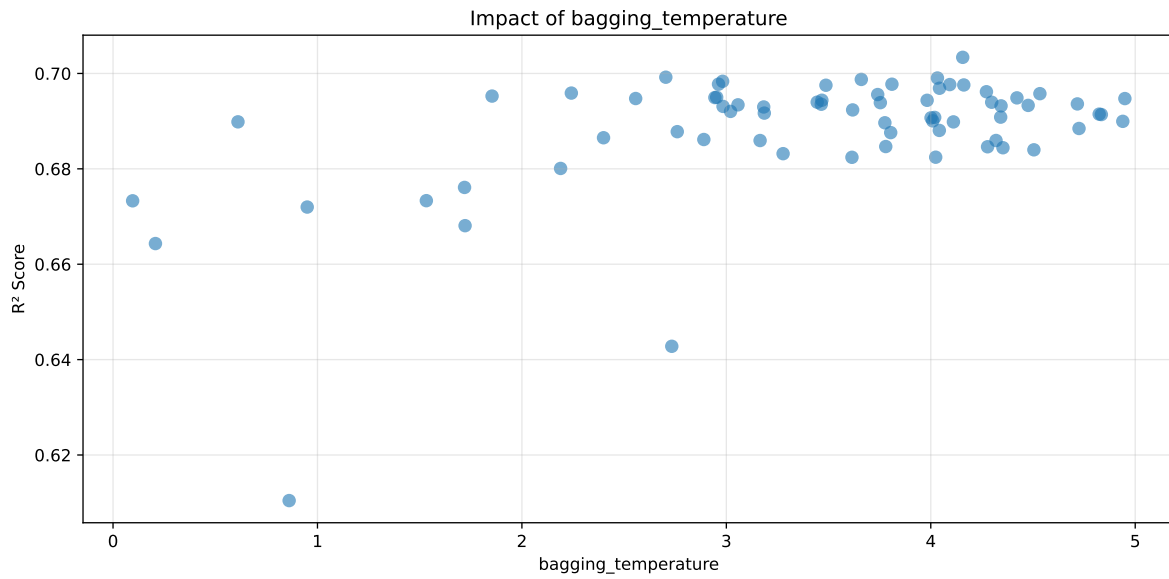
Len(group) : 1356, Len(training) : 950, Len(test) : 406

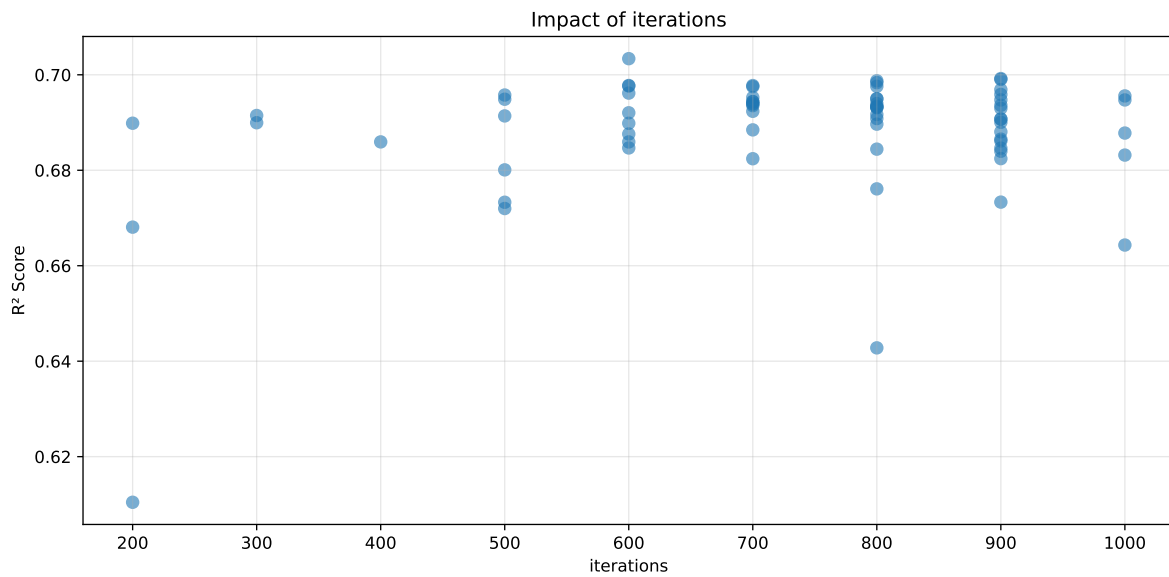
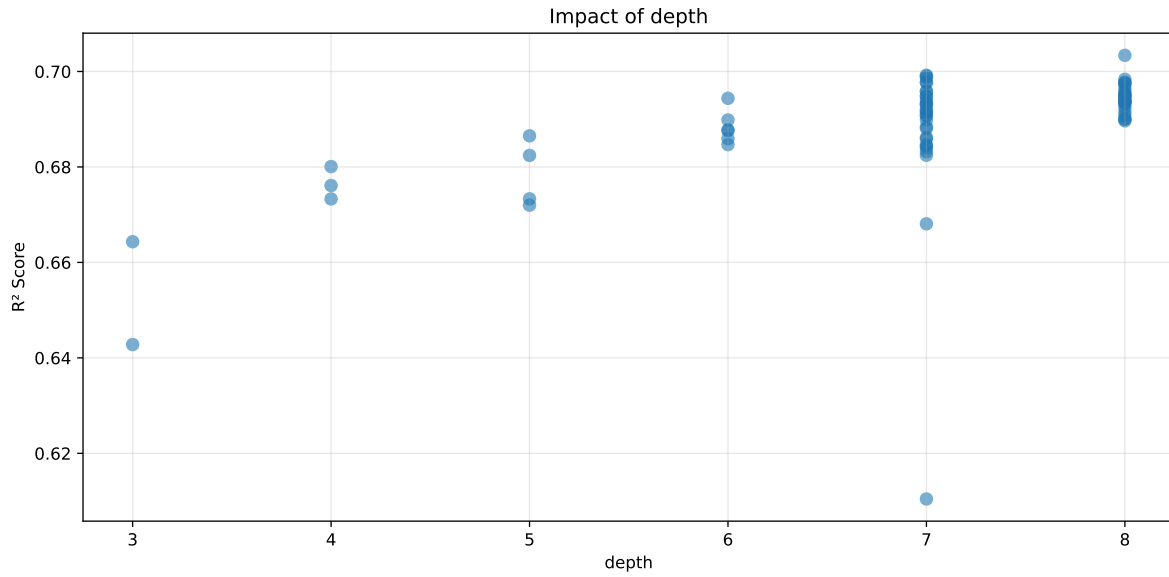
Distribution y_train vs y_test: y_train: mean=2.317, std=0.708, min=1.041, max=6.086
y_test: mean=2.318, std=0.740, min=1.068, max=5.447

===== TOP Importance FEATURES =====
feature importance r_umidity 54.109091 r_NH3 13.329538 r_temperature 12.905938 r_THI
11.902326 r_CO2 7.753107



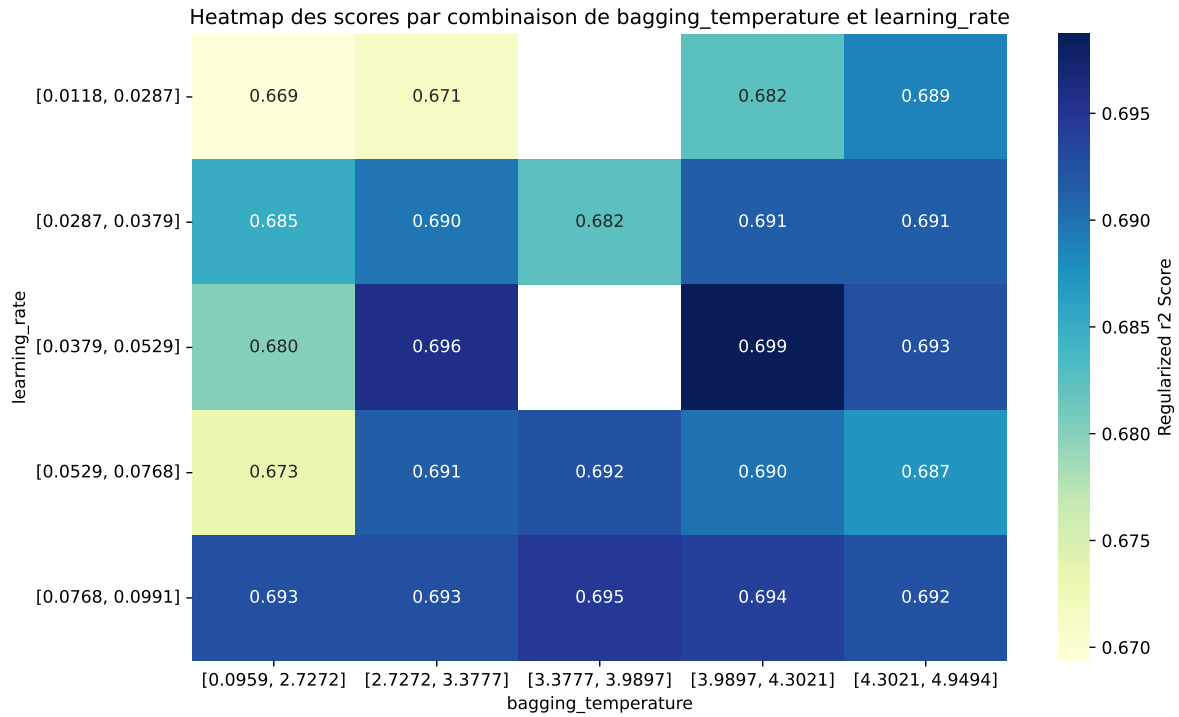
Paramètres avec >5% d'importance : ['bagging_temperature', 'learning_rate', 'depth', 'iterations']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7365	0.6428	0.7253	0.0338	0.3616	0.7324	0.0896	1.1394	1.1458
2	0.6966	0.6104	0.6896	0.034	0.3518	0.7017	0.0913	1.1495	1.1412
3	0.8039	0.6733	0.7484	0.0244	0.2815	0.7576	0.0843	1.1252	1.1939
4	0.8137	0.672	0.7486	0.0227	0.247	0.7649	0.0929	1.1383	1.2108
5	0.813	0.6681	0.743	0.0239	0.2333	0.7517	0.0837	1.1252	1.2169
6	0.8273	0.6761	0.7514	0.0188	0.2226	0.7634	0.0873	1.1291	1.2236
7	0.8386	0.6824	0.754	0.023	0.214	0.7684	0.086	1.126	1.2288
8	0.8427	0.6832	0.7567	0.0235	0.2045	0.7692	0.086	1.1259	1.2336
9	0.85	0.6846	0.7553	0.0249	0.1886	0.7675	0.0829	1.1211	1.2415
10	0.8624	0.684	0.7574	0.0223	0.1487	0.7676	0.0836	1.1222	1.2609
---	---	---	---	---	---	---	---	---	---
70	0.9962	0.6897	0.7438	0.0214	-0.23	0.7569	0.0672	1.0974	1.4445

Hyperparameters:

Rank 1: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.011782714825305397, 'reg__l2_leaf_reg': 14.481376875377752, 'reg__bagging_temperature': 2.7330881306777273, 'reg__random_strength': 2.2294252315333023, 'reg__subsample': 0.8428489497343556, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 2}

Rank 2: {'reg__iterations': 200, 'reg__depth': 7, 'reg__learning_rate': 0.01565048206399277,

'reg__l2_leaf_reg': 48.797866072105535, 'reg__bagging_temperature': 0.8614457265500342, 'reg__random_strength': 3.5388600145234395, 'reg__subsample': 0.7909033944450958, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 4}

Rank 3: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.02209566205614813, 'reg__l2_leaf_reg': 38.40986297682973, 'reg__bagging_temperature': 0.0958736718049974, 'reg__random_strength': 1.0777428083389688, 'reg__subsample': 0.5526190283806335, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 10}

Rank 4: {'reg__iterations': 500, 'reg__depth': 5, 'reg__learning_rate': 0.027773232642816927, 'reg__l2_leaf_reg': 5.621940524310117, 'reg__bagging_temperature': 0.9498554837745821, 'reg__random_strength': 3.7744444397115915, 'reg__subsample': 0.8623213552330125, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 1}

Rank 5: {'reg__iterations': 200, 'reg__depth': 7, 'reg__learning_rate': 0.02046359081094265, 'reg__l2_leaf_reg': 32.006917239777266, 'reg__bagging_temperature': 1.7220252700748717, 'reg__random_strength': 0.21725141058556108, 'reg__subsample': 0.5794423725646792, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 8}

Rank 6: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.017765915724587814, 'reg__l2_leaf_reg': 22.92227480698112, 'reg__bagging_temperature': 1.7193946411502181, 'reg__random_strength': 2.377123282306541, 'reg__subsample': 0.6372542268482394, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 10}

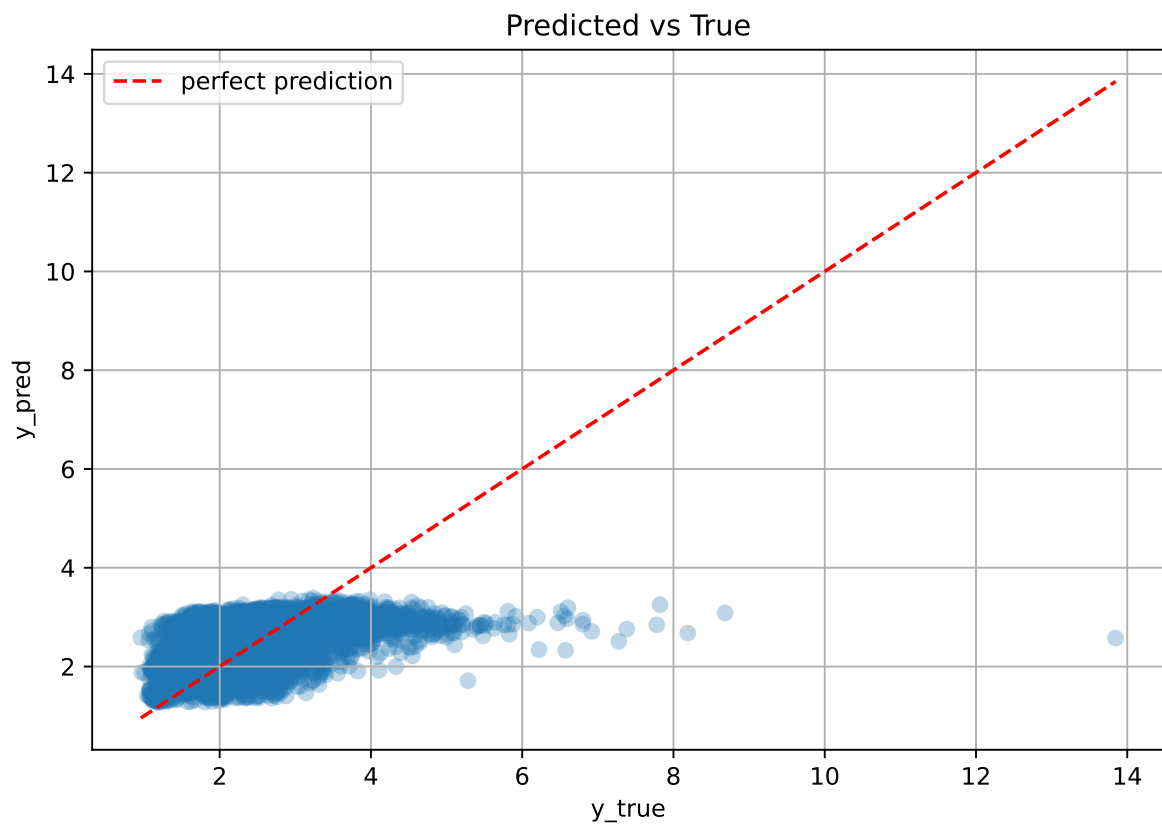
Rank 7: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.028979164412557212, 'reg__l2_leaf_reg': 23.36217889336371, 'reg__bagging_temperature': 3.6144217652036197, 'reg__random_strength': 4.490301369970898, 'reg__subsample': 0.5620086994102897, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 2}

Rank 8: {'reg__iterations': 1000, 'reg__depth': 7, 'reg__learning_rate': 0.01294036606928637, 'reg__l2_leaf_reg': 30.915446829310245, 'reg__bagging_temperature': 3.277563747347908, 'reg__random_strength': 1.2101733097970473, 'reg__subsample': 0.8509529614300657, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 3}

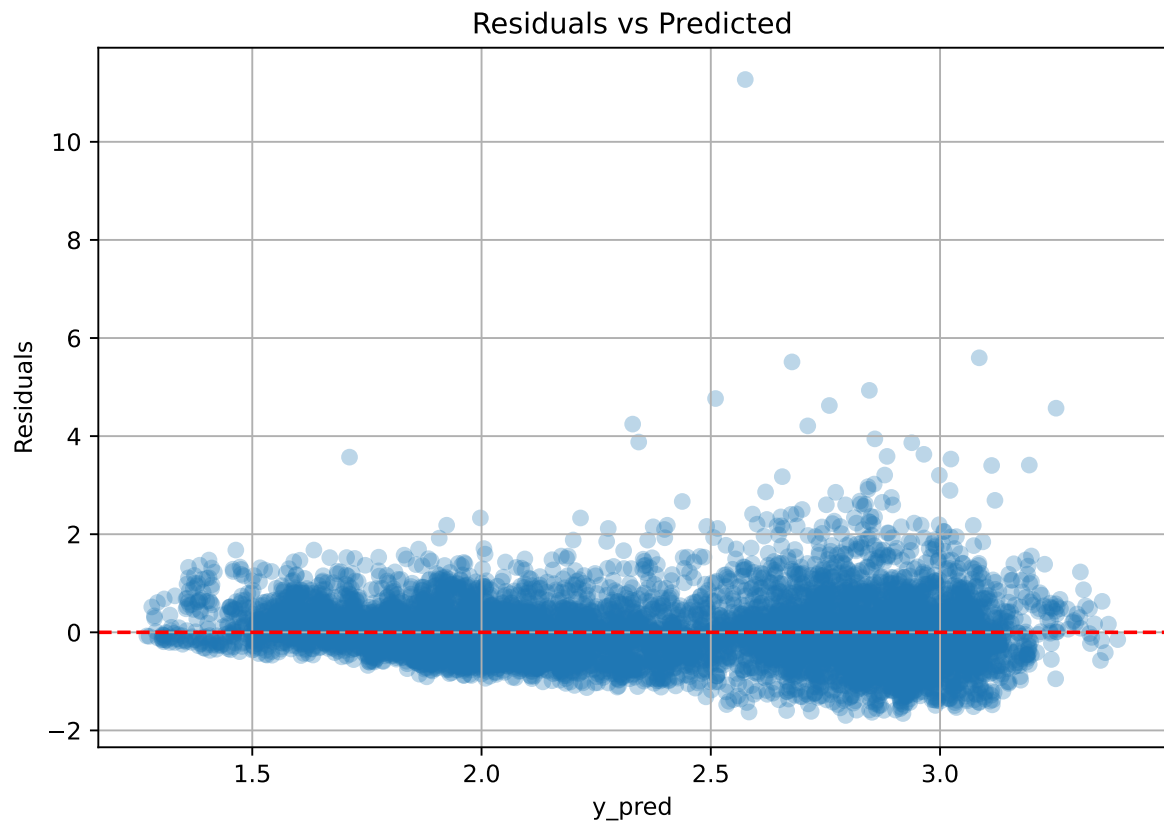
Rank 9: {'reg__iterations': 900, 'reg__depth': 7, 'reg__learning_rate': 0.03720706785797831, 'reg__l2_leaf_reg': 42.75463418353792, 'reg__bagging_temperature': 4.277815023454596, 'reg__random_strength': 3.500496913964555, 'reg__subsample': 0.6563529222506675, 'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 1}

Rank 10: {'reg__iterations': 900, 'reg__depth': 7, 'reg__learning_rate': 0.02482675862820895, 'reg__l2_leaf_reg': 48.103815846667544, 'reg__bagging_temperature': 4.504370123044369, 'reg__random_strength': 0.39924756970915587, 'reg__subsample': 0.6572068482508985, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 2}

2.5.2 Scatter



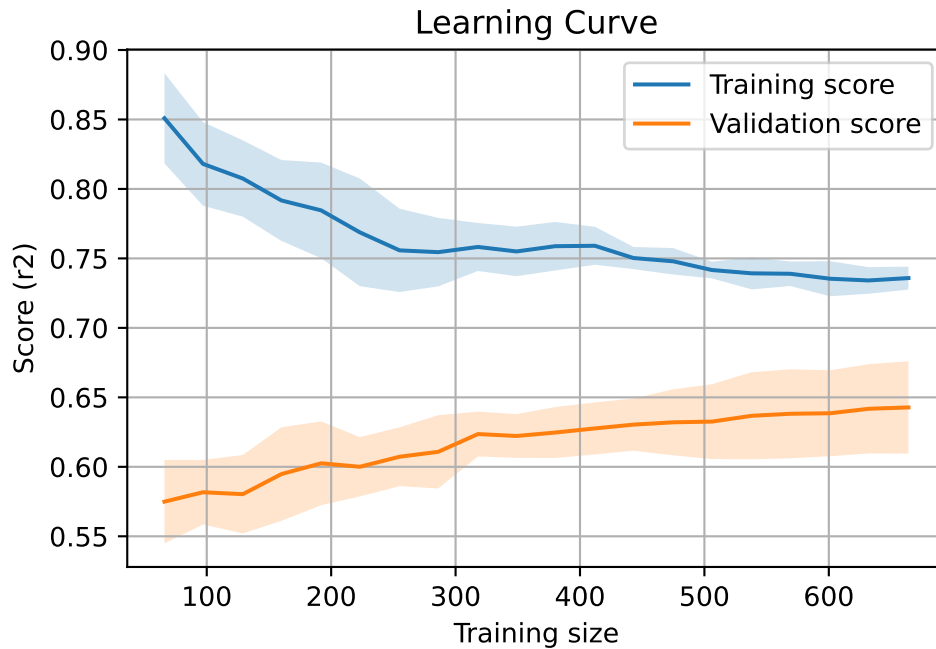
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 3

`Len(group)` : 1356, `Len(training)` : 950,

`Len(training_real)` : 665, `Len(test_of_training)` : 285, `Len(test)` : 406

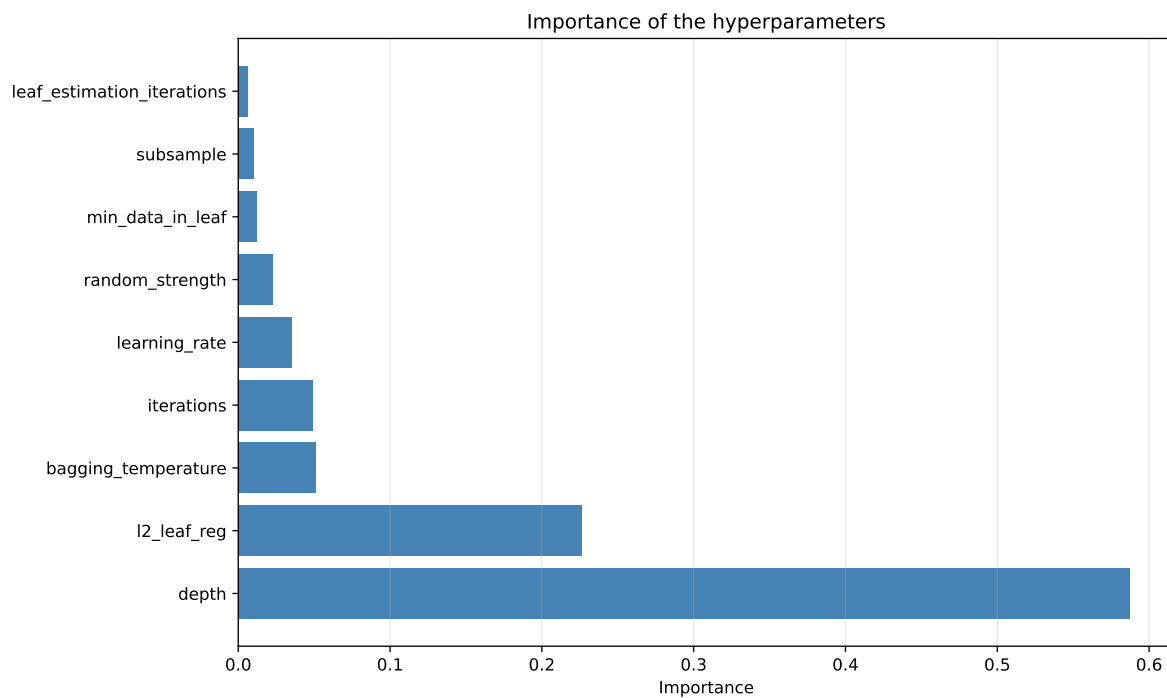
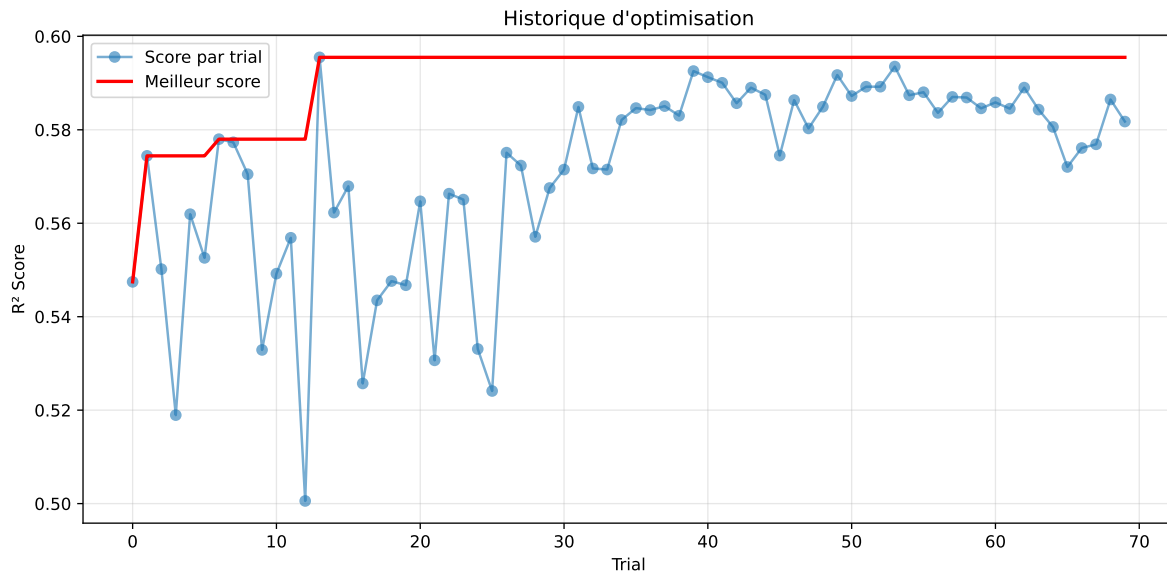


2.5.5 Gridsearch — Group: 4

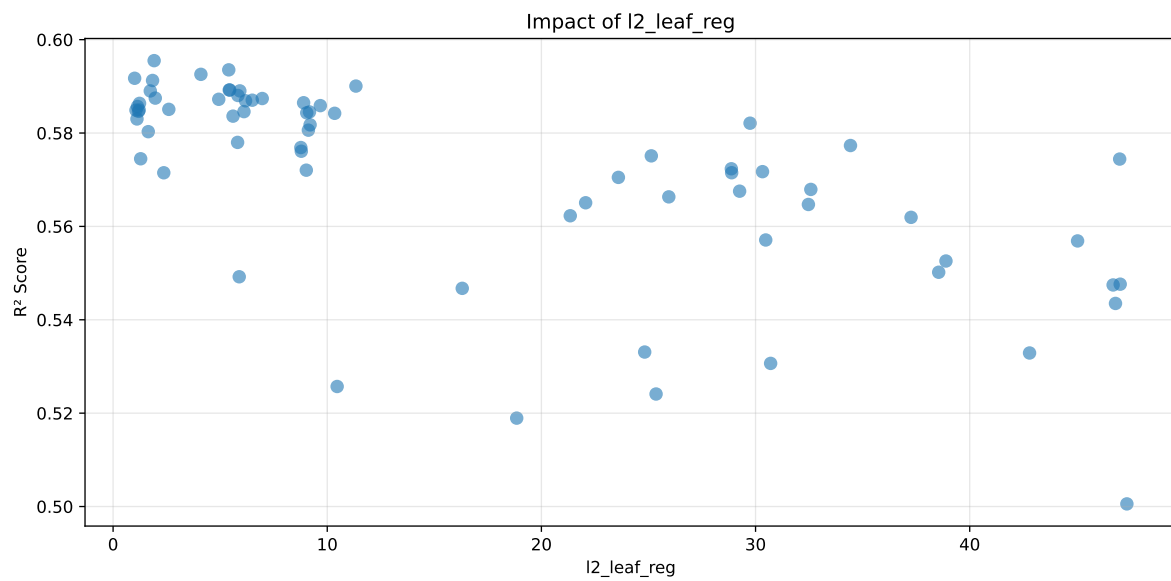
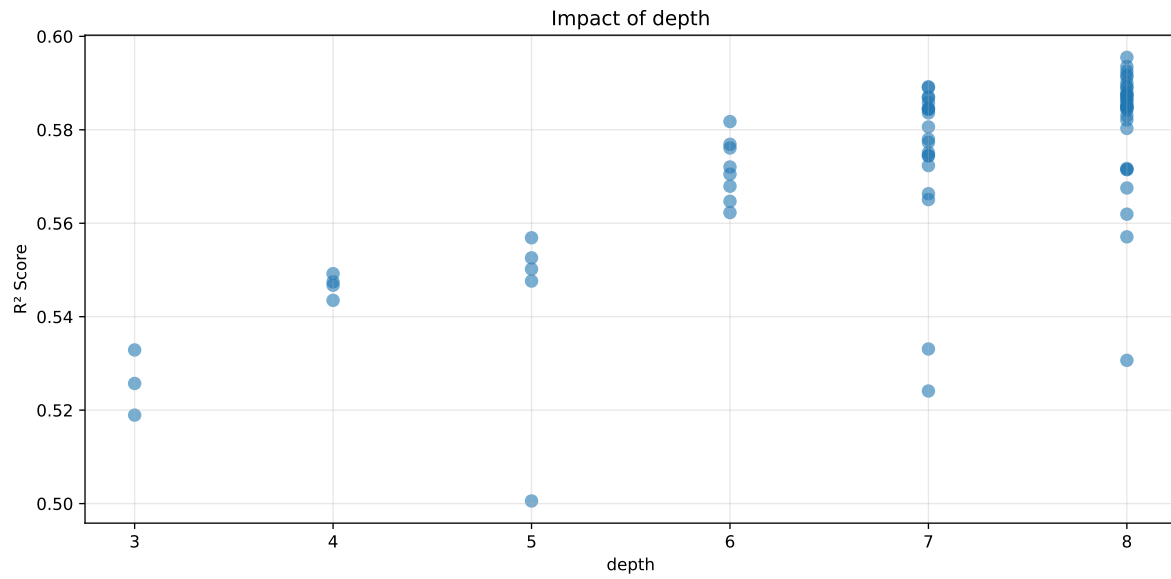
Len(group) : 1360, Len(training) : 952, Len(test) : 408

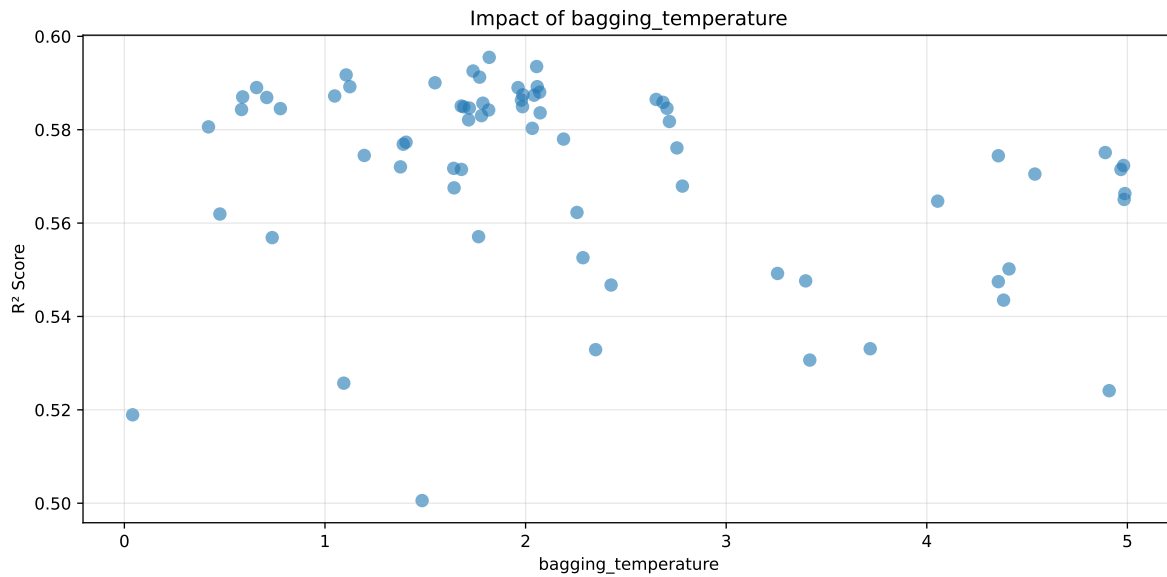
Distribution y_train vs y_test: y_train: mean=2.239, std=0.831, min=0.959, max=7.383
y_test: mean=2.202, std=0.791, min=1.062, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 31.087246 r_THI 26.964228 r_NH3 19.095999 r_CO2 14.958915
r_temperature 7.893613



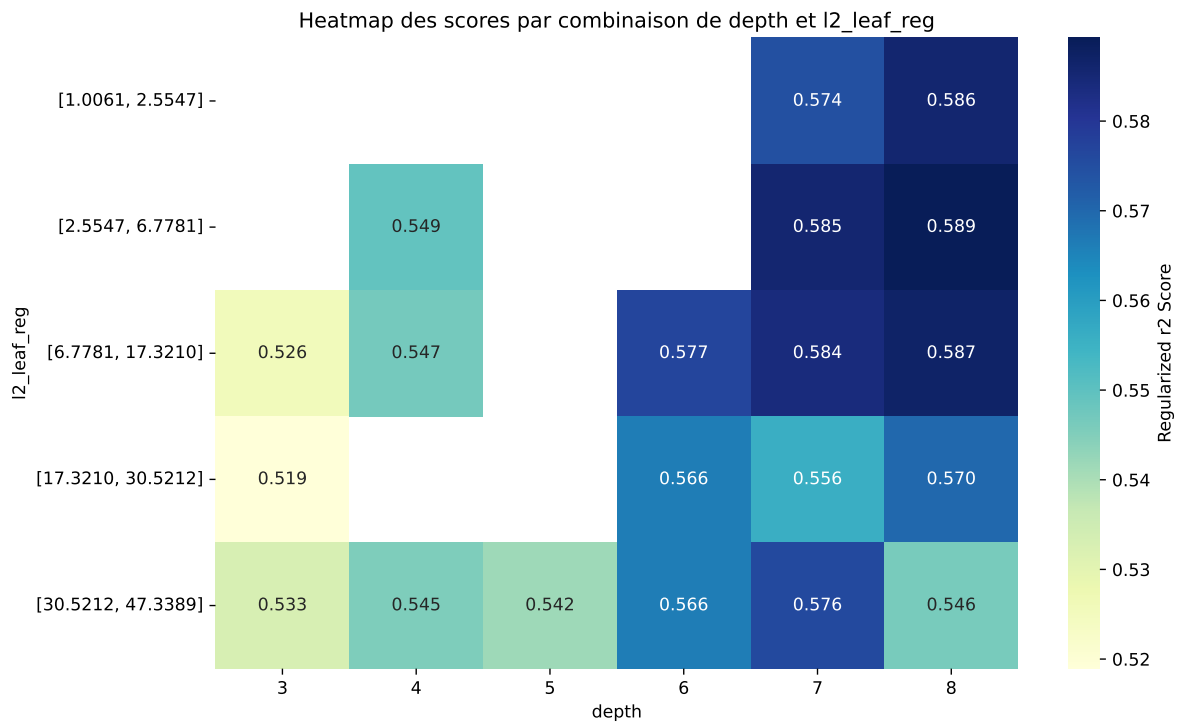
Paramètres avec >5% d'importance : ['depth', 'l2_leaf_reg', 'bagging_temperature']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6561	0.5241	0.586	0.0345	0.128	0.5949	0.0708	1.1351	1.2519
2	0.6717	0.5331	0.5915	0.0341	0.1174	0.6045	0.0714	1.1339	1.2599
3	0.6316	0.5006	0.5484	0.0361	0.1074	0.5653	0.0648	1.1294	1.2618
4	0.6831	0.5307	0.5896	0.0367	0.0734	0.5994	0.0688	1.1296	1.2872
5	0.7265	0.5571	0.6101	0.0346	0.0488	0.6174	0.0603	1.1082	1.3041
6	0.7757	0.5715	0.6147	0.0321	-0.0411	0.6302	0.0587	1.1027	1.3573
7	0.7388	0.5435	0.5693	0.0356	-0.0425	0.5852	0.0417	1.0766	1.3594
8	0.7737	0.5675	0.615	0.0308	-0.0508	0.63	0.0624	1.11	1.3632
9	0.7766	0.5651	0.6094	0.0362	-0.0697	0.6233	0.0582	1.1031	1.3744
10	0.7785	0.5663	0.6079	0.0327	-0.0701	0.6244	0.0581	1.1025	1.3746
---	---	---	---	---	---	---	---	---	---
70	0.9988	0.5715	0.6011	0.0206	-0.7104	0.6507	0.0792	1.1386	1.7477

Hyperparameters:

Rank 1: {'reg__iterations': 500, 'reg__depth': 7, 'reg__learning_rate': 0.01025054709285605, 'reg__l2_leaf_reg': 25.354584115533264, 'reg__bagging_temperature': 4.909228644491314, 'reg__random_strength': 4.987717636657373, 'reg__subsample': 0.7811585965107514, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 4}

Rank 2: {'reg__iterations': 500, 'reg__depth': 7, 'reg__learning_rate': 0.011385018416863488, 'reg__l2_leaf_reg': 24.823028215810712, 'reg__bagging_temperature': 3.7180010757550024, 'reg__random_strength': 4.94942462971731, 'reg__subsample': 0.7844129621669512, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 4}

Rank 3: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.05703446534682218, 'reg__l2_leaf_reg': 47.33892490667613, 'reg__bagging_temperature': 1.48415039758379, 'reg__random_strength': 2.4848682538713502, 'reg__subsample': 0.6155800468520755, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 1}

Rank 4: {'reg__iterations': 400, 'reg__depth': 8, 'reg__learning_rate': 0.010304977919225468, 'reg__l2_leaf_reg': 30.703781998090168, 'reg__bagging_temperature': 3.416759151708309, 'reg__random_strength': 4.133528156557912, 'reg__subsample': 0.7456103059592616, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 10}

Rank 5: {'reg__iterations': 600, 'reg__depth': 8, 'reg__learning_rate': 0.010457762625909828, 'reg__l2_leaf_reg': 30.475553458867044, 'reg__bagging_temperature': 1.7657088570148052, 'reg__random_strength': 4.647897874504919, 'reg__subsample': 0.7556889899847283, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 7}

Rank 6: {'reg__iterations': 700, 'reg__depth': 8, 'reg__learning_rate': 0.0112220637425982, 'reg__l2_leaf_reg': 28.884069800474535, 'reg__bagging_temperature': 1.6802249877541946, 'reg__random_strength': 3.8897431930067947, 'reg__subsample': 0.7372704046442345, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 7}

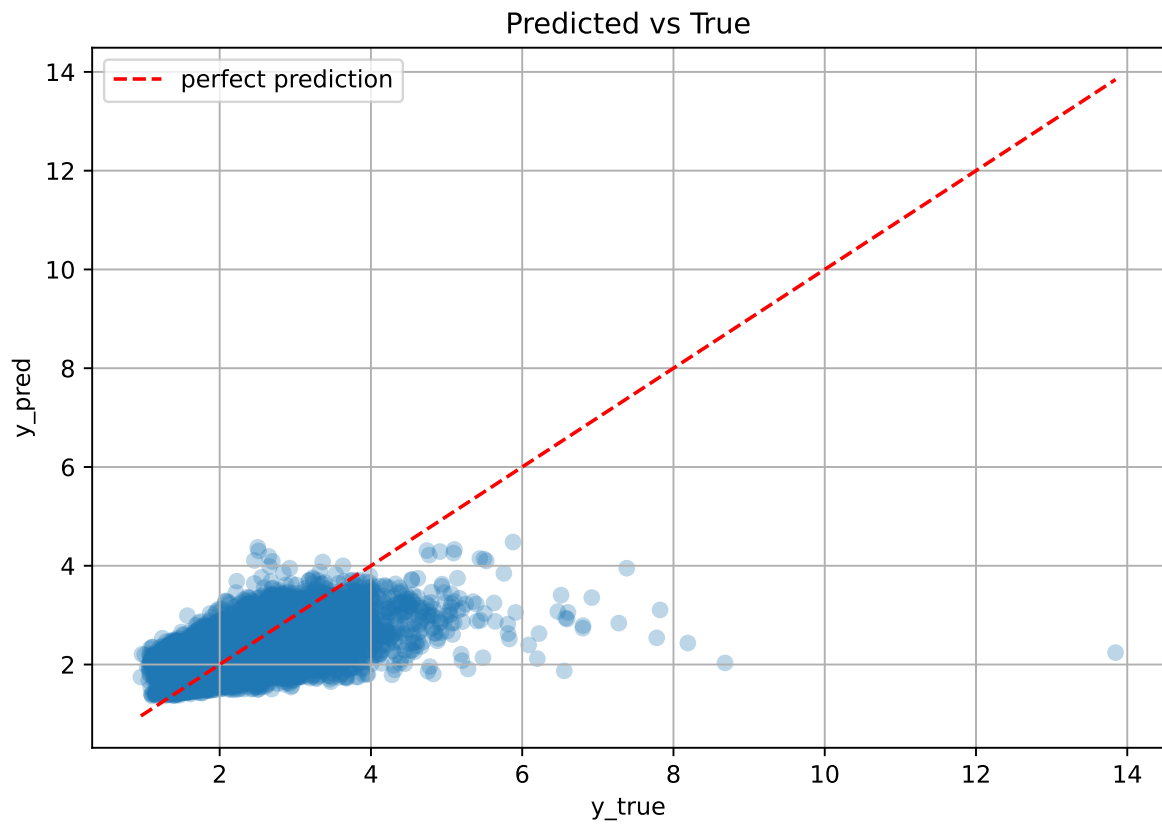
Rank 7: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.019798835556025893, 'reg__l2_leaf_reg': 46.80723207571126, 'reg__bagging_temperature': 4.382810269448198, 'reg__random_strength': 3.365350717797763, 'reg__subsample': 0.7017105586735293, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 4}

Rank 8: {'reg__iterations': 700, 'reg__depth': 8, 'reg__learning_rate': 0.011036062889648979, 'reg__l2_leaf_reg': 29.25562995466064, 'reg__bagging_temperature': 1.643596628406251, 'reg__random_strength': 4.290436219872411, 'reg__subsample': 0.7579927199147897, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 8}

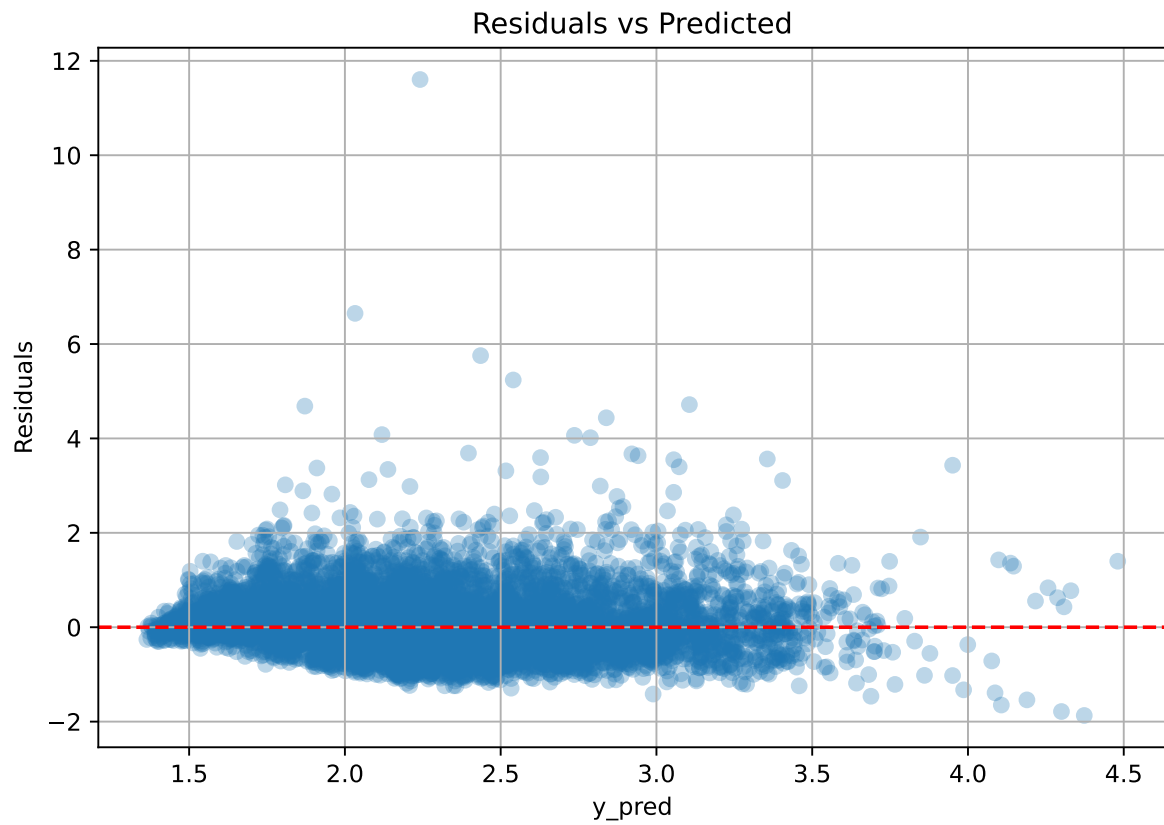
Rank 9: {'reg__iterations': 500, 'reg__depth': 7, 'reg__learning_rate': 0.020119859150390842, 'reg__l2_leaf_reg': 22.06595392577666, 'reg__bagging_temperature': 4.983969594044155, 'reg__random_strength': 4.955194502518941, 'reg__subsample': 0.780622739831923, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 4}

Rank 10: {'reg__iterations': 500, 'reg__depth': 7, 'reg__learning_rate': 0.021341941224406517, 'reg__l2_leaf_reg': 25.94912842984386, 'reg__bagging_temperature': 4.987458011397483, 'reg__random_strength': 4.713374966360652, 'reg__subsample': 0.779011868887069, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 4}

2.5.6 Scatter



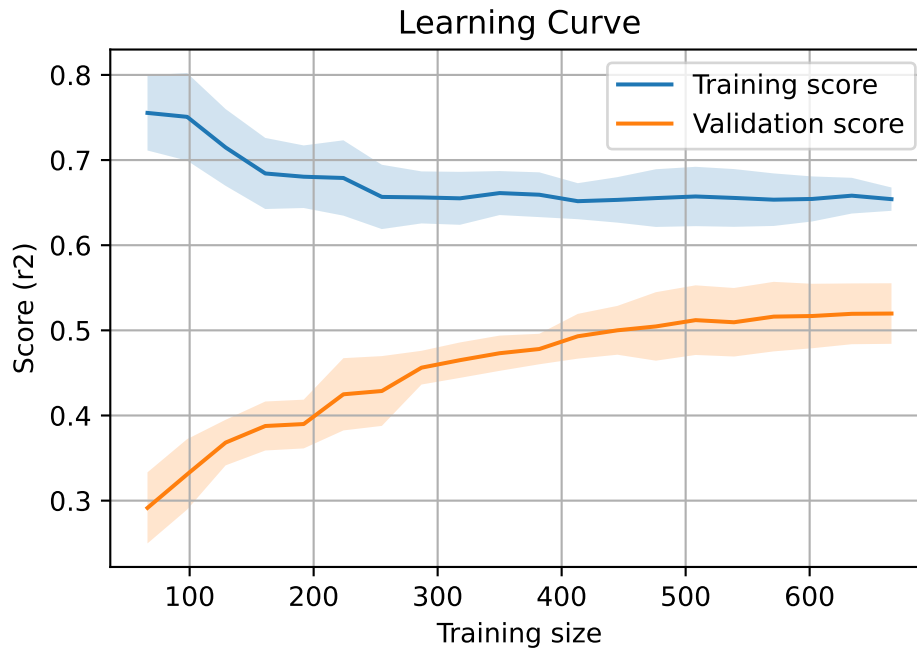
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 4

`Len(group)` : 1360, `Len(training)` : 952,

`Len(training_real)` : 667, `Len(test_of_training)` : 285, `Len(test)` : 408

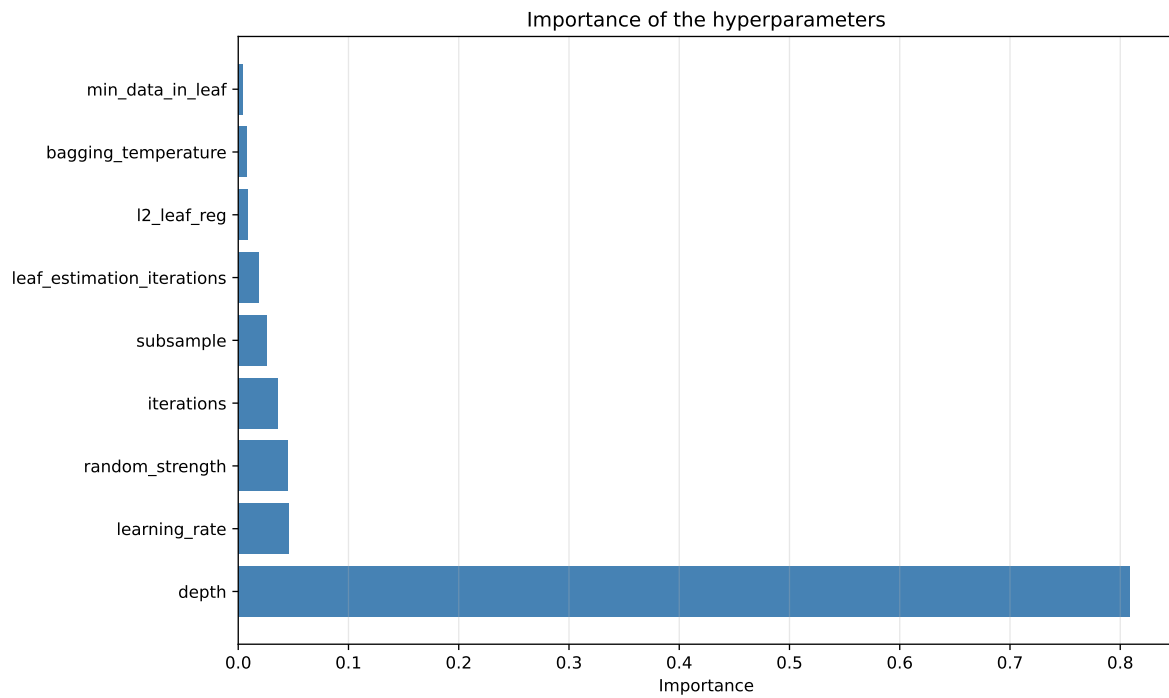
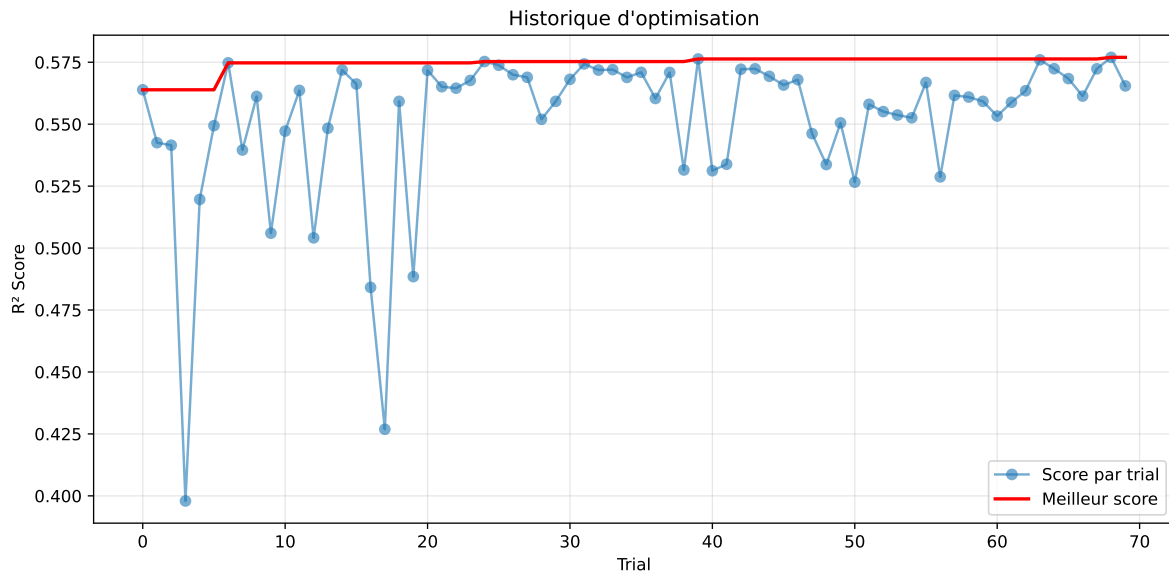


2.5.9 Gridsearch — Group: 5

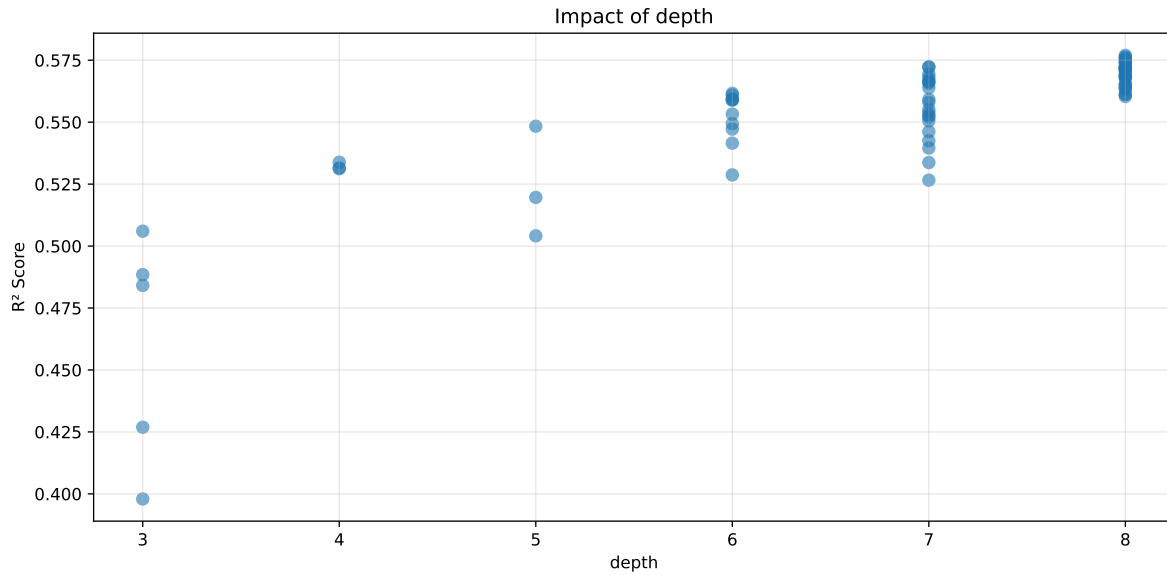
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

===== TOP Importance FEATURES =====
feature importance r_umidity 63.148585 r_temperature 16.540521 r_NH3 8.483620 r_CO2
7.178927 r_THI 4.648348



Paramètres avec >5% d'importance : ['depth']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4605	0.4269	0.4927	0.0208	0.3261	0.4947	0.0678	1.1588	1.0787
2	0.4231	0.3979	0.4572	0.0195	0.3225	0.4598	0.0618	1.1553	1.0632
3	0.582	0.5041	0.5655	0.0264	0.2705	0.5692	0.0651	1.1291	1.1544
4	0.5597	0.4841	0.5485	0.0253	0.2576	0.5566	0.0724	1.1496	1.156
5	0.6261	0.5266	0.5849	0.0221	0.2282	0.5927	0.0661	1.1254	1.1889
6	0.6422	0.5287	0.5864	0.0226	0.1882	0.5962	0.0675	1.1276	1.2147
7	0.6322	0.5196	0.5798	0.0235	0.182	0.5937	0.074	1.1425	1.2166
8	0.6511	0.5337	0.5912	0.0224	0.1816	0.6002	0.0665	1.1245	1.2199
9	0.6599	0.5396	0.5914	0.0223	0.1785	0.5989	0.0593	1.11	1.2231
10	0.6062	0.4885	0.5536	0.0296	0.1354	0.5671	0.0786	1.161	1.2409
---	---	---	---	---	---	---	---	---	---
70	0.9705	0.5604	0.6031	0.0327	-0.67	0.622	0.0616	1.1099	1.7319

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.017729193671165096, 'reg__l2_leaf_reg': 41.779766951573784, 'reg__bagging_temperature': 1.5411705074235278, 'reg__random_strength': 3.957625258390458, 'reg__subsample': 0.7365743026301124, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 2}

Rank 2: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.010626362574117055, 'reg__l2_leaf_reg': 25.533133448110572, 'reg__bagging_temperature': 3.2945106394688772, 'reg__random_strength': 4.41795326972654, 'reg__subsample': 0.8639473479502346, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 2}

Rank 3: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.02334676878708254, 'reg__l2_leaf_reg': 9.016494518459309, 'reg__bagging_temperature': 4.618346938849993, 'reg__random_strength': 4.608107194558206, 'reg__subsample': 0.7985835648546624, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 10}

Rank 4: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.02555837486279626, 'reg__l2_leaf_reg': 37.47979847656484, 'reg__bagging_temperature': 0.39582095471303536, 'reg__random_strength': 2.910017637268847, 'reg__subsample': 0.7784152005926088, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 1}

Rank 5: {'reg__iterations': 400, 'reg__depth': 7, 'reg__learning_rate': 0.01372510257864038, 'reg__l2_leaf_reg': 13.993956602111227, 'reg__bagging_temperature': 3.510051703746133, 'reg__random_strength': 2.6514824117935305, 'reg__subsample': 0.7351575107098333, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 7}

Rank 6: {'reg__iterations': 600, 'reg__depth': 6, 'reg__learning_rate': 0.010196387521429037, 'reg__l2_leaf_reg': 8.689162685848318, 'reg__bagging_temperature': 3.0408897867231, 'reg__random_strength': 1.0647818853625073, 'reg__subsample': 0.6769402173040259, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 3}

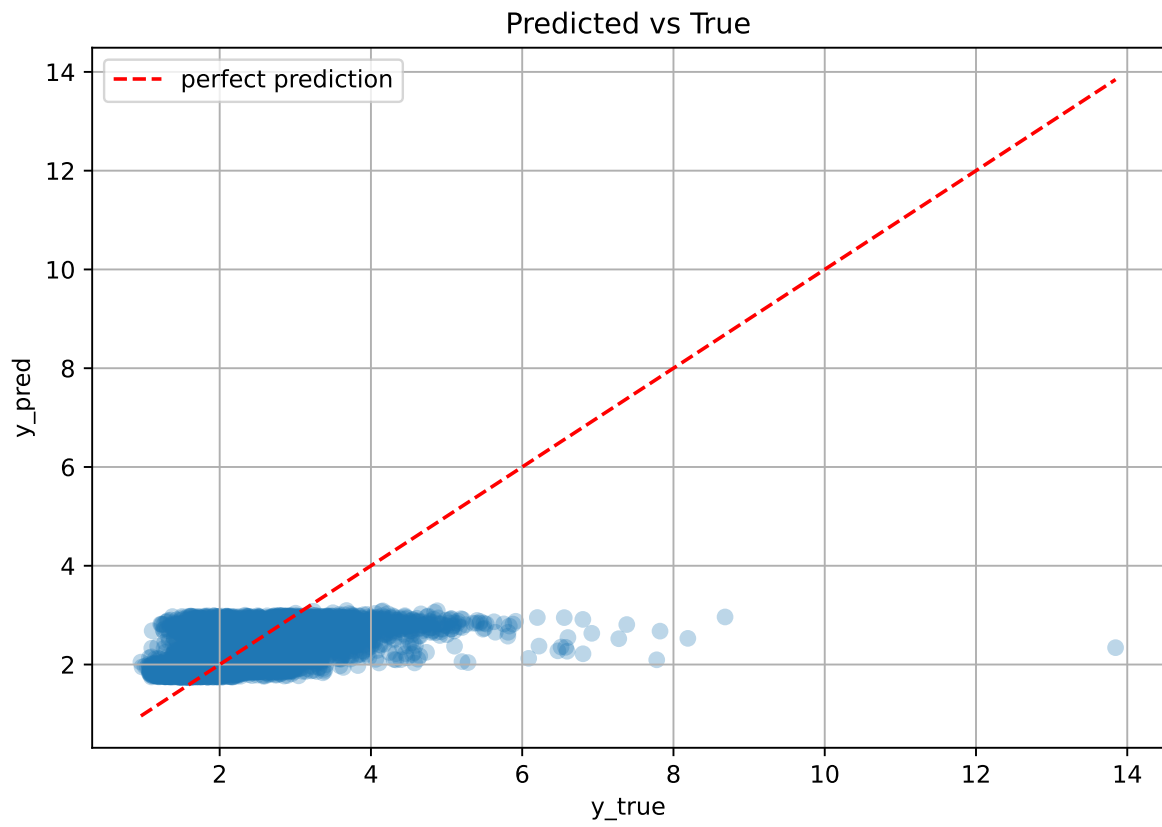
Rank 7: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.022153149878656284, 'reg__l2_leaf_reg': 42.59839365519779, 'reg__bagging_temperature': 0.08644397808668325, 'reg__random_strength': 0.5761450442543908, 'reg__subsample': 0.8636063109600225, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 9}

Rank 8: {'reg__iterations': 400, 'reg__depth': 7, 'reg__learning_rate': 0.013589048703644843, 'reg__l2_leaf_reg': 13.343032128234478, 'reg__bagging_temperature': 2.172530985590783, 'reg__random_strength': 1.431705649355368, 'reg__subsample': 0.741740017557129, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 8}

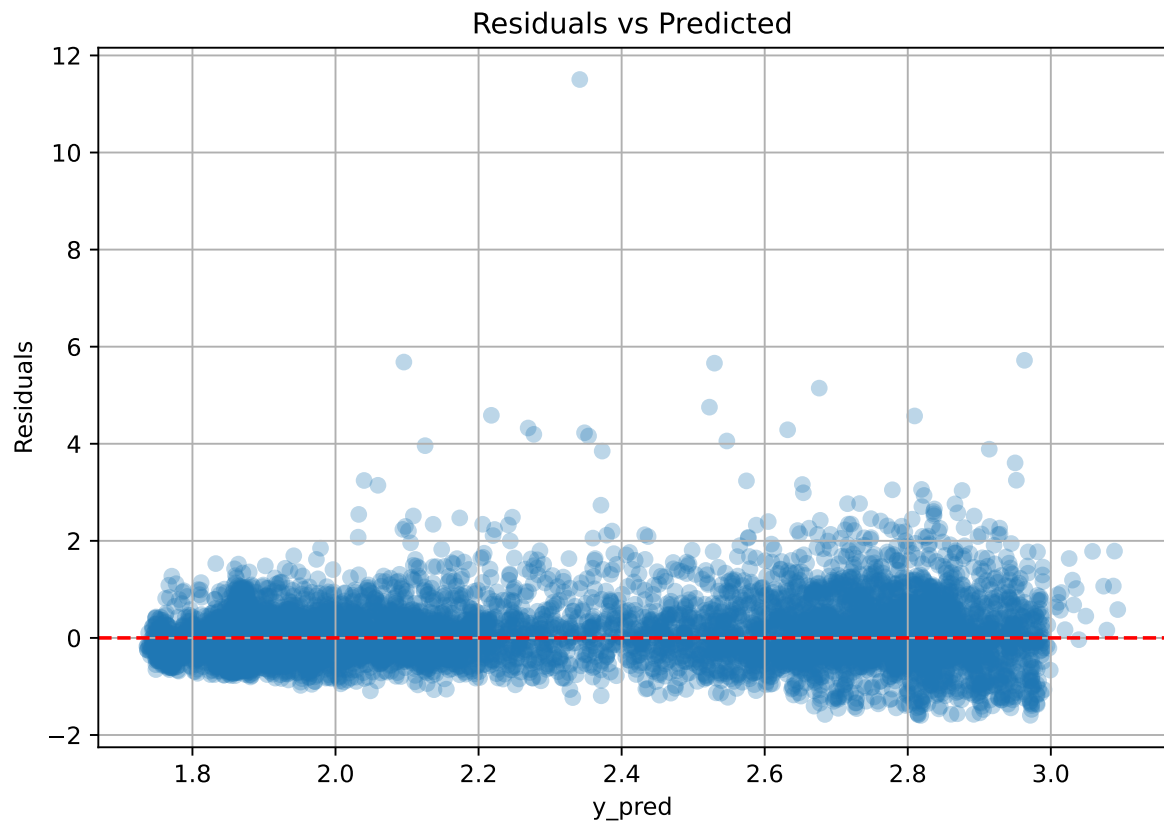
Rank 9: {'reg__iterations': 200, 'reg__depth': 7, 'reg__learning_rate': 0.02501157151227807, 'reg__l2_leaf_reg': 11.366697688310403, 'reg__bagging_temperature': 4.562542313401905, 'reg__random_strength': 1.024568391608001, 'reg__subsample': 0.8785438976550435, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 9}

Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.06318047578376135, 'reg__l2_leaf_reg': 19.141562007939832, 'reg__bagging_temperature': 1.774059007237554, 'reg__random_strength': 0.1412644481078995, 'reg__subsample': 0.7737635064643427, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 1}

2.5.10 Scatter



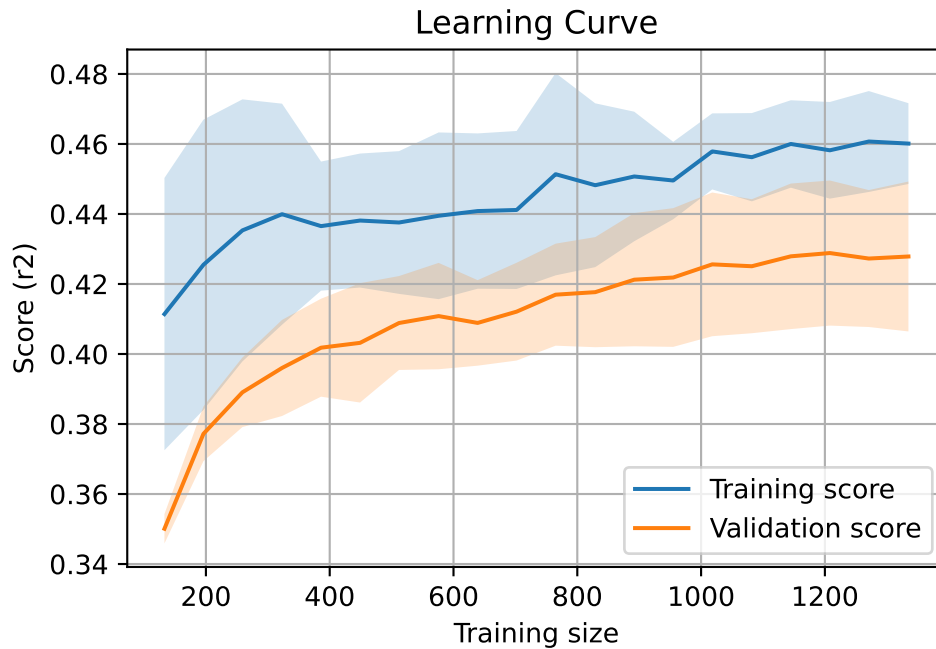
2.5.11 Scatter residuals



2.5.12 Learning curve — Group: 5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

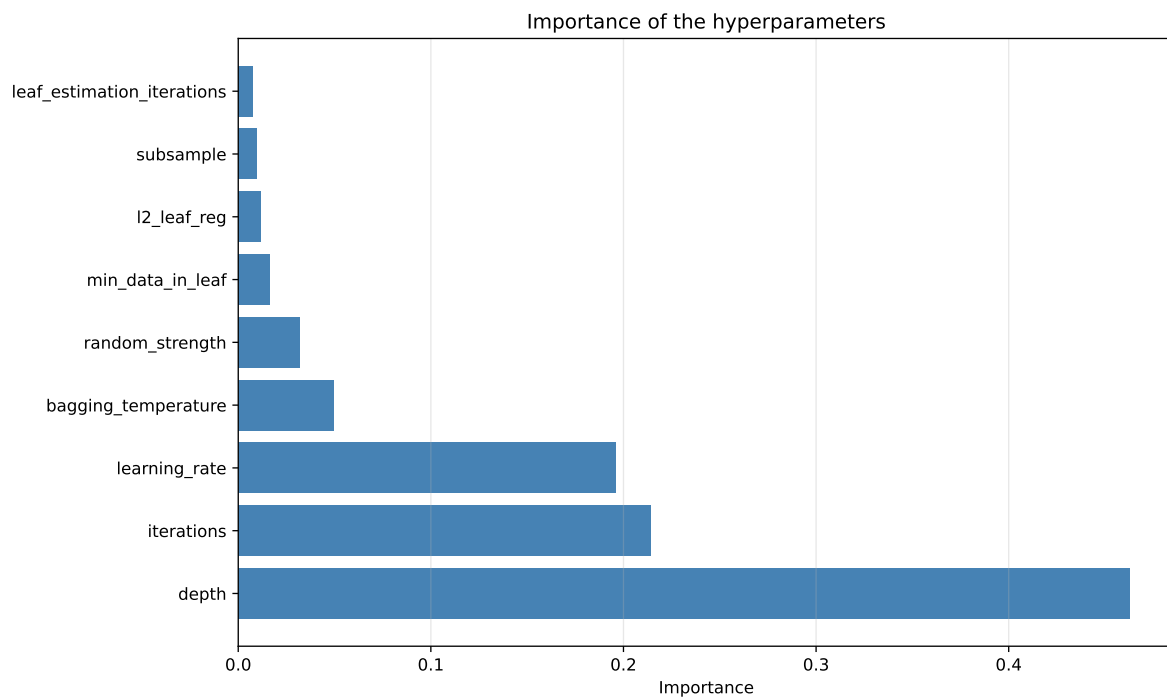
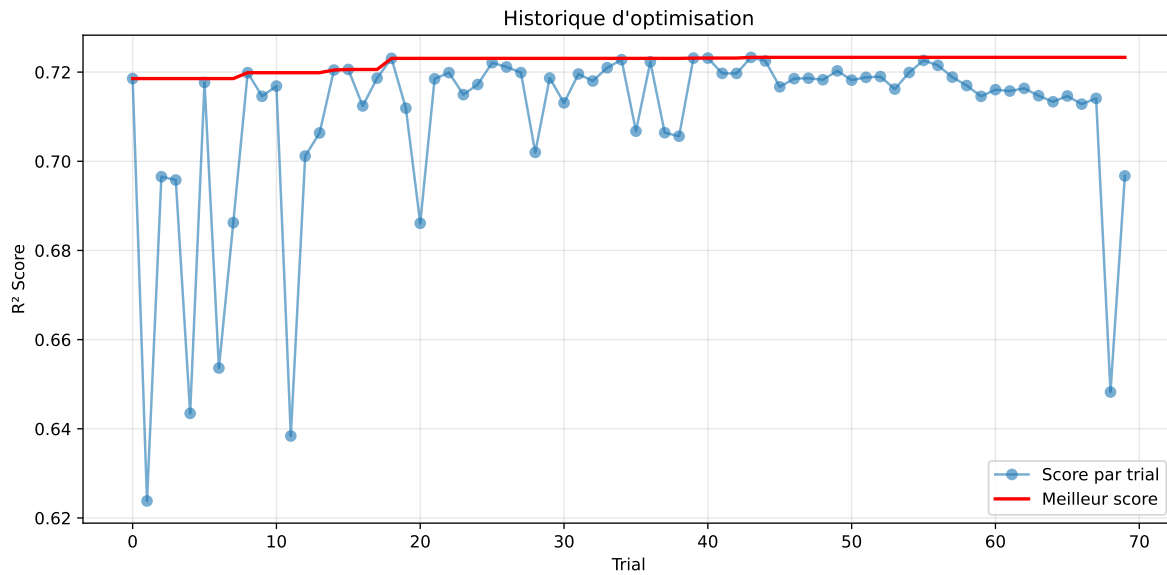


2.5.13 Gridsearch — Group: 11

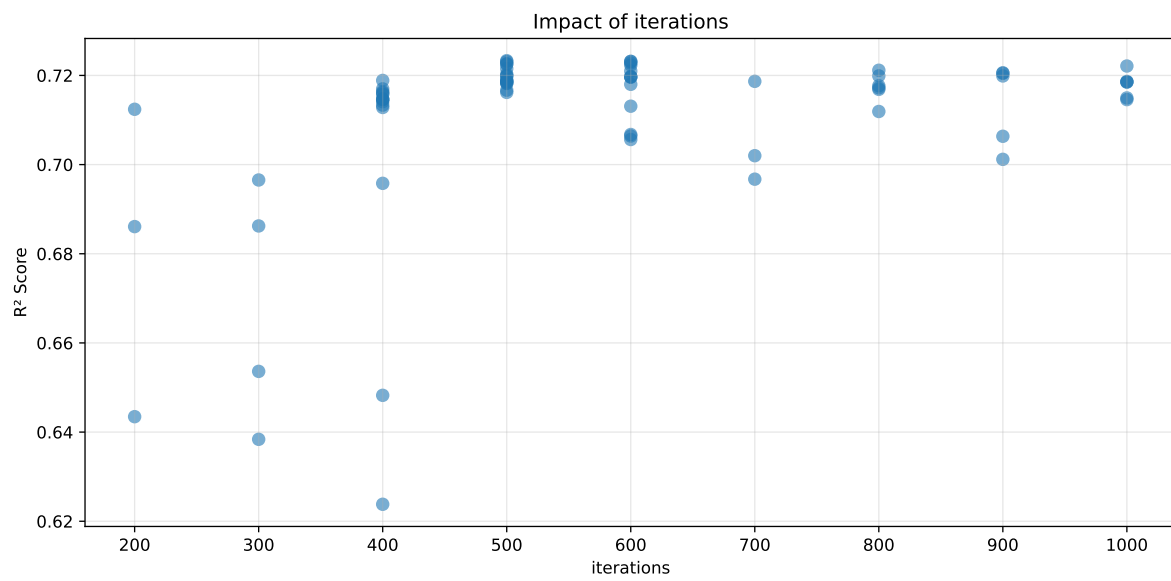
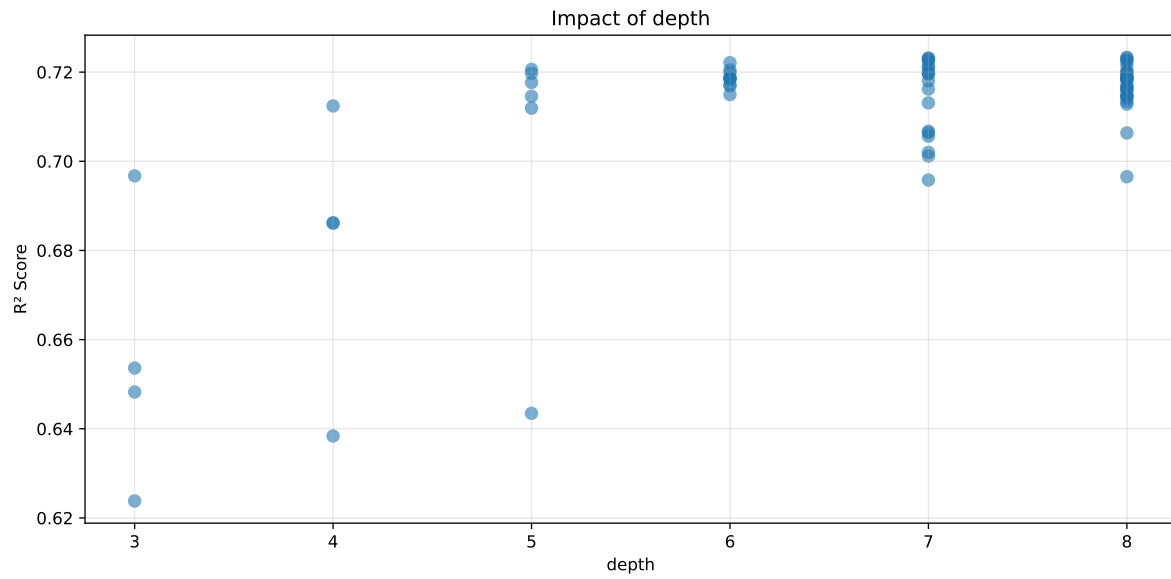
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

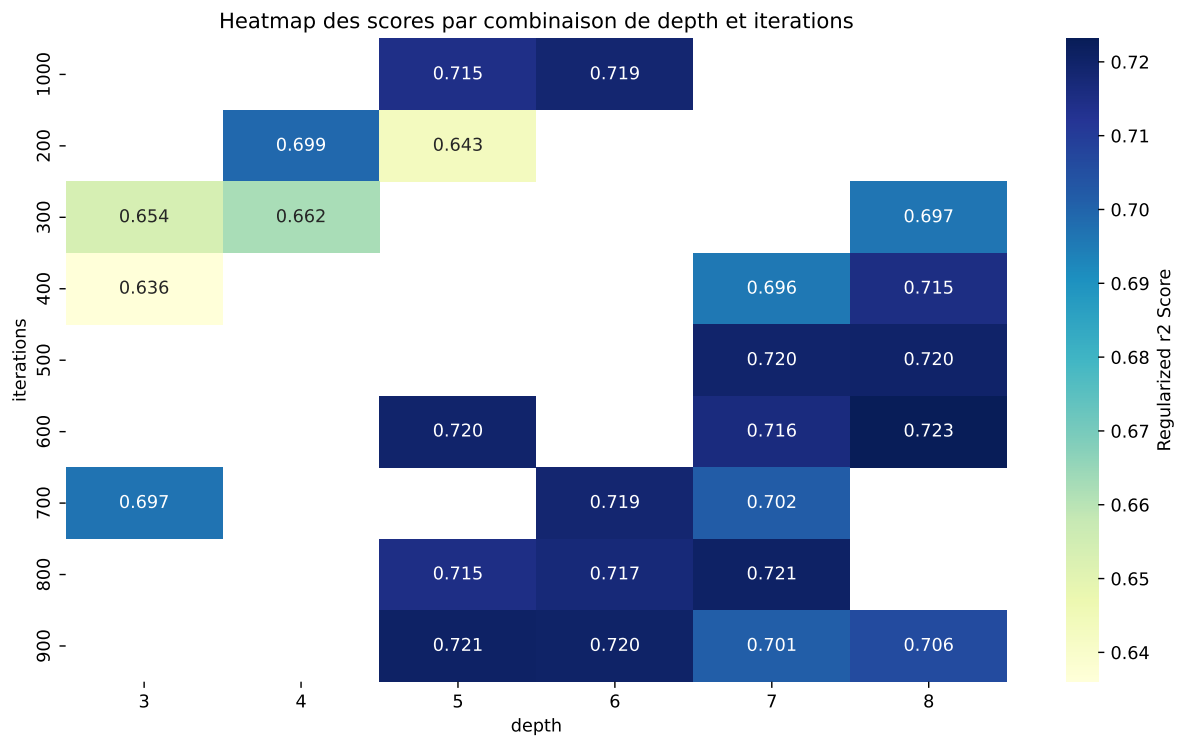
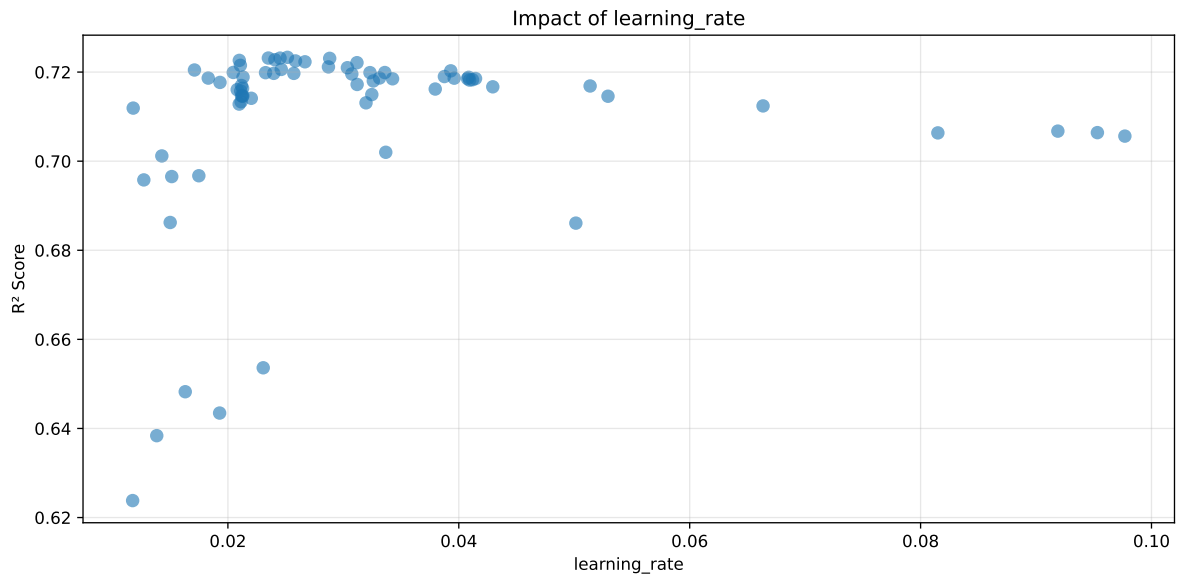
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

===== TOP Importance FEATURES =====
feature importance r_umidity 35.644254 r_NH3 34.955022 r_CO2 14.231788 r_temperature
8.599972 r_THI 6.568964



Paramètres avec >5% d'importance : ['depth', 'iterations', 'learning_rate']





rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7158	0.6861	0.592	0.0162	0.6861	0.5973	-0.0888	0.8705	1.0433
2	0.6634	0.6536	0.5666	0.015	0.6536	0.5716	-0.082	0.8746	1.015
3	0.6566	0.6482	0.5633	0.0124	0.6482	0.5699	-0.0783	0.8791	1.0129
4	0.6593	0.6435	0.5511	0.0119	0.6435	0.5648	-0.0787	0.8777	1.0246
5	0.647	0.6384	0.5494	0.0117	0.6384	0.5574	-0.081	0.8731	1.0135
6	0.6303	0.6238	0.5412	0.0129	0.6238	0.5475	-0.0763	0.8777	1.0104
7	0.7365	0.6967	0.5962	0.0211	0.5774	0.5958	-0.1009	0.8551	1.0571
8	0.7278	0.6862	0.5878	0.0185	0.5617	0.5859	-0.1003	0.8538	1.0605
9	0.7437	0.6958	0.5988	0.0129	0.5522	0.6044	-0.0914	0.8686	1.0688
10	0.7519	0.7012	0.6018	0.015	0.5491	0.6079	-0.0933	0.867	1.0723
---	---	---	---	---	---	---	---	---	---
70	0.982	0.7063	0.5876	0.017	-0.1207	0.5993	-0.107	0.8485	1.3903

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.050140074609450906, 'reg__l2_leaf_reg': 32.66396697180134, 'reg__bagging_temperature': 1.8280389928221275, 'reg__random_strength': 1.4247872176238747, 'reg__subsample': 0.8127427223234602, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 2}

Rank 2: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.023060230163632203, 'reg__l2_leaf_reg': 33.523474694863495, 'reg__bagging_temperature': 4.983545040799367, 'reg__random_strength': 3.3555106396921475, 'reg__subsample': 0.7073108121683627, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 7}

Rank 3: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.016305846498414062, 'reg__l2_leaf_reg': 29.640015337998065, 'reg__bagging_temperature': 3.4984587022285742, 'reg__random_strength': 3.727866584819302, 'reg__subsample': 0.7189642030050104, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 10}

Rank 4: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.019282904155148915, 'reg__l2_leaf_reg': 16.3365597757007, 'reg__bagging_temperature': 3.585514043027644, 'reg__random_strength': 1.1965814722287478, 'reg__subsample': 0.5969323452056201, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 1}

Rank 5: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.013837697763629475, 'reg__l2_leaf_reg': 25.66657967246398, 'reg__bagging_temperature': 1.9057580154788138, 'reg__random_strength': 2.9141635313202174, 'reg__subsample': 0.6999900564452286, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 3}

Rank 6: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.011748537315093071, 'reg__l2_leaf_reg': 11.242501422220487, 'reg__bagging_temperature': 4.560509624032452, 'reg__random_strength': 4.162388542454705, 'reg__subsample': 0.5640859300829362, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 10}

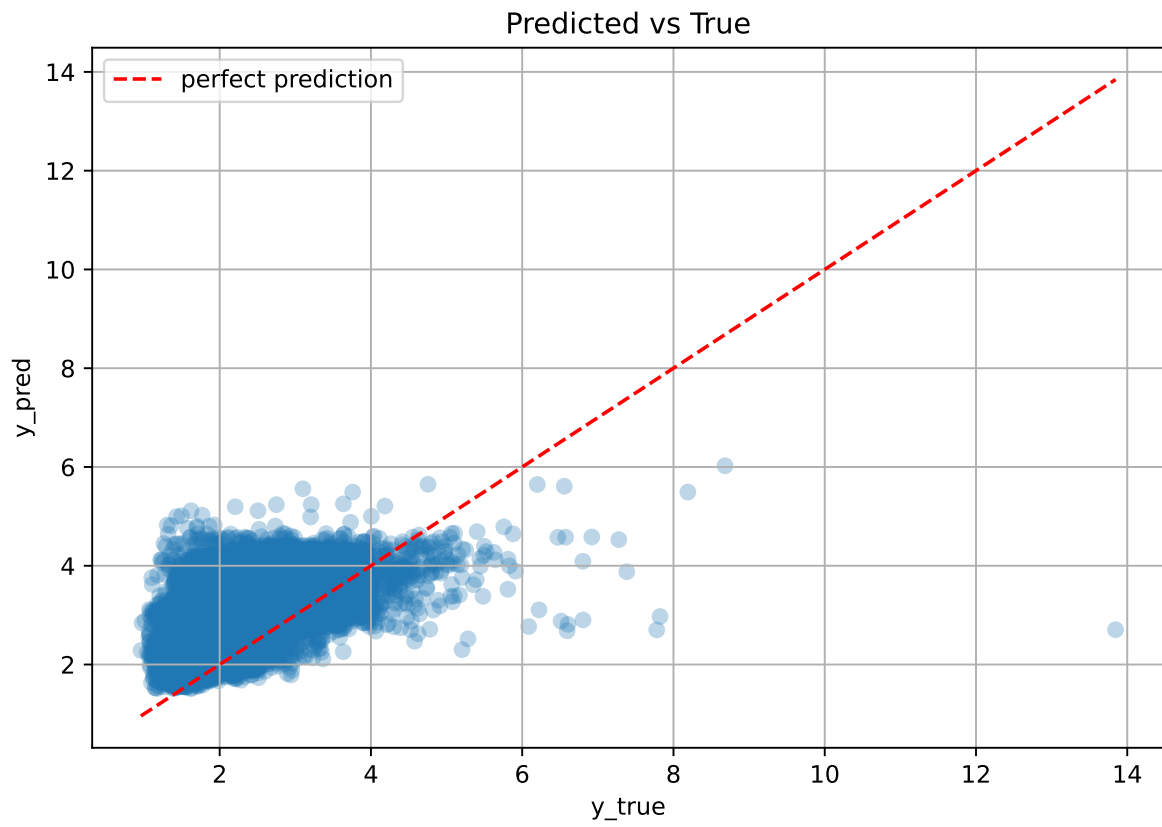
Rank 7: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.01748250420853108, 'reg__l2_leaf_reg': 13.85202825974061, 'reg__bagging_temperature': 3.5540141785061183, 'reg__random_strength': 3.632242039522379, 'reg__subsample': 0.6990542333016007, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 10}

Rank 8: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.015001265574685271, 'reg__l2_leaf_reg': 11.631384271502043, 'reg__bagging_temperature': 1.1630661813703096, 'reg__random_strength': 0.14078492860458536, 'reg__subsample': 0.7173650180151812, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 10}

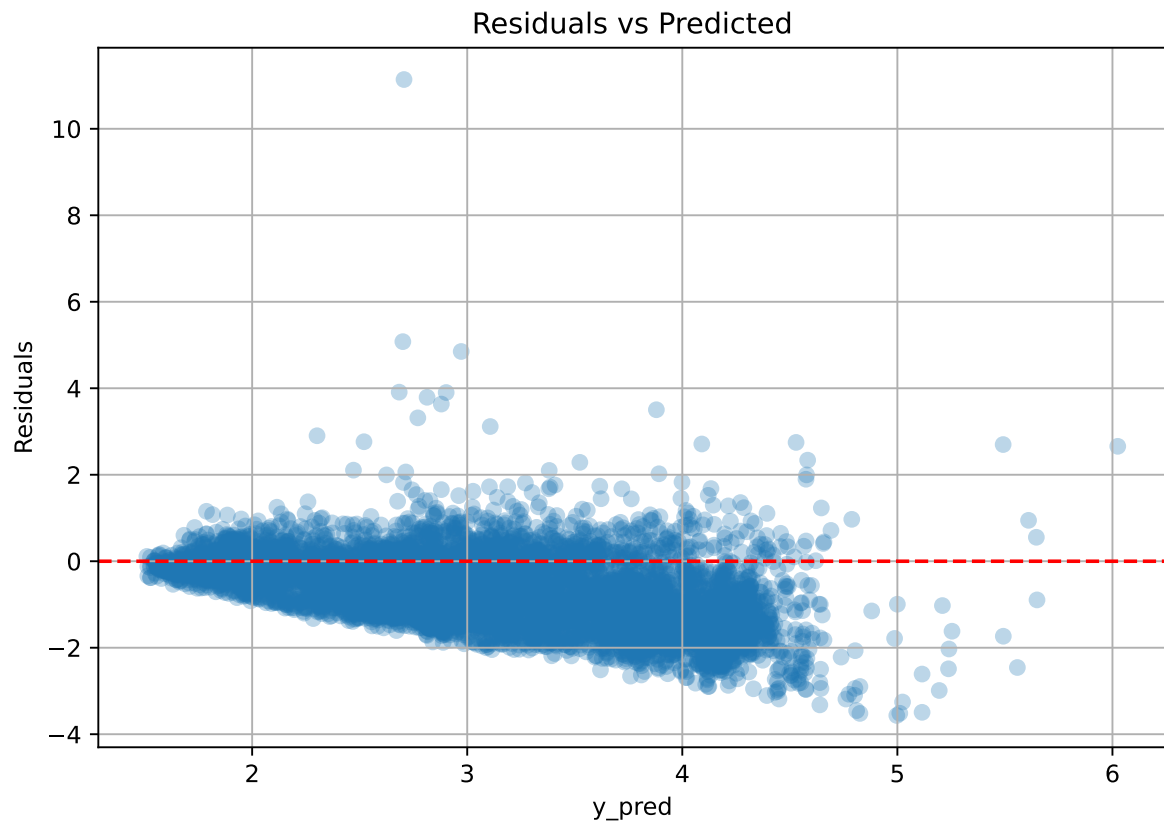
Rank 9: {'reg__iterations': 400, 'reg__depth': 7, 'reg__learning_rate': 0.012717319202156869, 'reg__l2_leaf_reg': 27.920301377459026, 'reg__bagging_temperature': 1.1604820792087196, 'reg__random_strength': 2.95941790459491, 'reg__subsample': 0.6421139507446395, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 10}

Rank 10: {'reg__iterations': 900, 'reg__depth': 7, 'reg__learning_rate': 0.014276910888602886, 'reg__l2_leaf_reg': 38.52674663832102, 'reg__bagging_temperature': 4.043018338665819, 'reg__random_strength': 3.890733847876589, 'reg__subsample': 0.7815230782577034, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 2}

2.5.14 Scatter



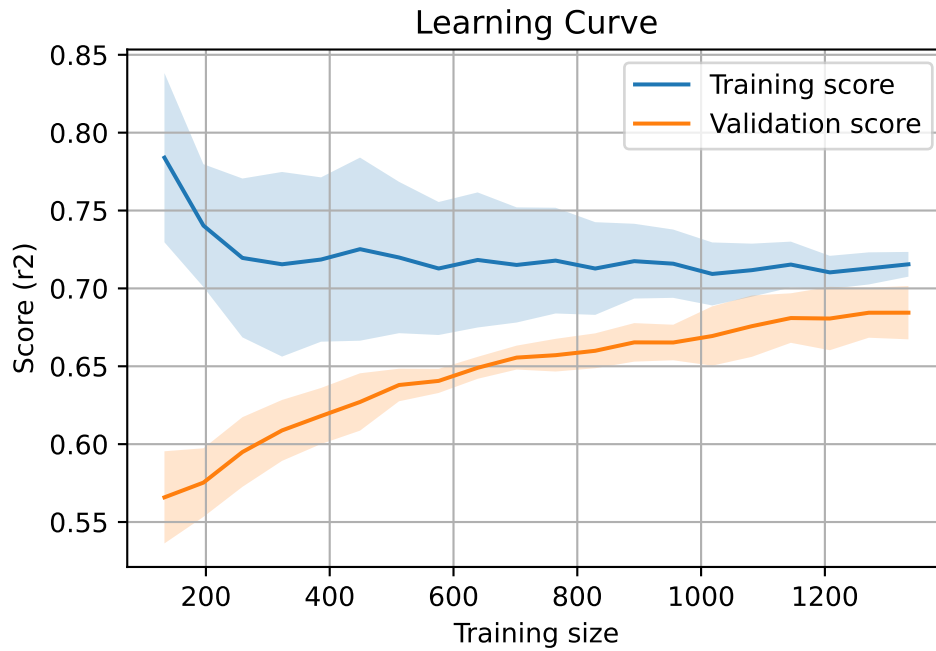
2.5.15 Scatter residuals



2.5.16 Learning curve — Group: 11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.6 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	3	950	406	1356	0.596	0.55	0.04	0.64	0.641	0.091	1.165	1.083	nan
Linear	4	952	408	1360	0.302	0.278	0.049	0.276	0.277	-0.001	0.997	1.087	nan
Linear	5	1909	817	2726	0.431	0.404	0.026	0.457	0.457	0.054	1.133	1.069	nan
Linear	11	1909	817	2726	0.463	0.463	0.017	0.394	0.394	-0.069	0.852	1.0	nan
Linear	MEAN	5720	2448	8168	0.448	0.424	0.033	0.442	0.442	0.019	1.037	1.06	nan
LGBM	3	950	406	1356	0.644	0.571	0.038	0.656	0.669	0.098	1.171	1.127	0.3541
LGBM	4	952	408	1360	0.405	0.341	0.041	0.357	0.406	0.064	1.188	1.187	0.1494
LGBM	5	1909	817	2726	0.424	0.392	0.024	0.446	0.467	0.074	1.19	1.081	0.2968
LGBM	11	1909	817	2726	0.685	0.653	0.014	0.552	0.572	-0.081	0.876	1.048	0.6531
LGBM	MEAN	5720	2448	8168	0.539	0.49	0.029	0.502	0.528	0.039	1.106	1.111	0.363
CatBoost	3	950	406	1356	0.736	0.643	0.034	0.725	0.732	0.09	1.139	1.146	0.3616
CatBoost	4	952	408	1360	0.656	0.524	0.034	0.586	0.595	0.071	1.135	1.252	0.128
CatBoost	5	1909	817	2726	0.46	0.427	0.021	0.493	0.495	0.068	1.159	1.079	0.3261
CatBoost	11	1909	817	2726	0.716	0.686	0.016	0.592	0.597	-0.089	0.87	1.043	0.6861
CatBoost	MEAN	5720	2448	8168	0.642	0.57	0.026	0.599	0.605	0.035	1.076	1.13	0.375

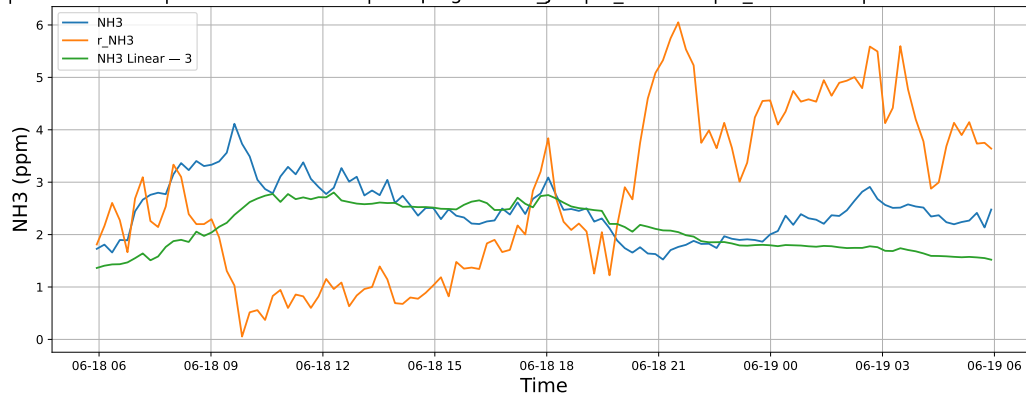
3 Plot

3.1 Standalone plots

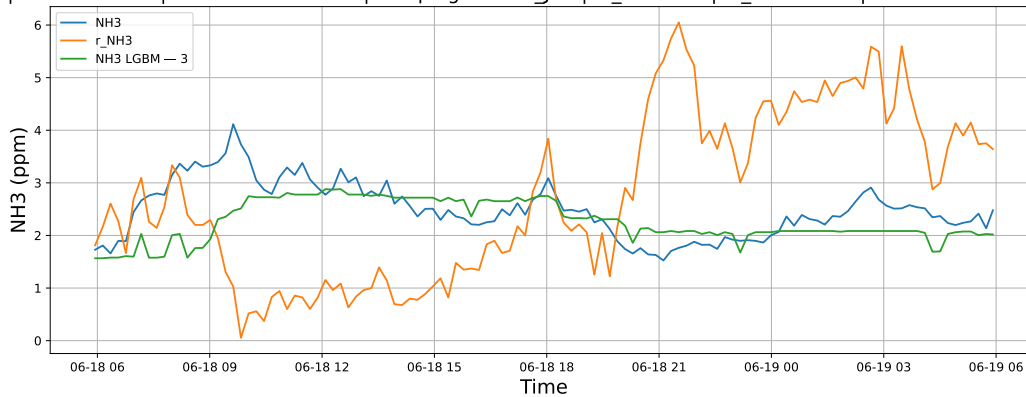
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

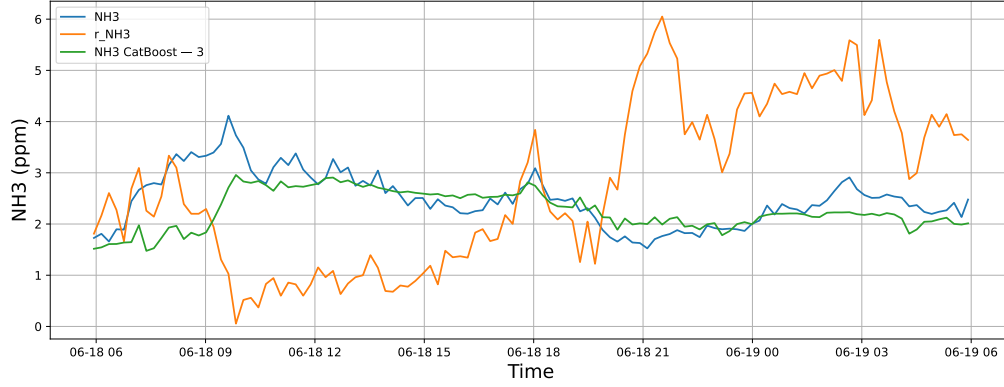
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

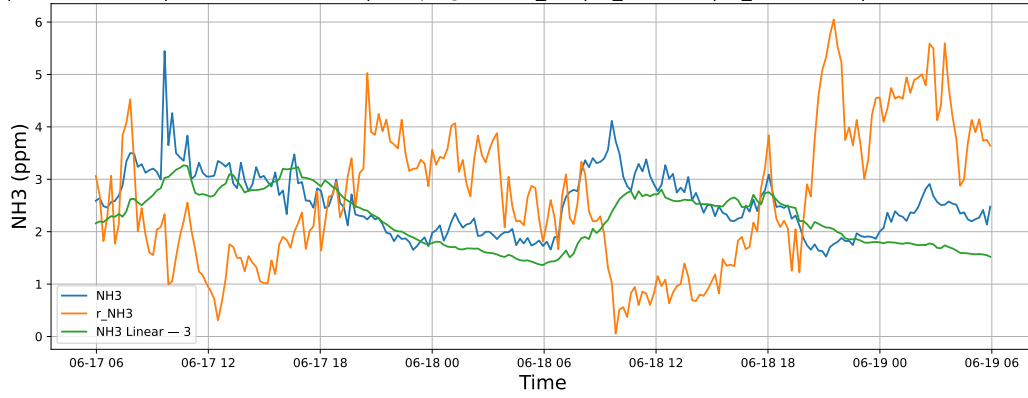


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

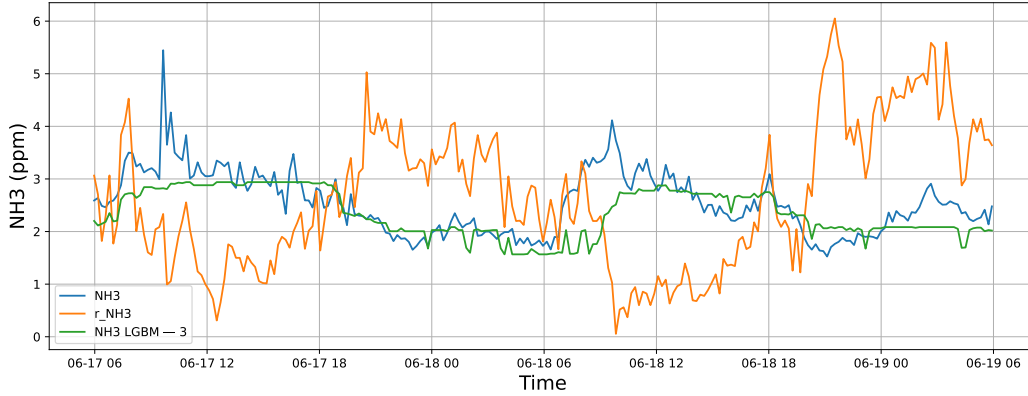


3.1.1.2 Time window : 48

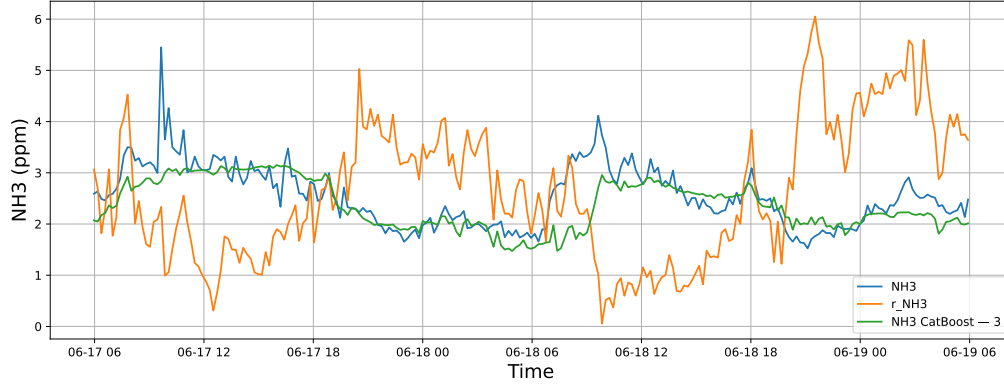
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

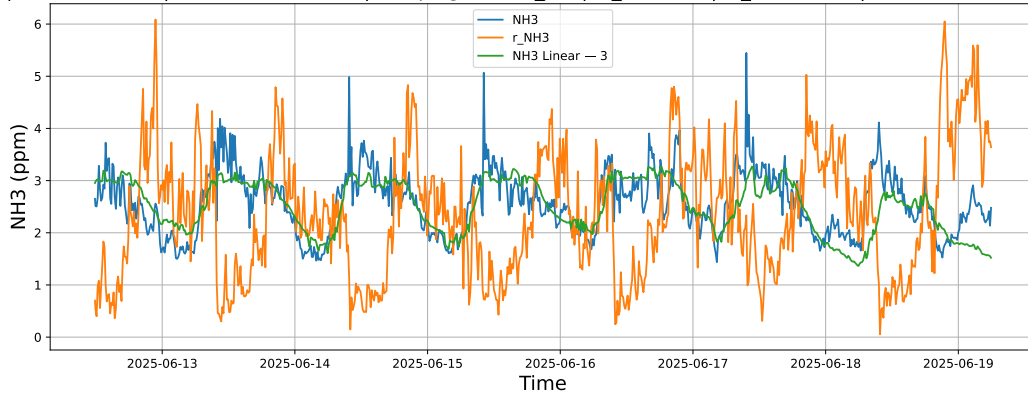


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

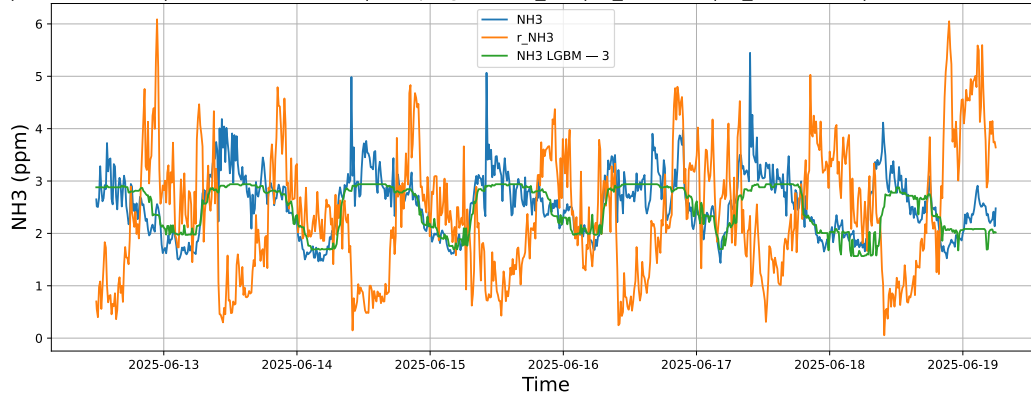


3.1.1.3 Time window : ALL

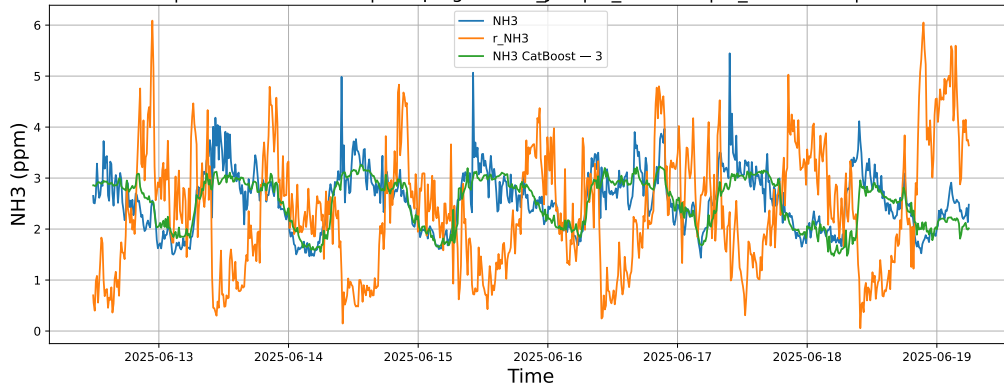
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



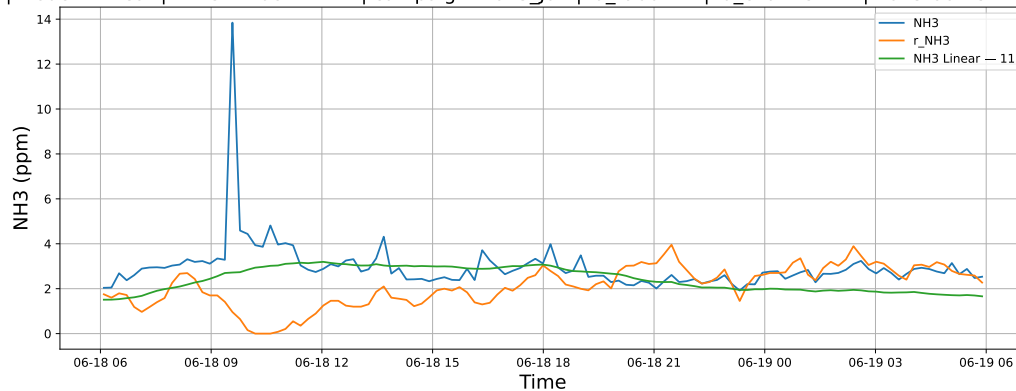
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



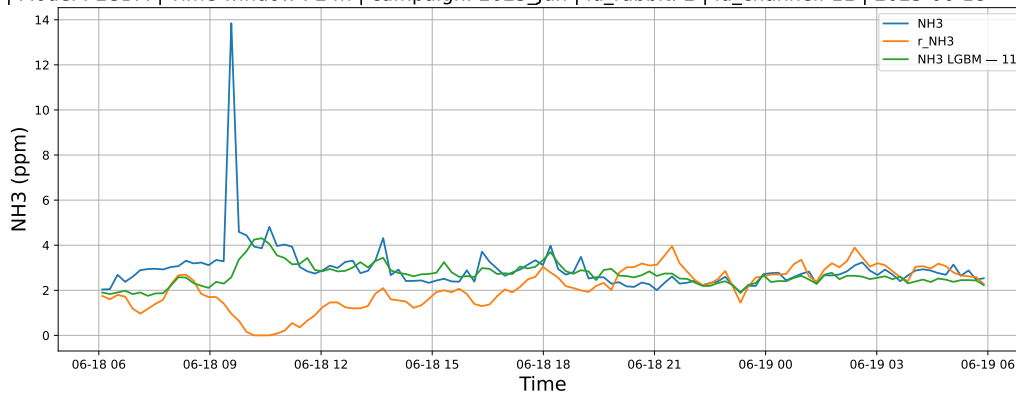
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

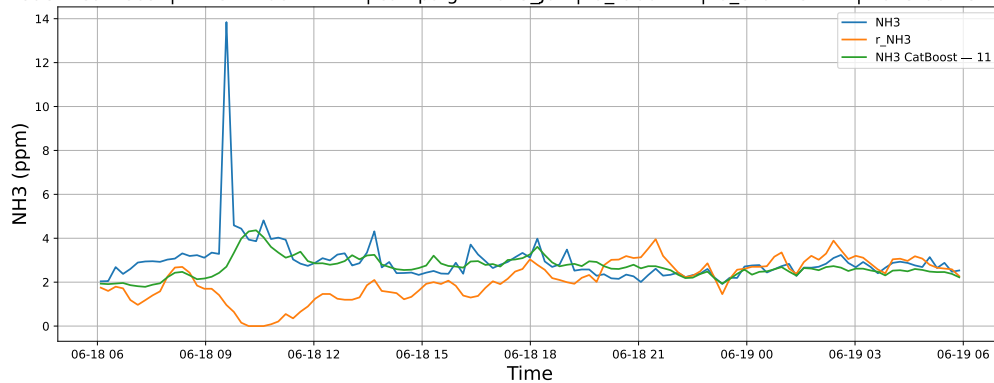
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

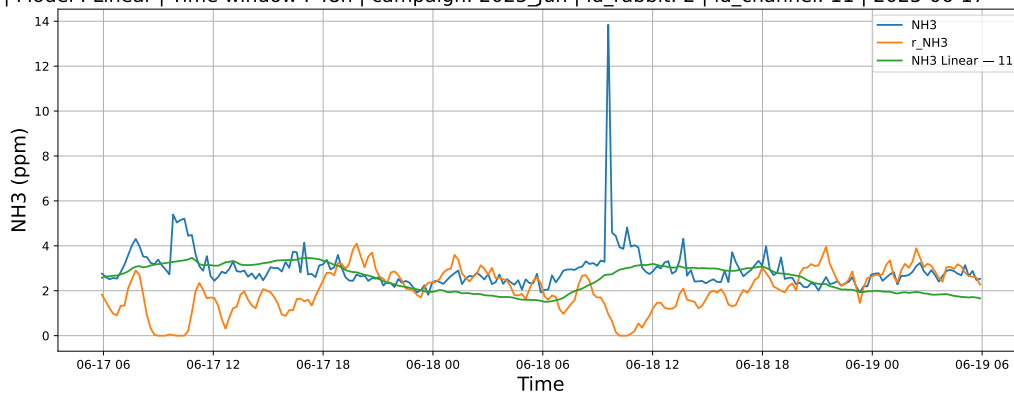


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

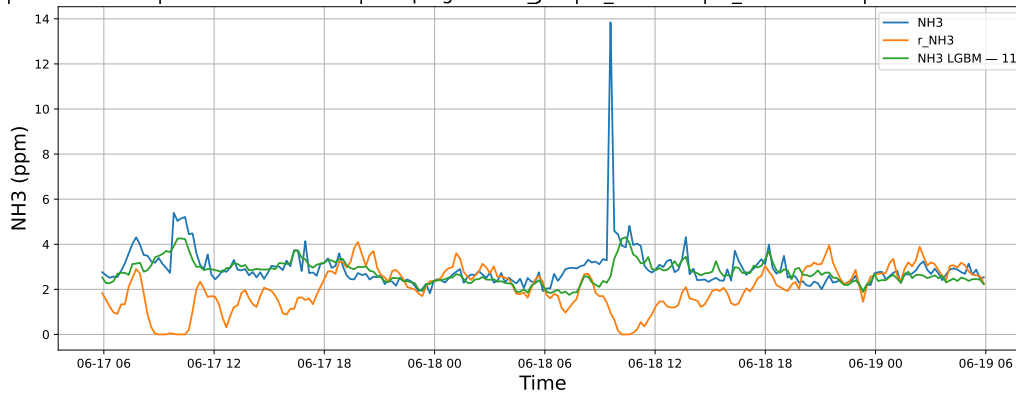


3.1.2.2 Time window : 48

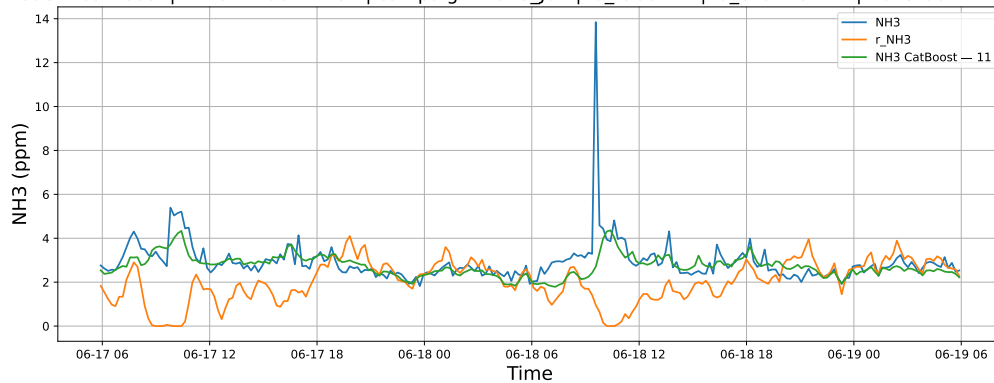
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

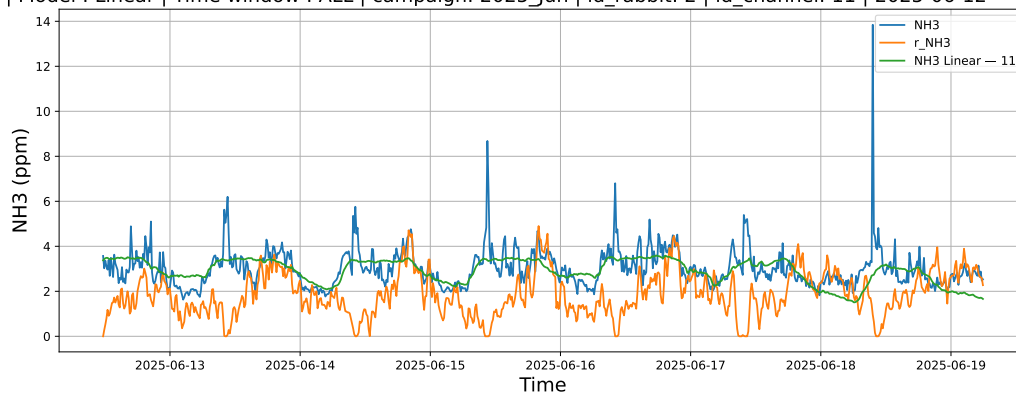


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

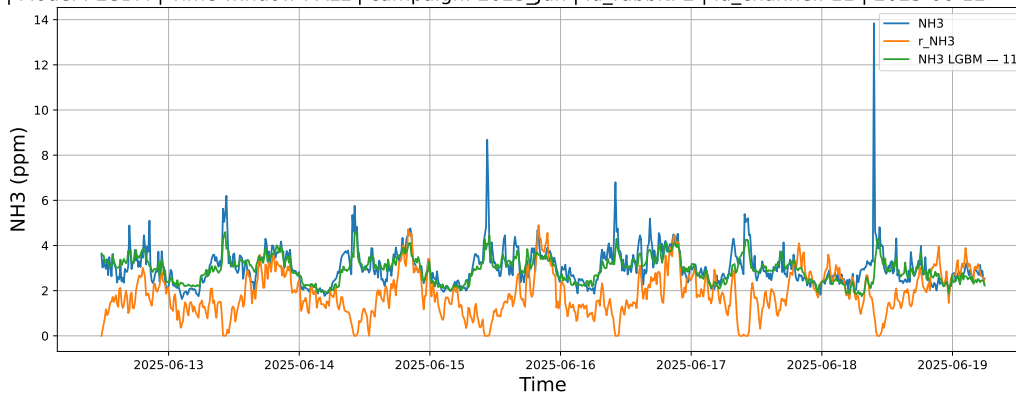


3.1.2.3 Time window : ALL

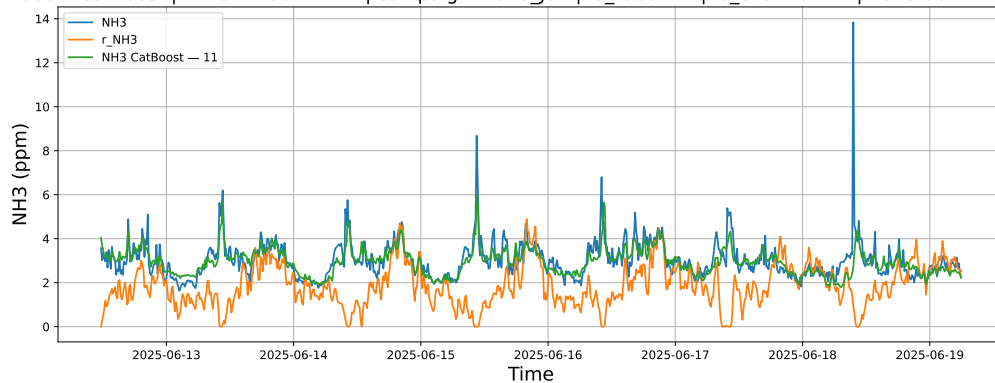
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



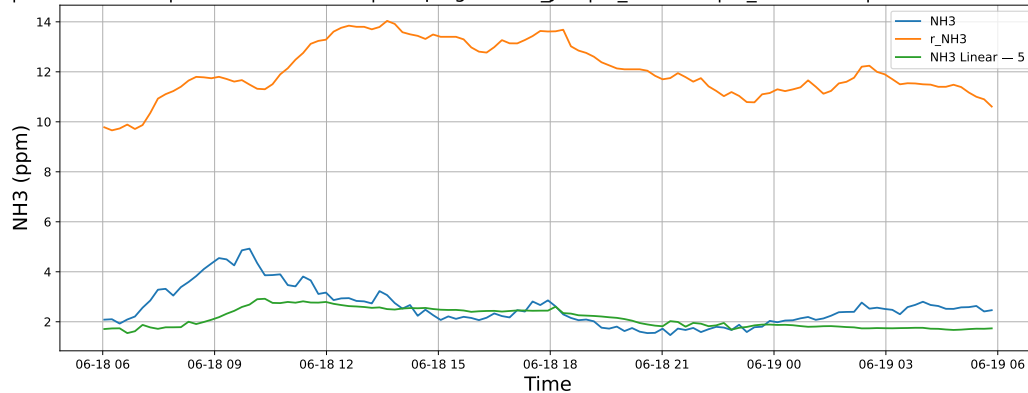
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



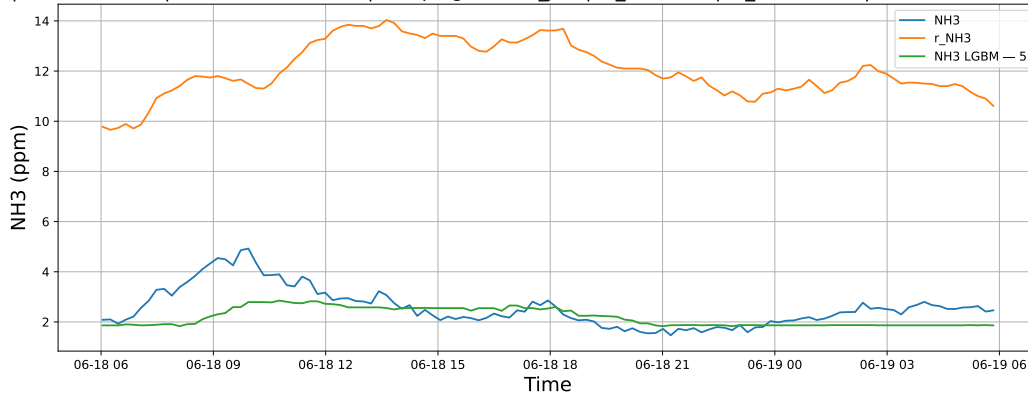
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

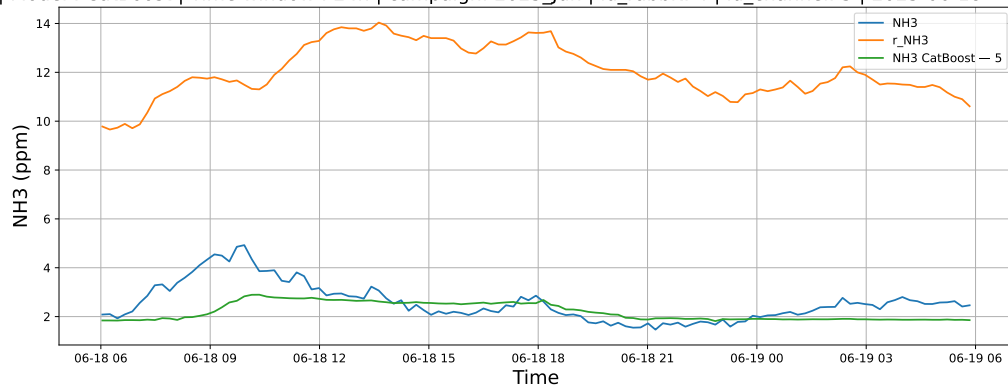
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

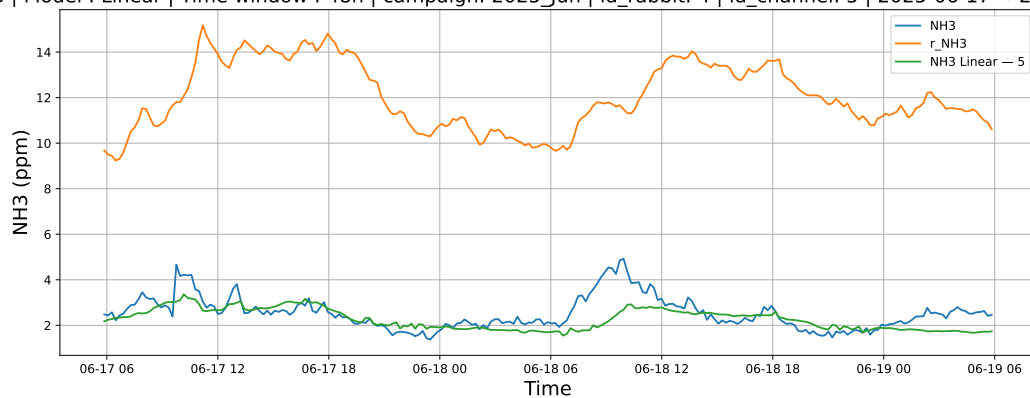


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

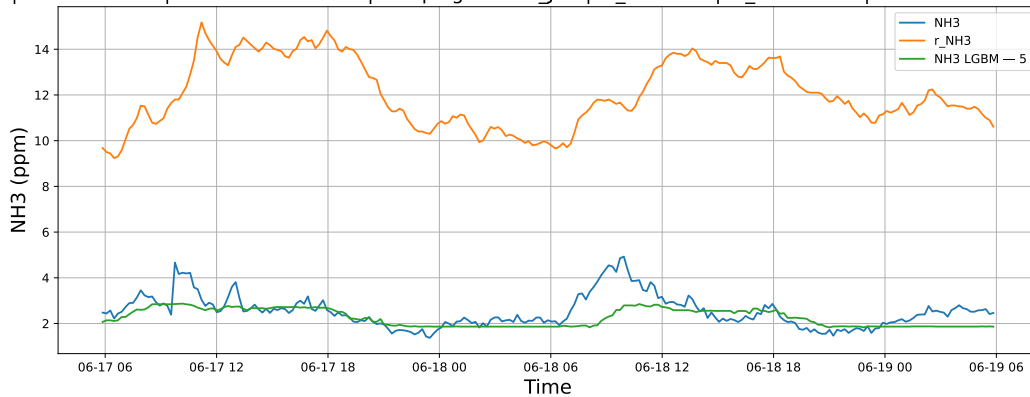


3.1.3.2 Time window : 48

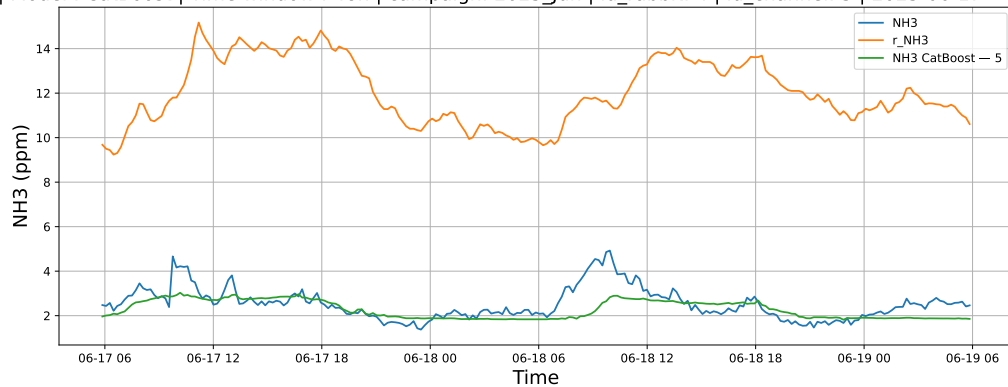
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

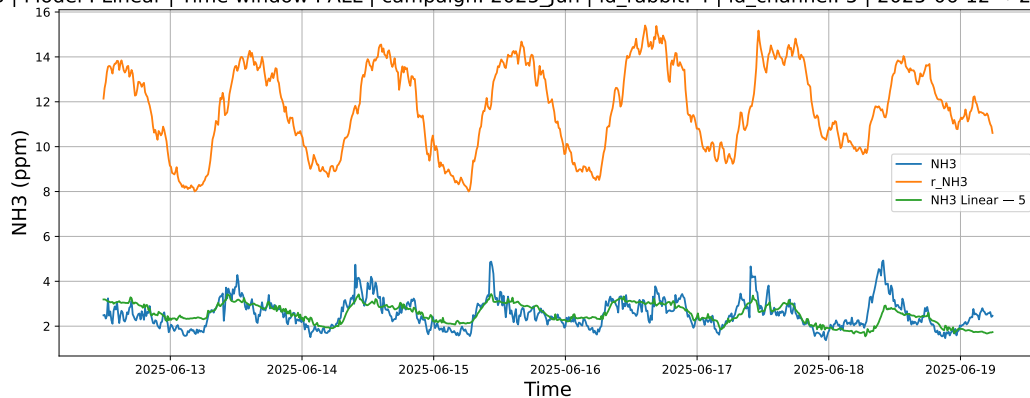


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

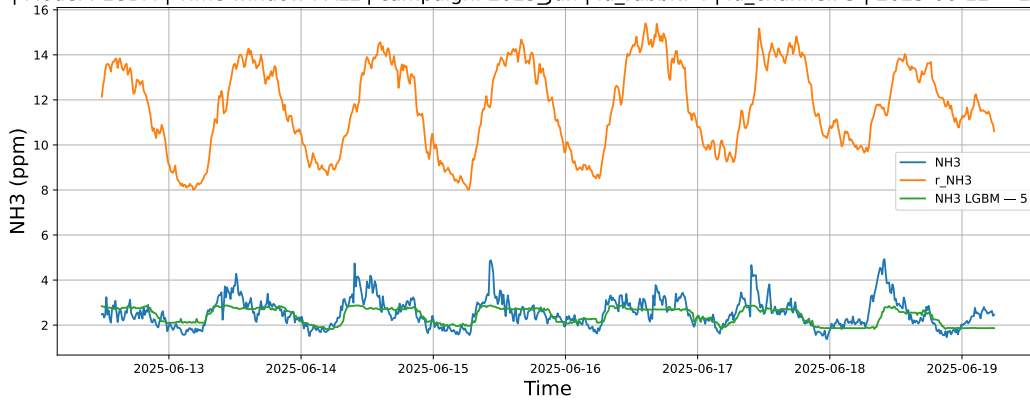


3.1.3.3 Time window : ALL

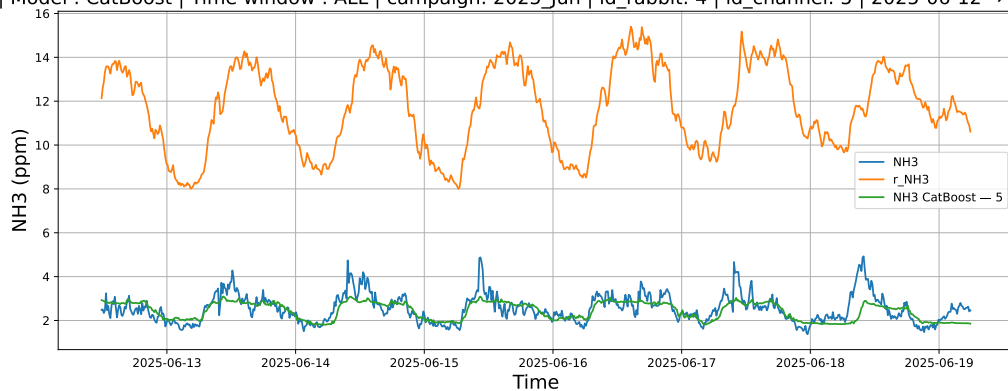
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



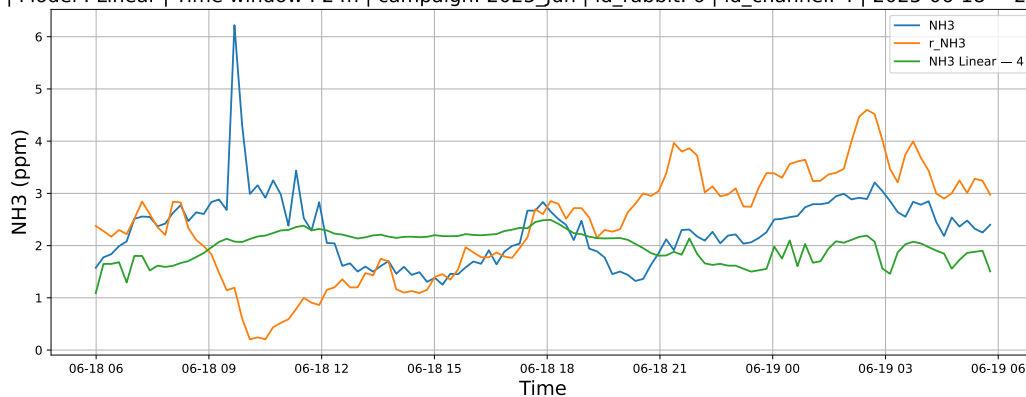
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



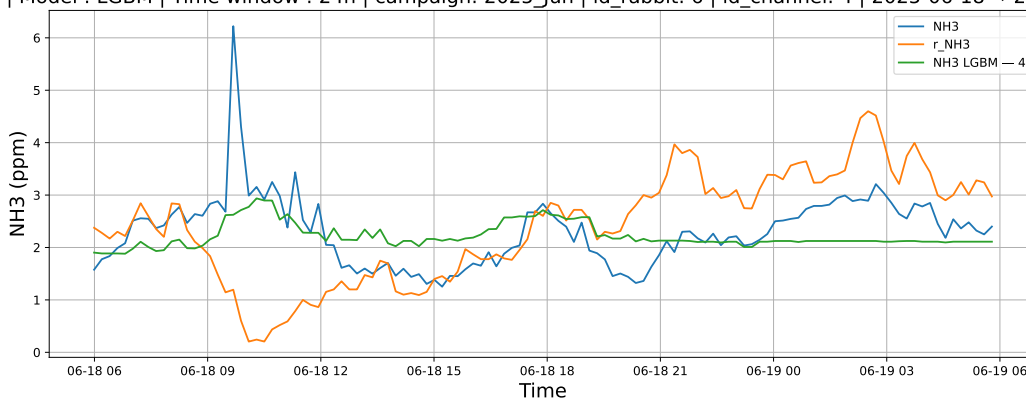
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

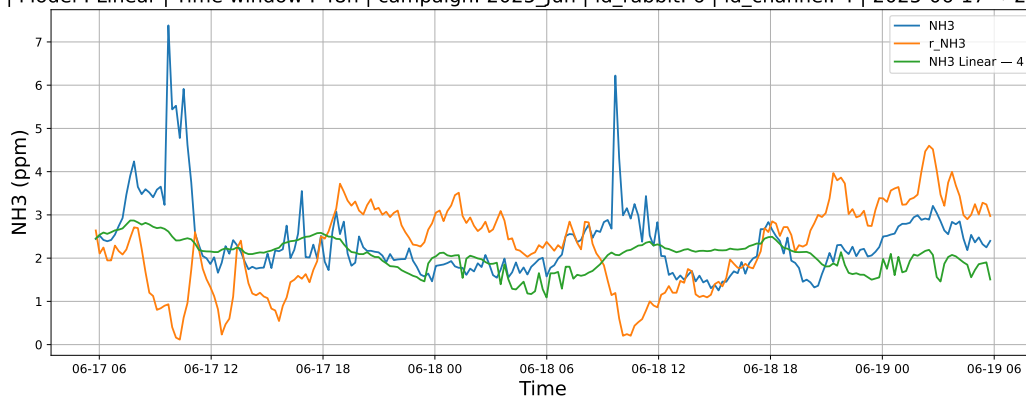


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

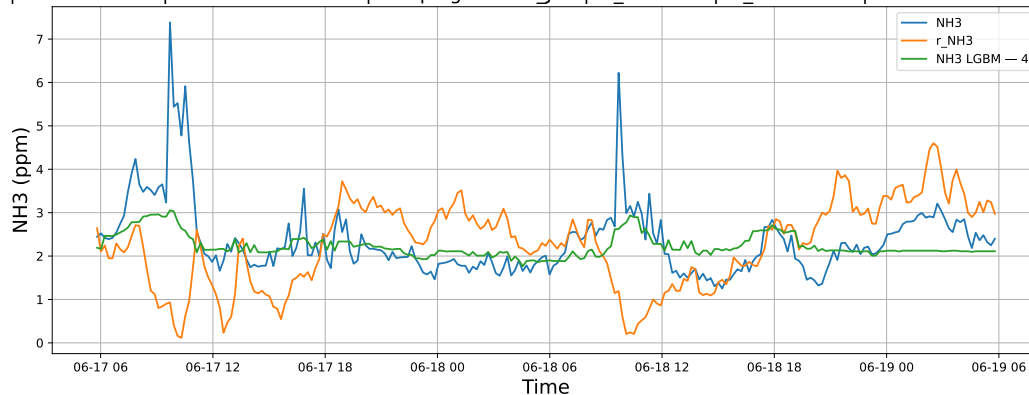


3.1.4.2 Time window : 48

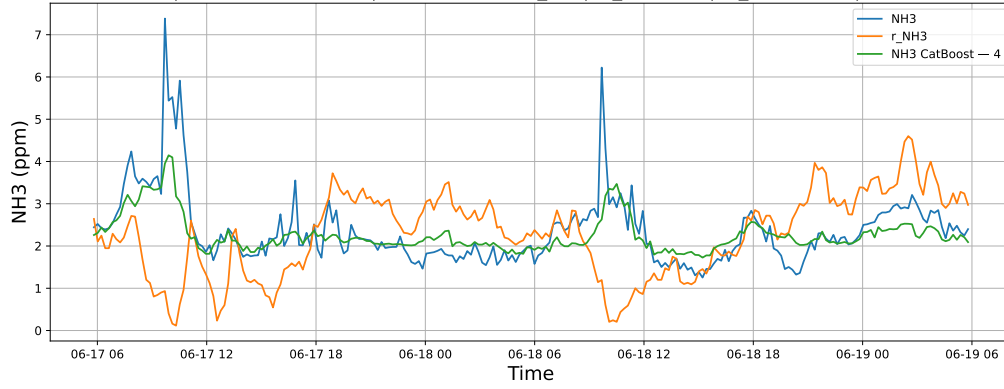
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

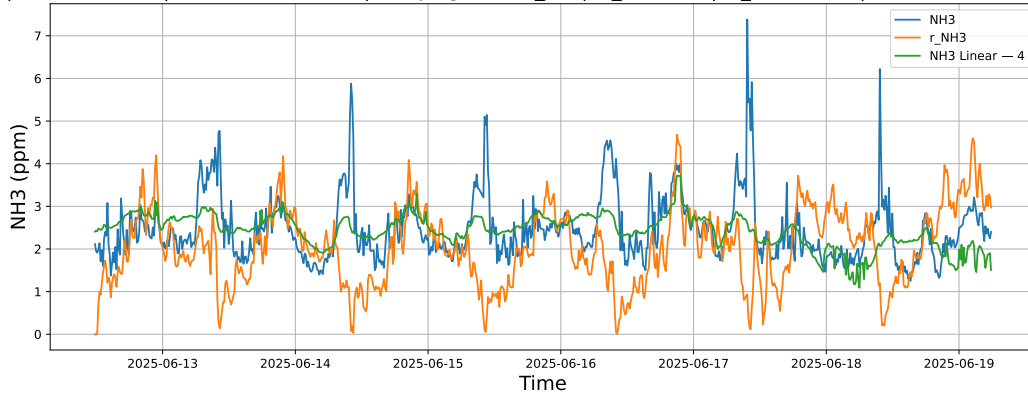


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

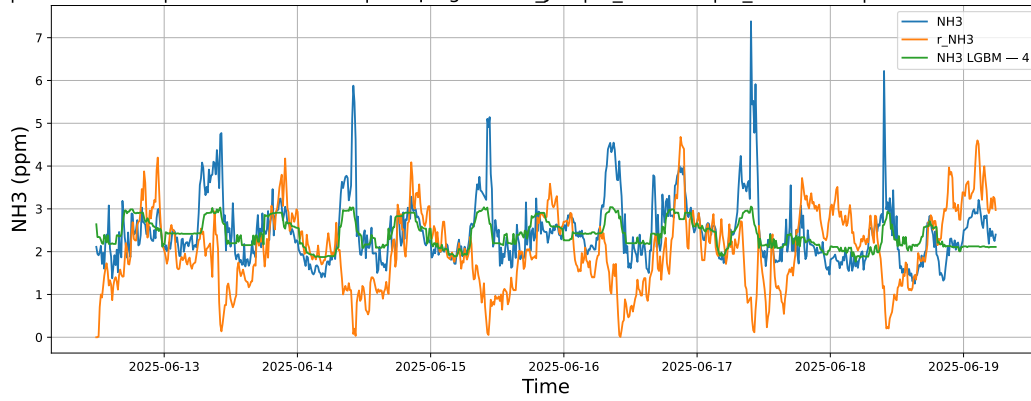


3.1.4.3 Time window : ALL

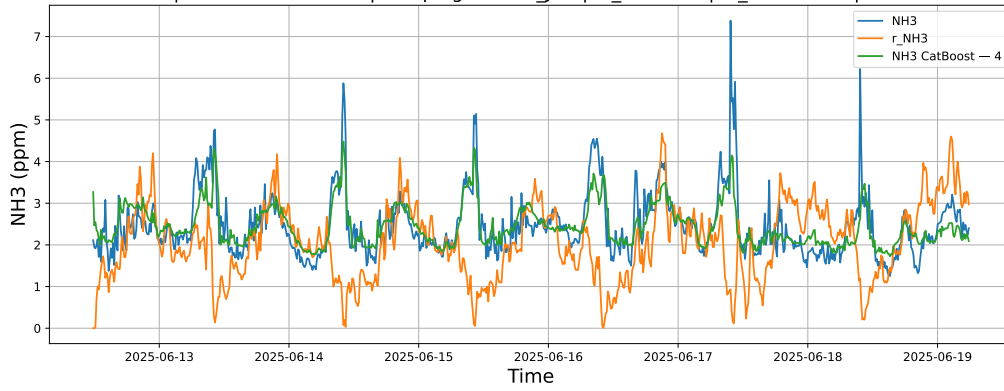
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



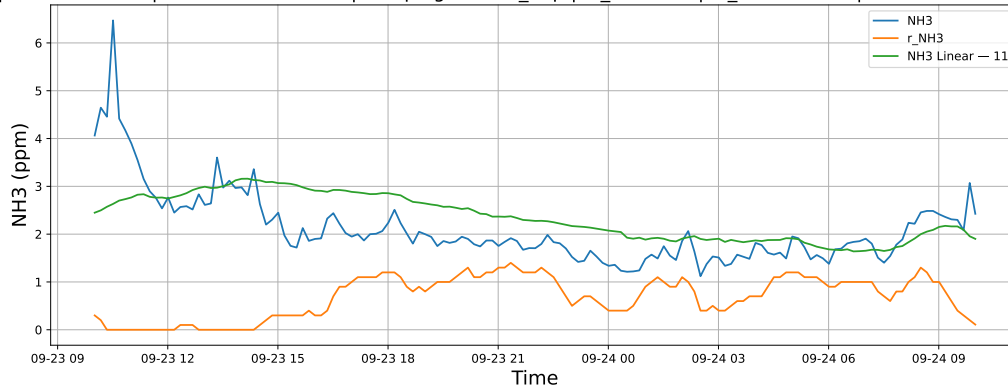
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



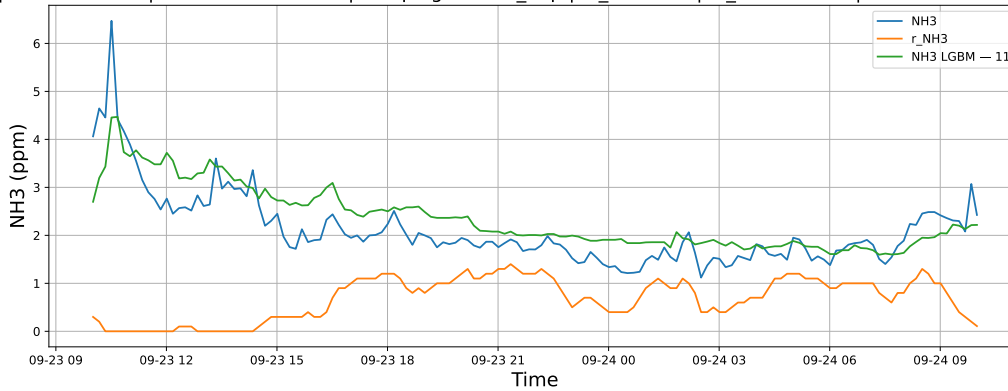
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

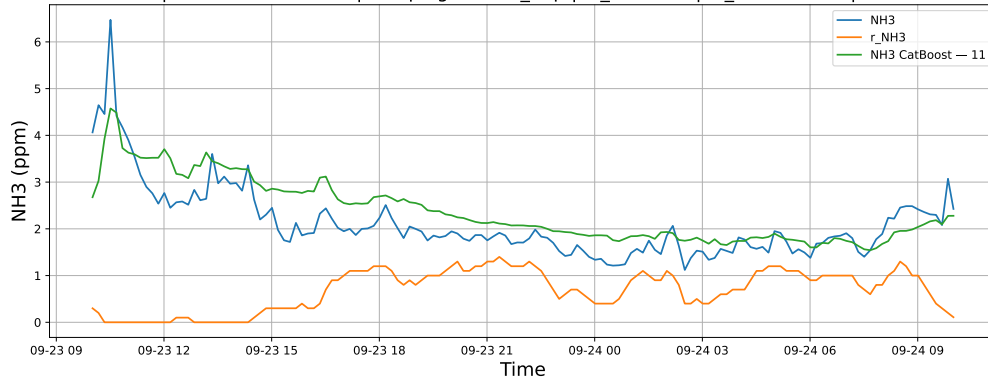
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

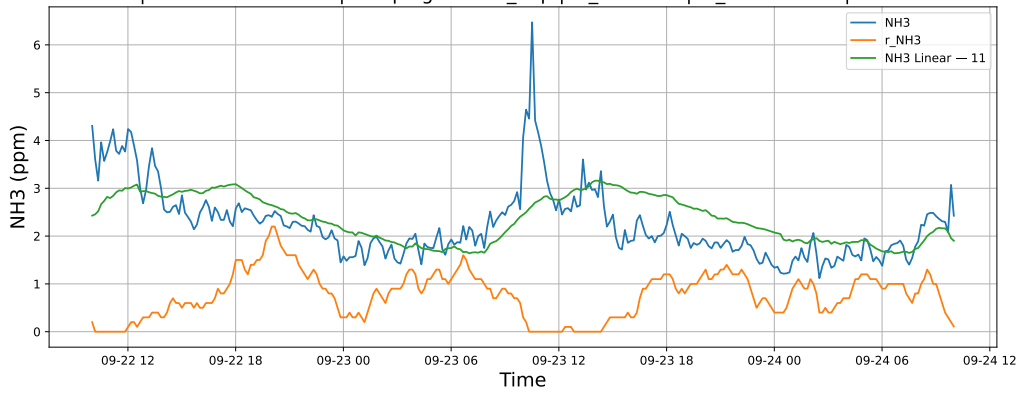


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

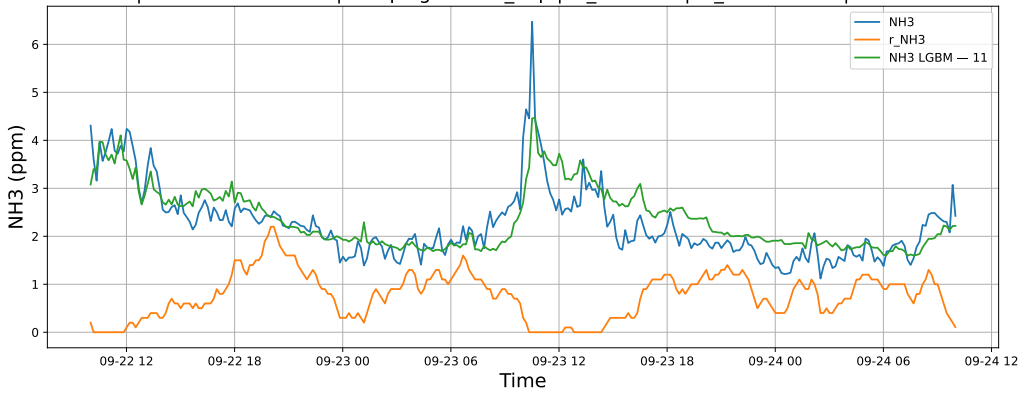


3.1.5.2 Time window : 48

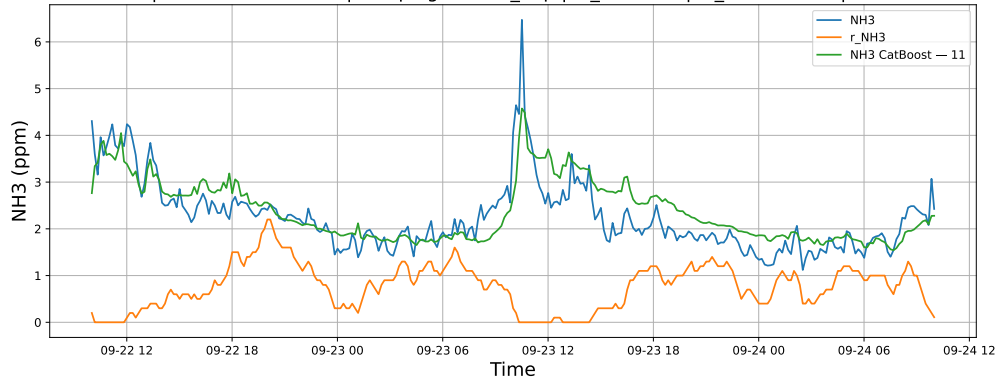
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

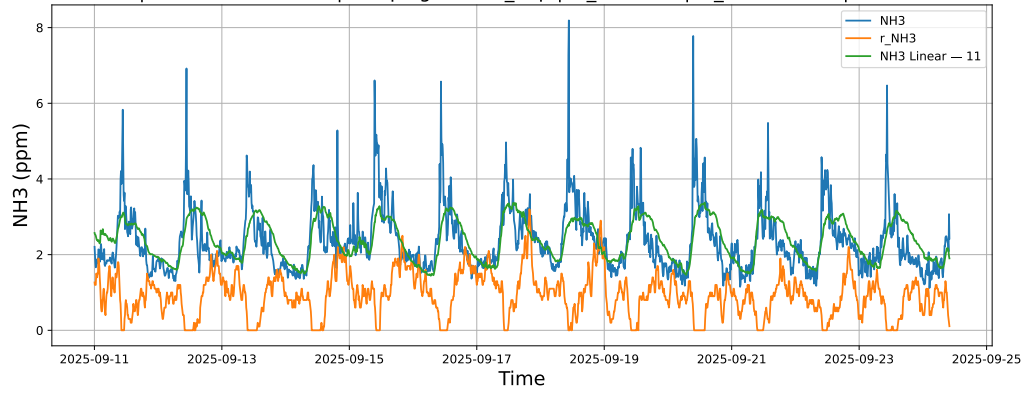


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

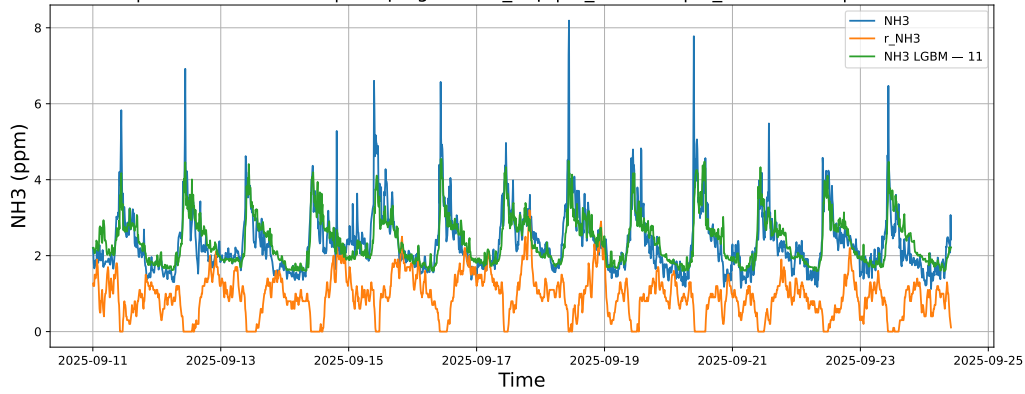


3.1.5.3 Time window : ALL

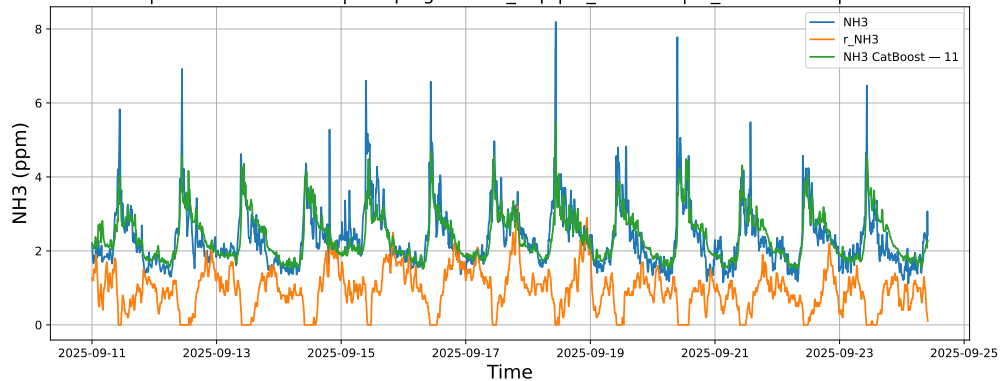
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



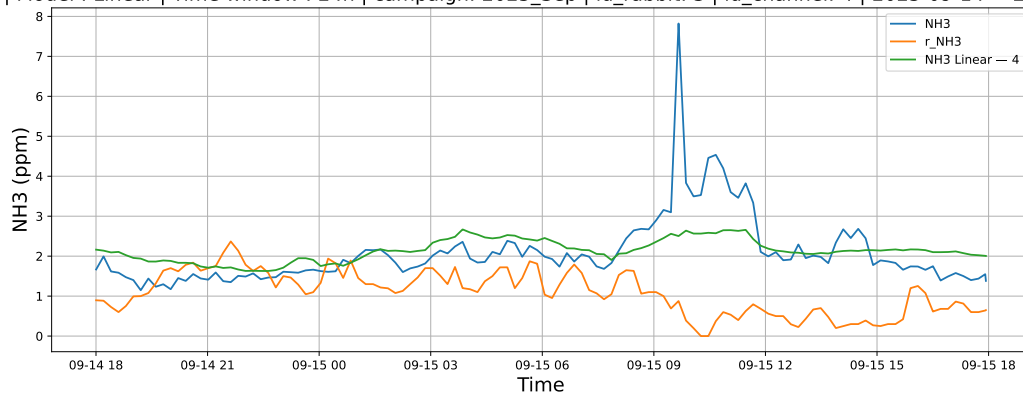
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



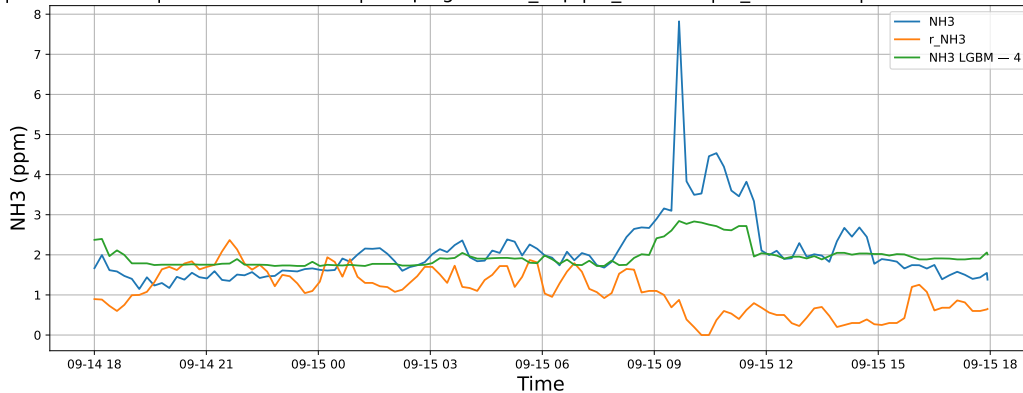
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

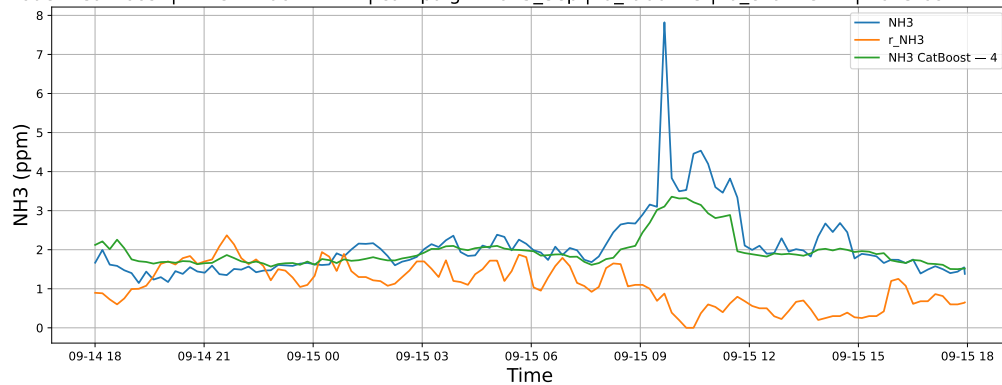
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

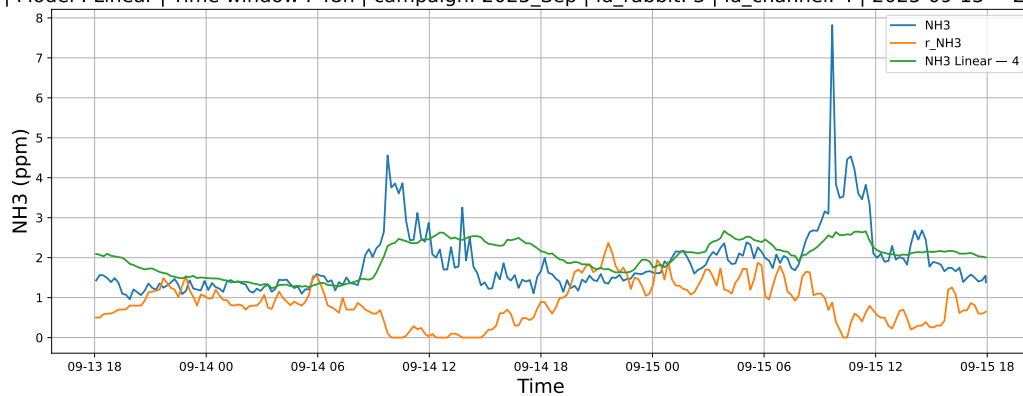


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

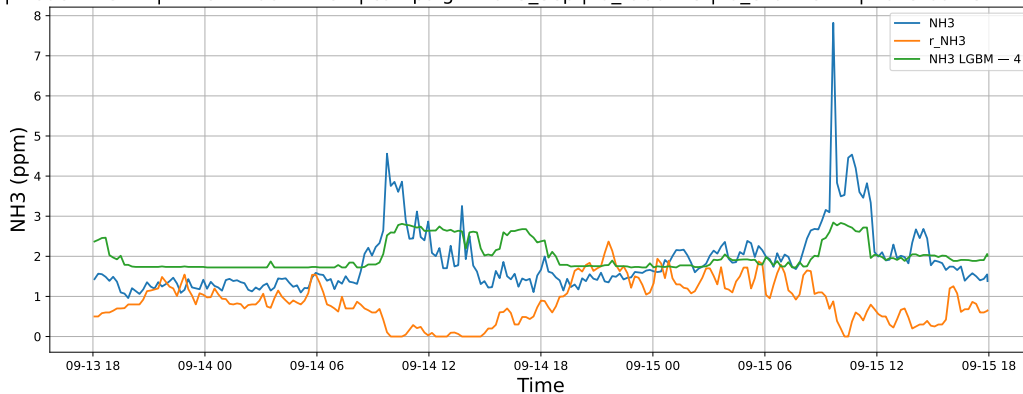


3.1.6.2 Time window : 48

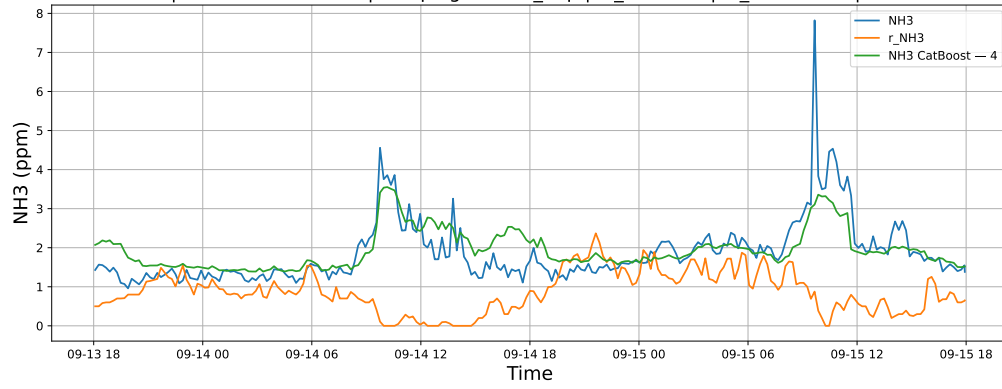
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

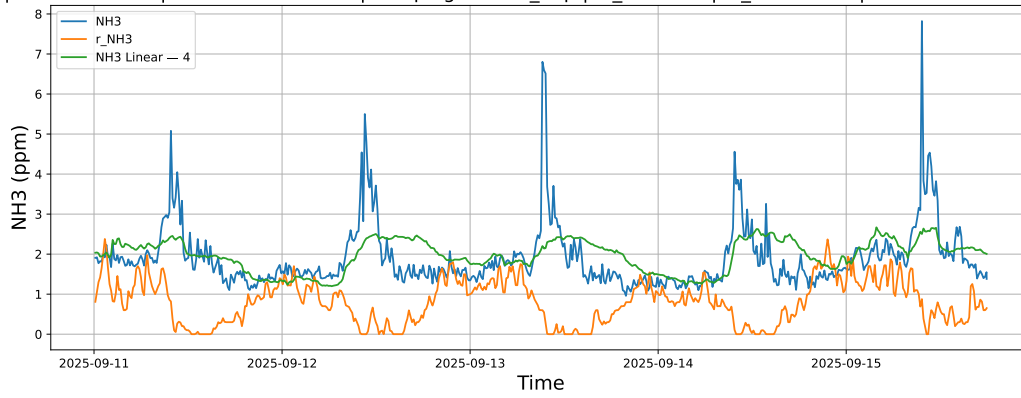


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

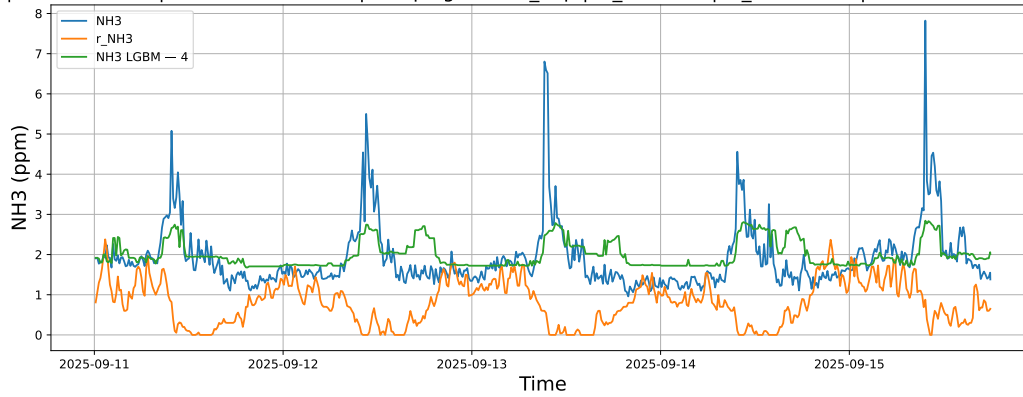


3.1.6.3 Time window : ALL

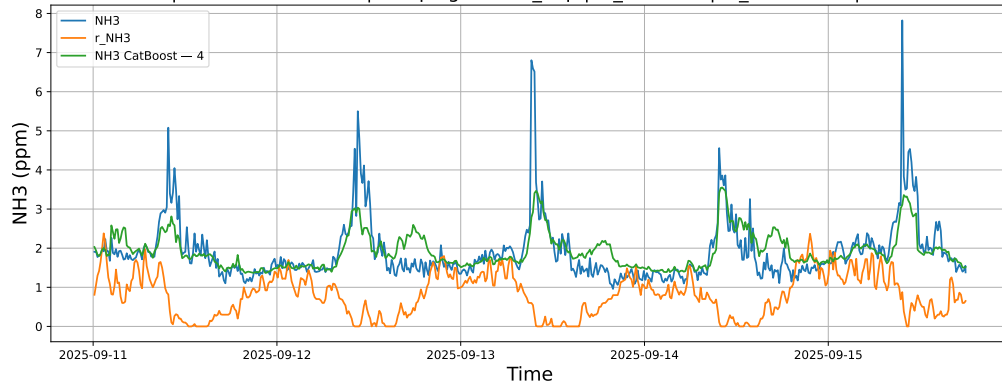
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



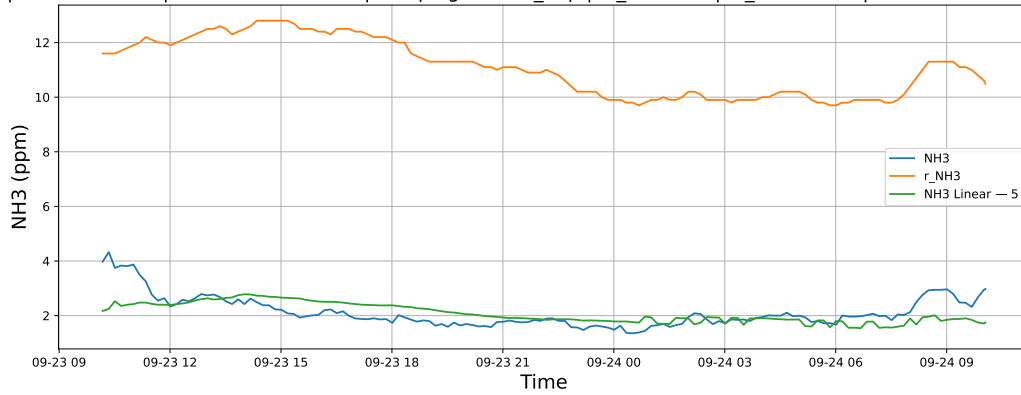
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



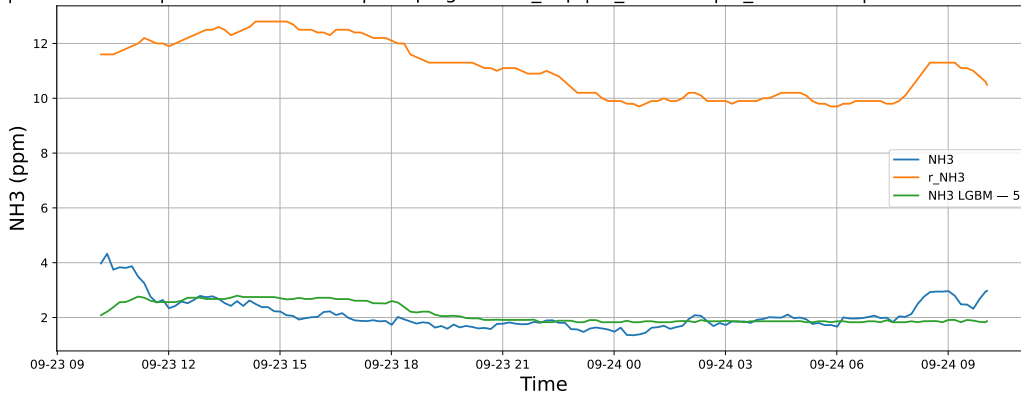
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

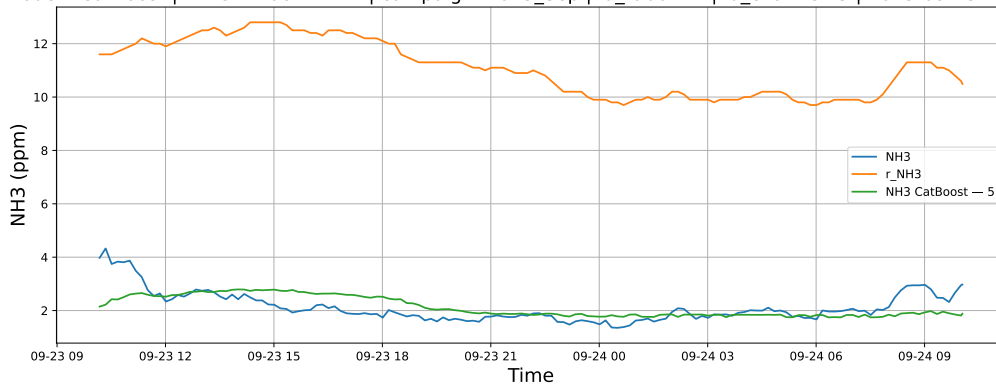
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

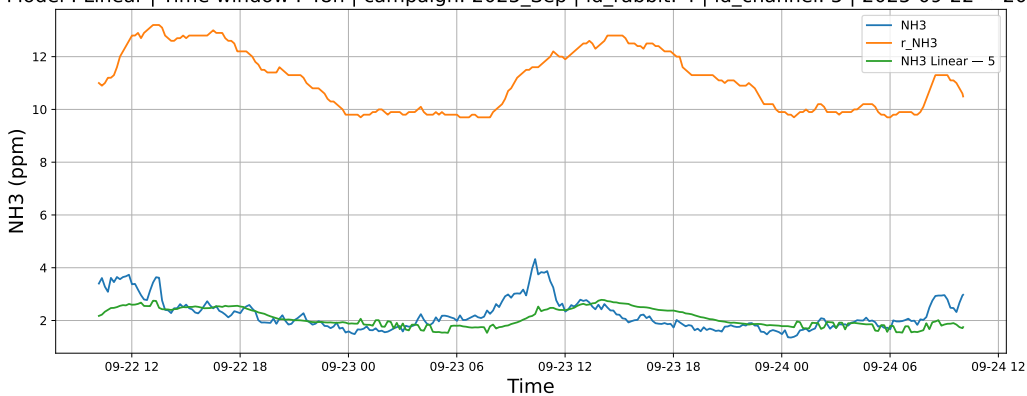


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

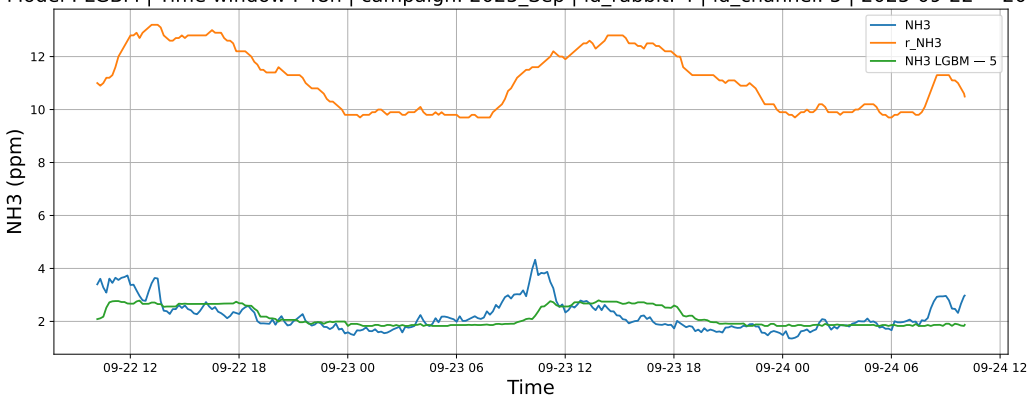


3.1.7.2 Time window : 48

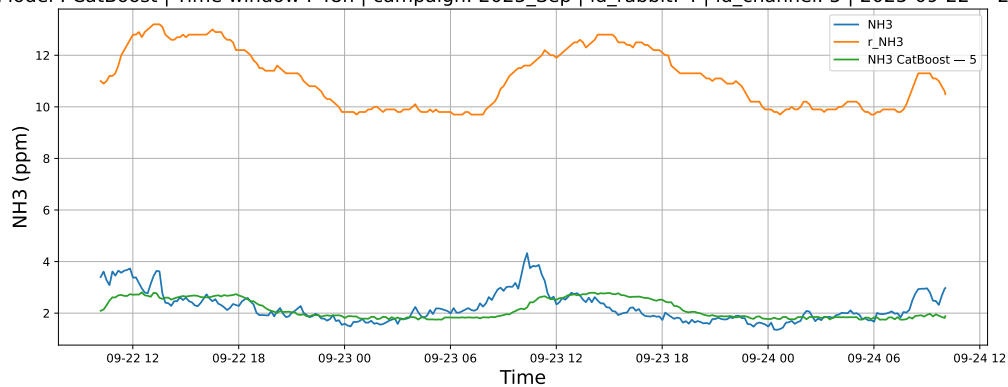
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

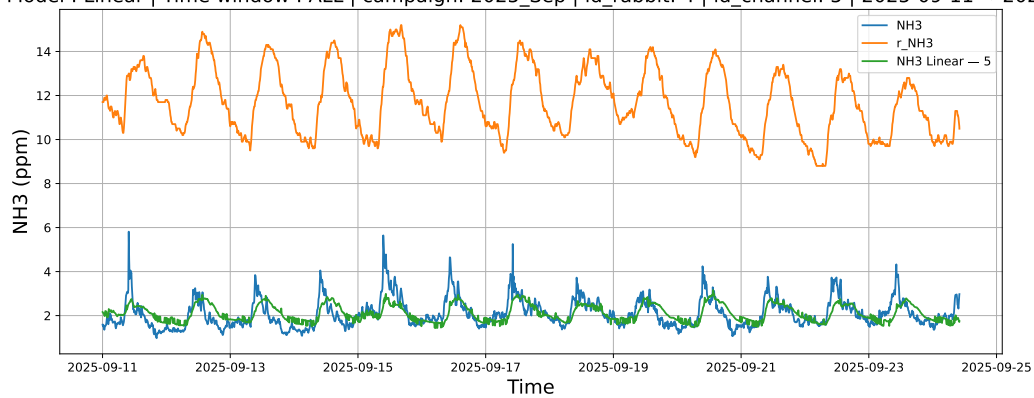


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

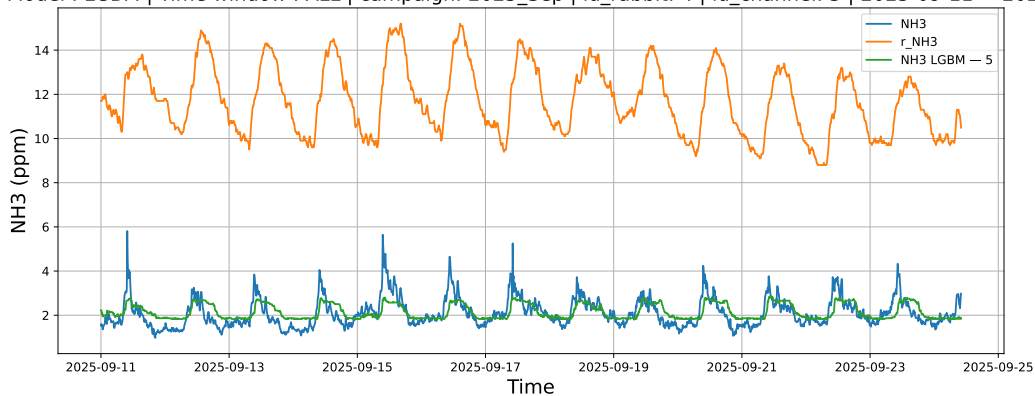


3.1.7.3 Time window : ALL

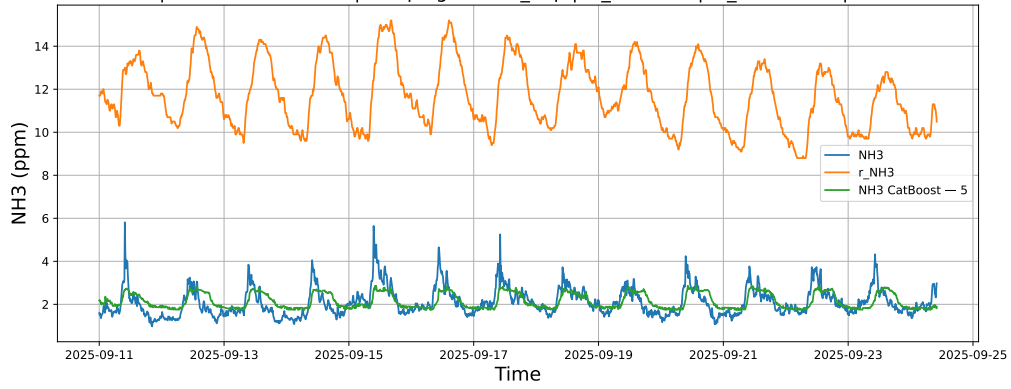
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



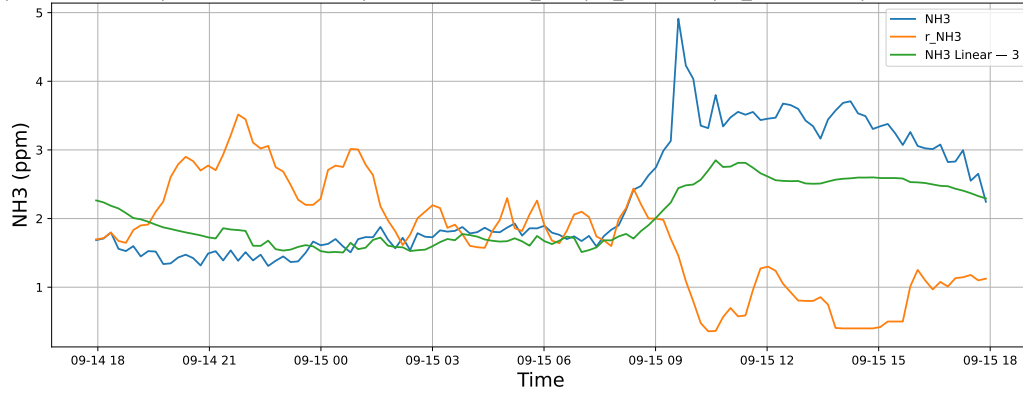
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



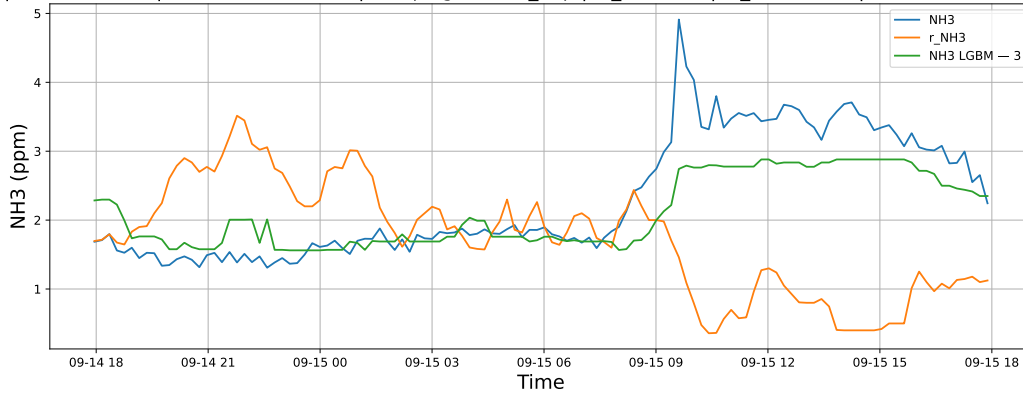
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

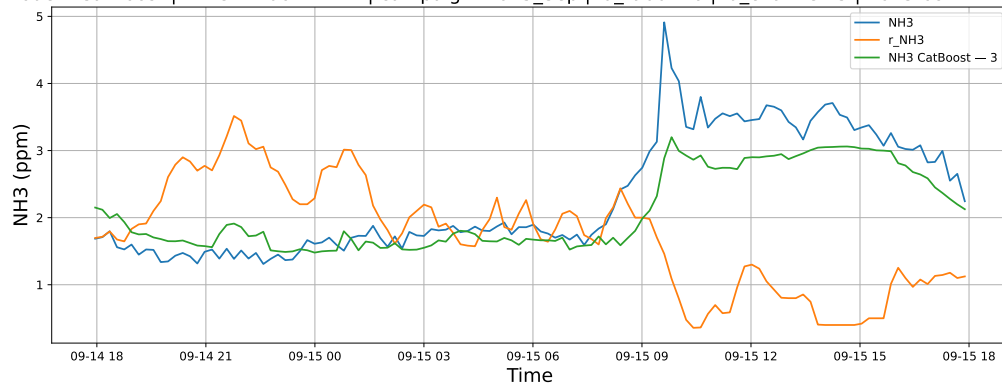
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

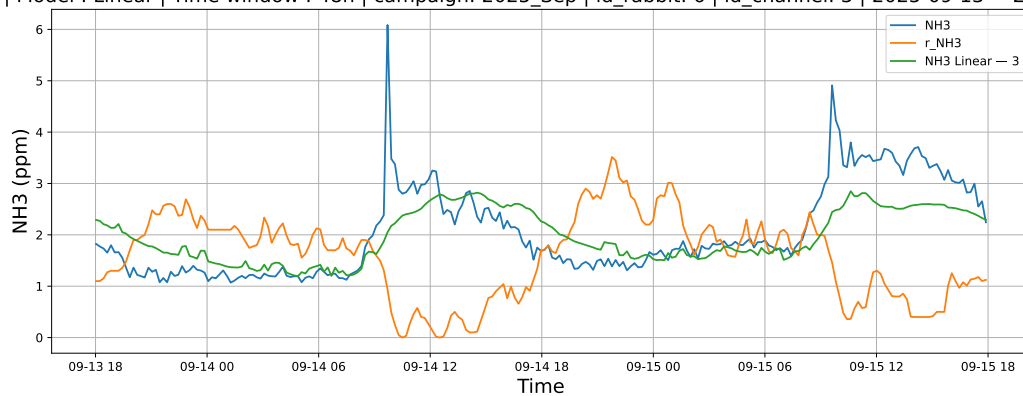


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

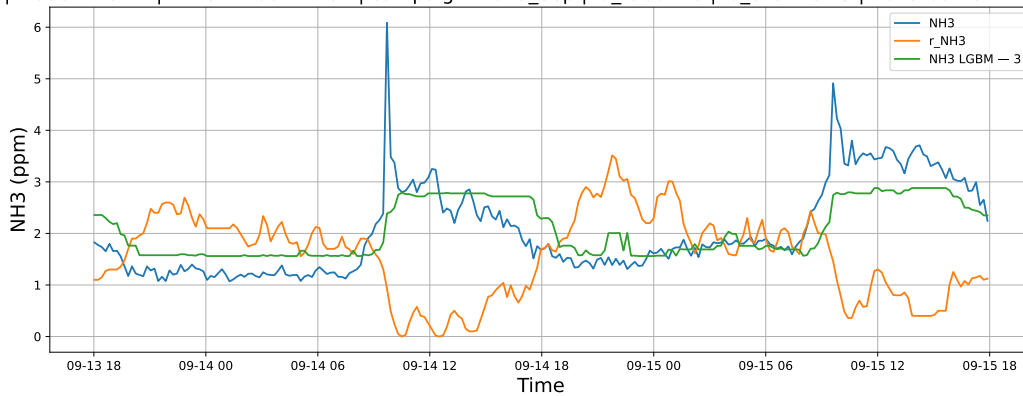


3.1.8.2 Time window : 48

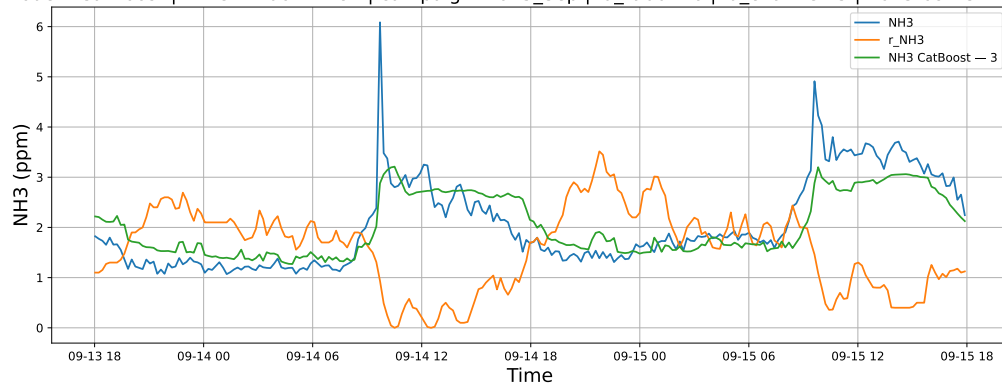
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

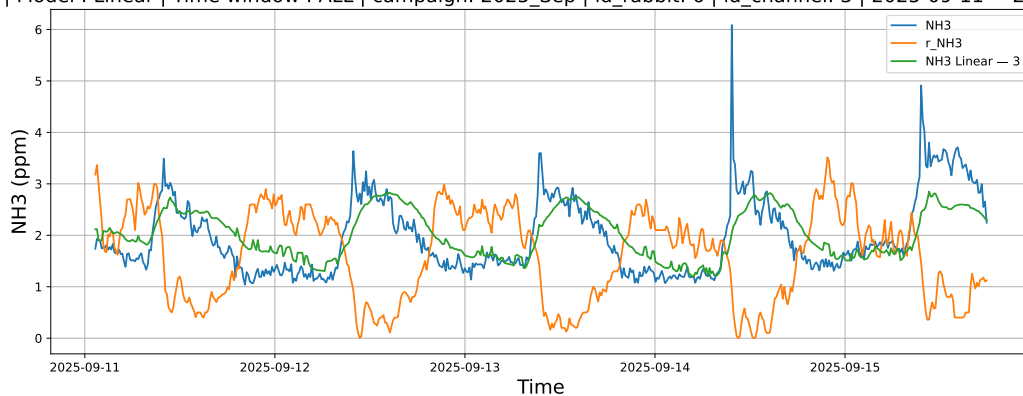


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

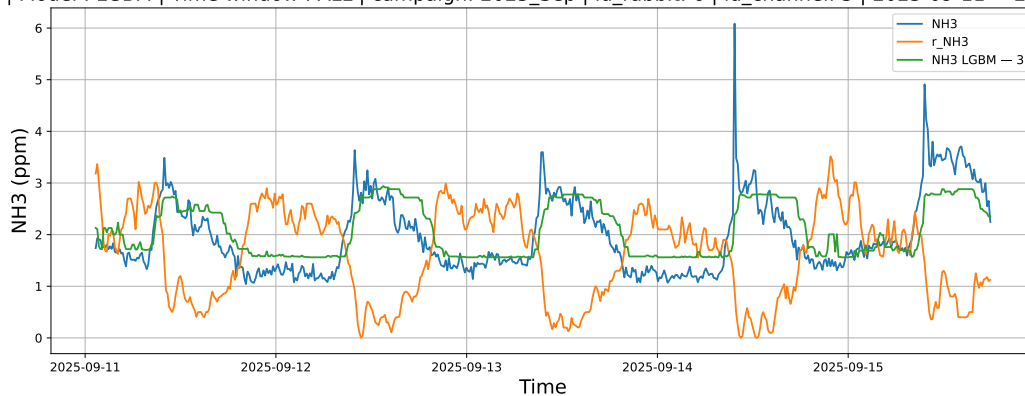


3.1.8.3 Time window : ALL

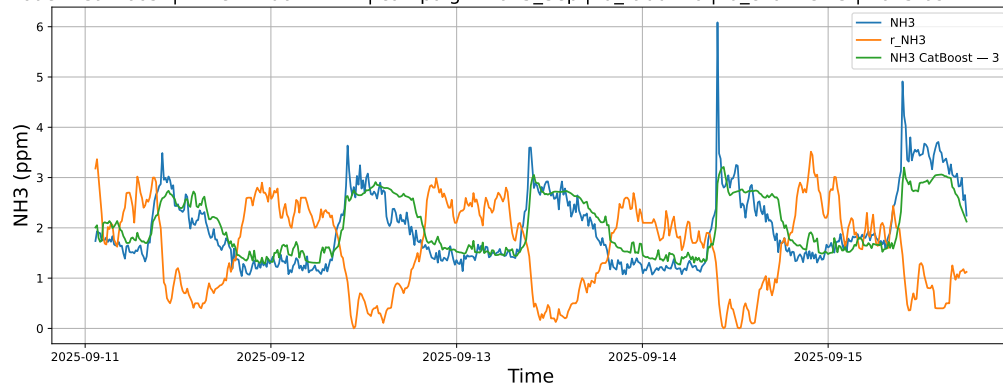
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

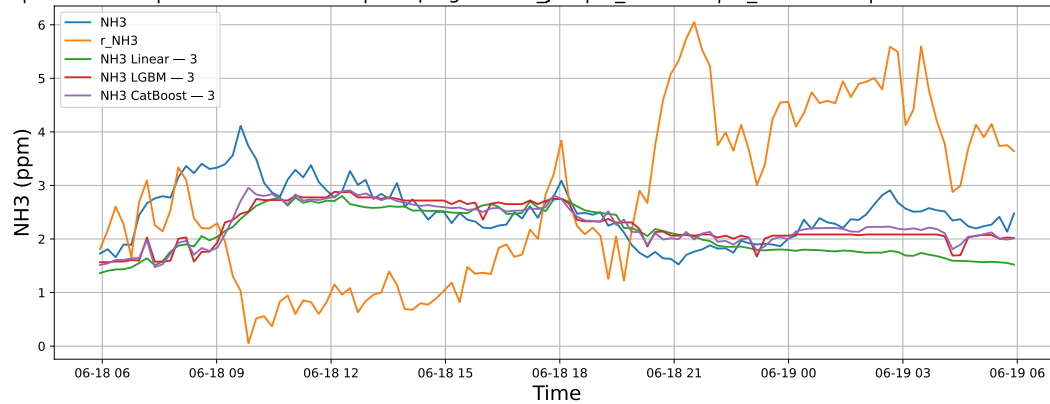


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

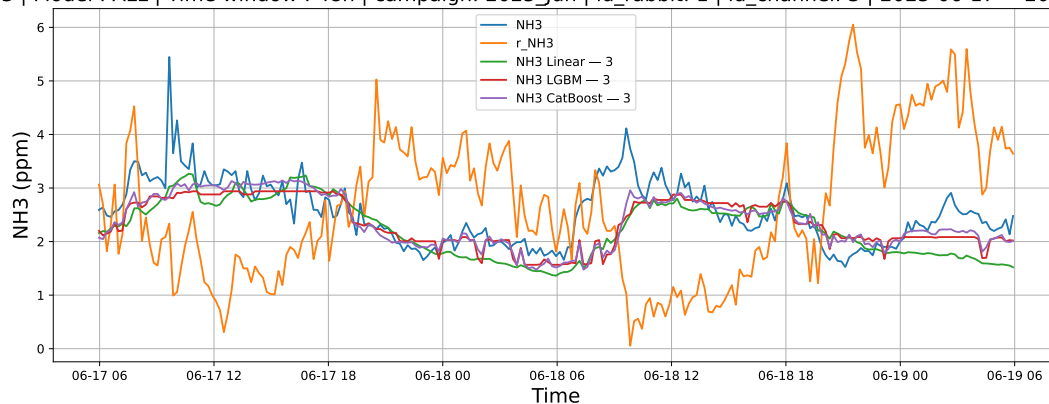
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



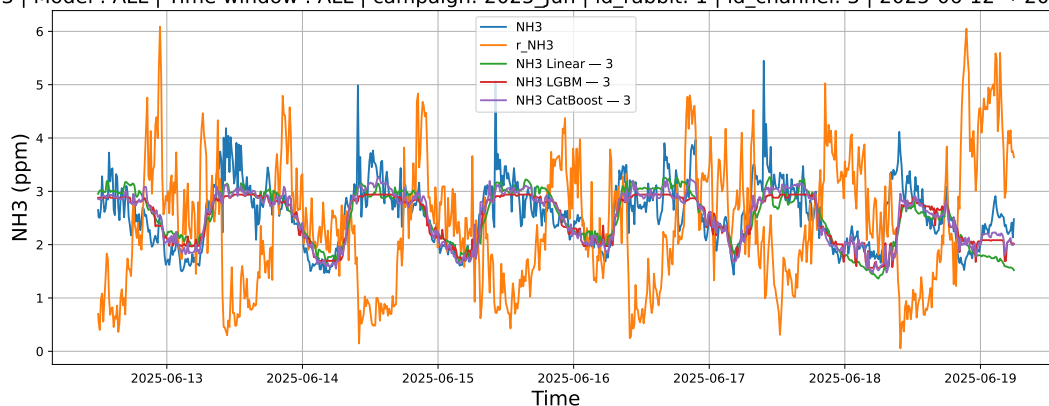
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

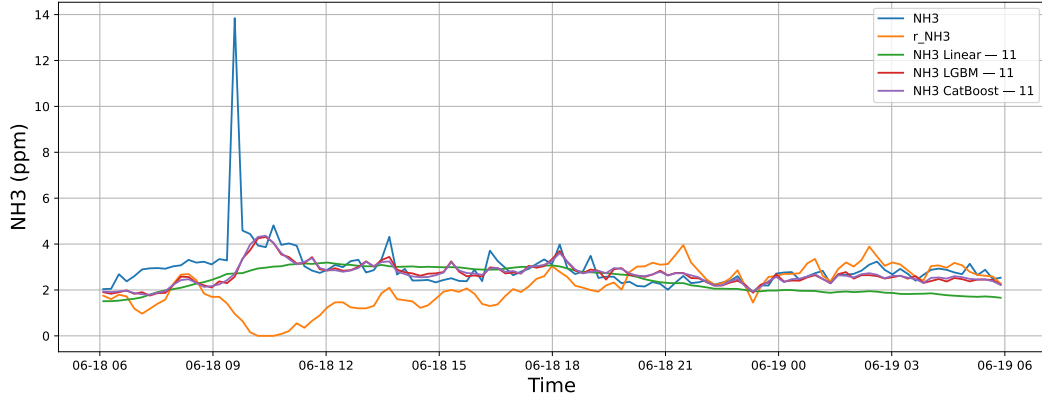
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

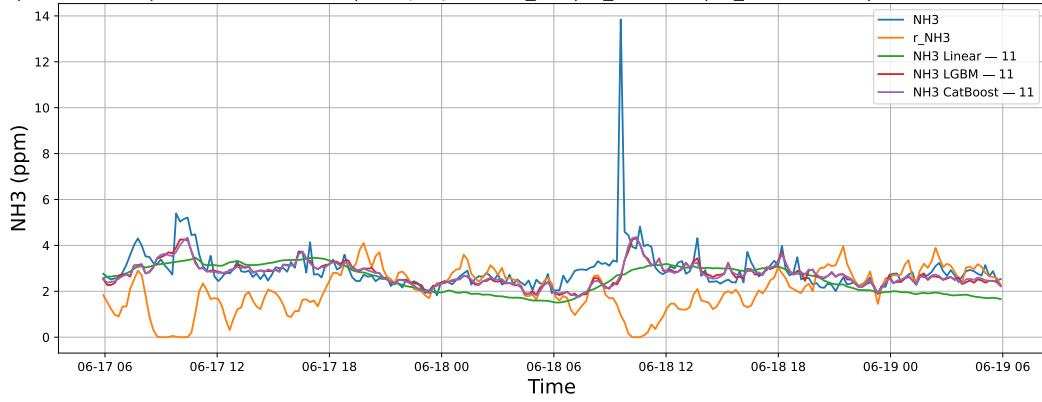
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



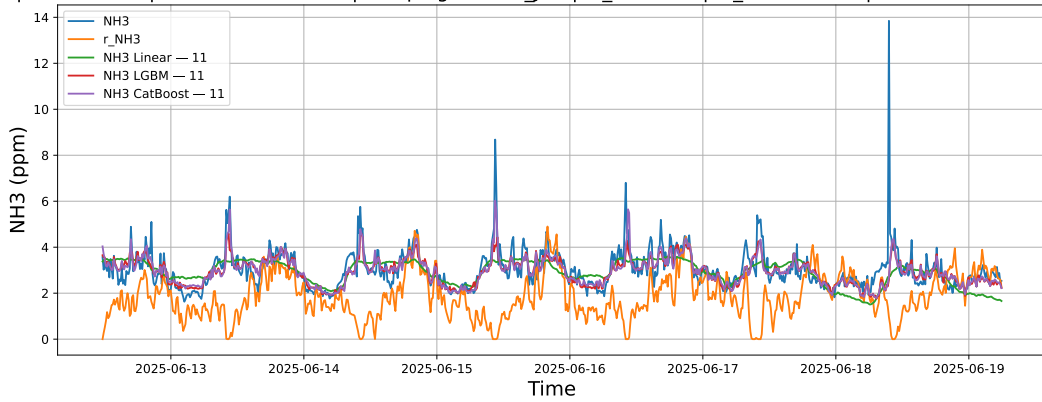
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

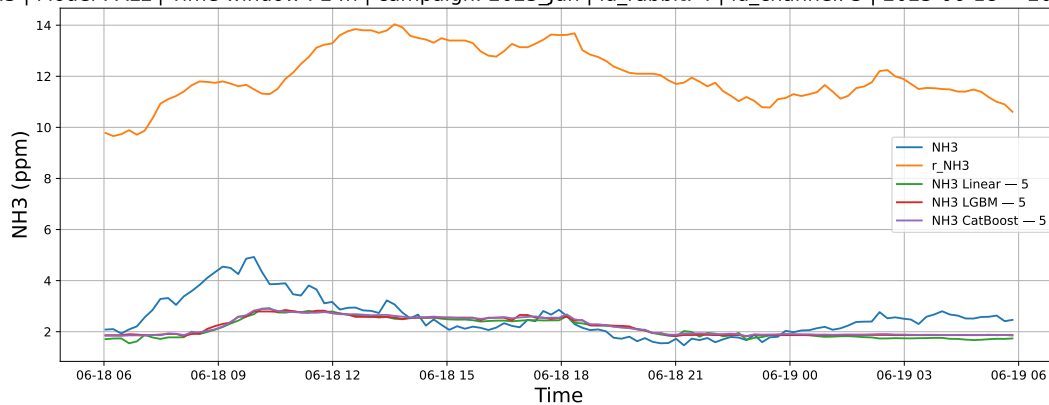
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

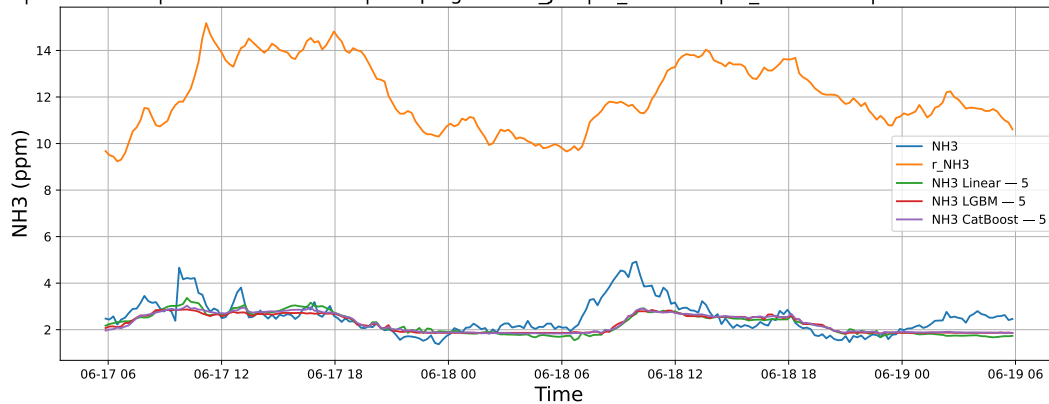
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



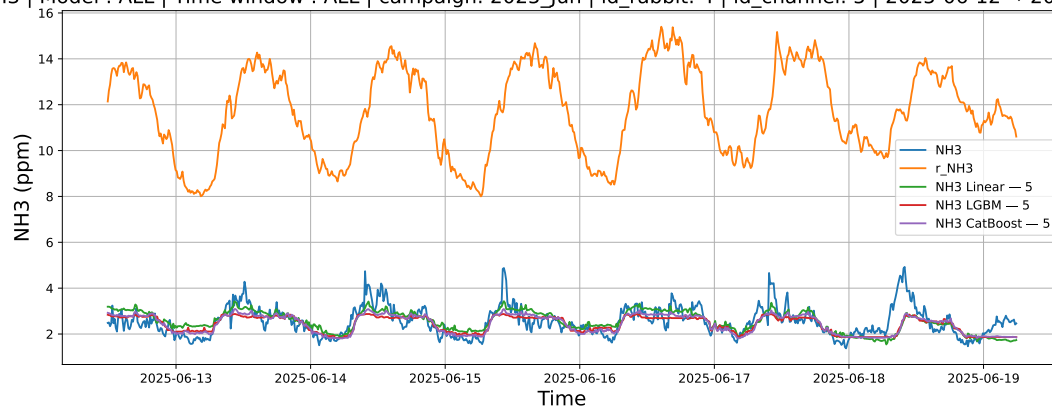
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

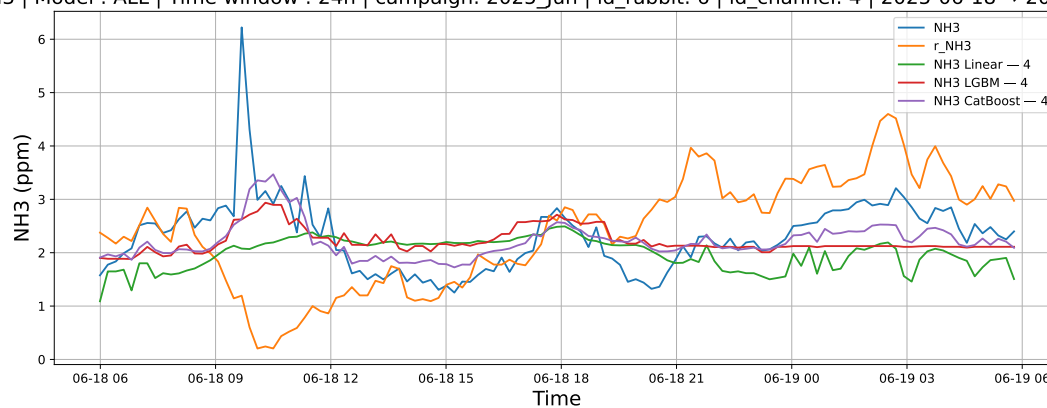
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

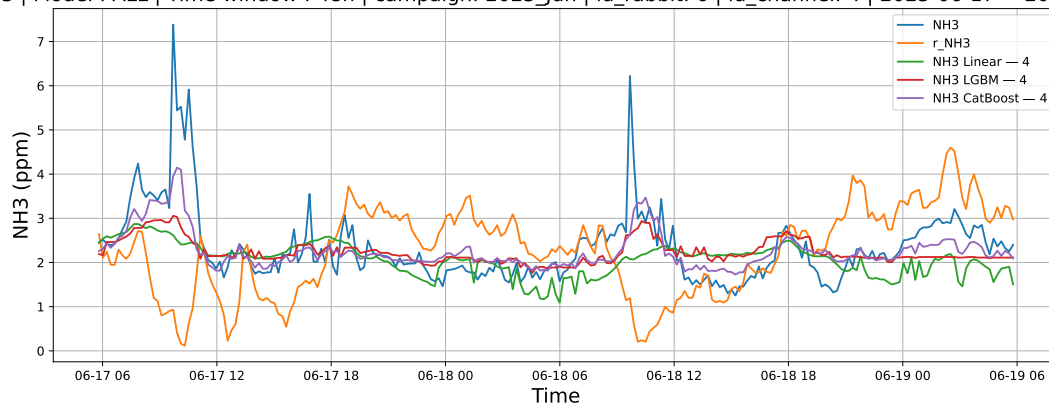
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



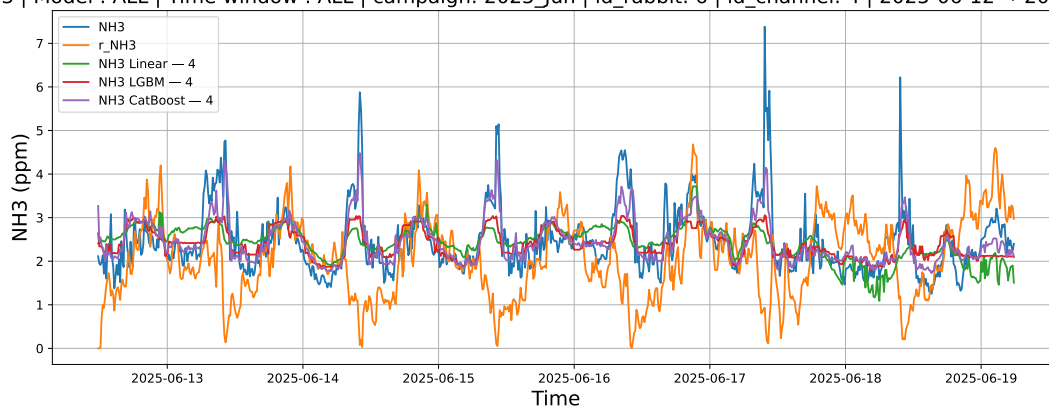
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

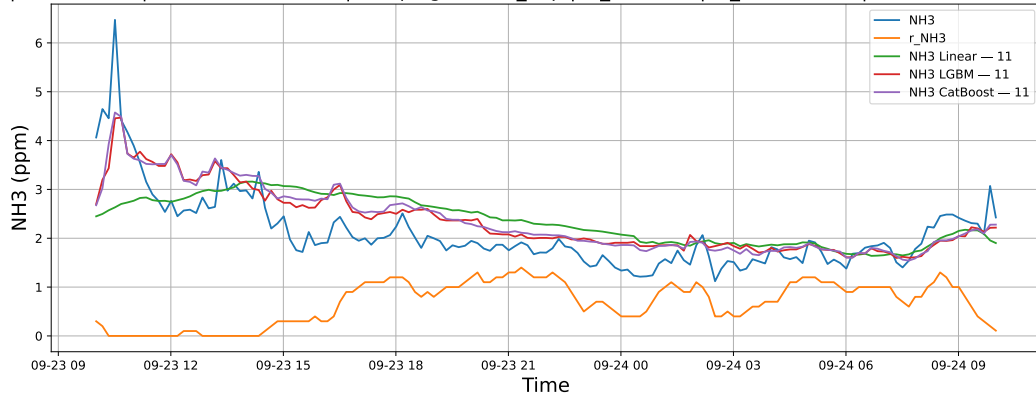
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

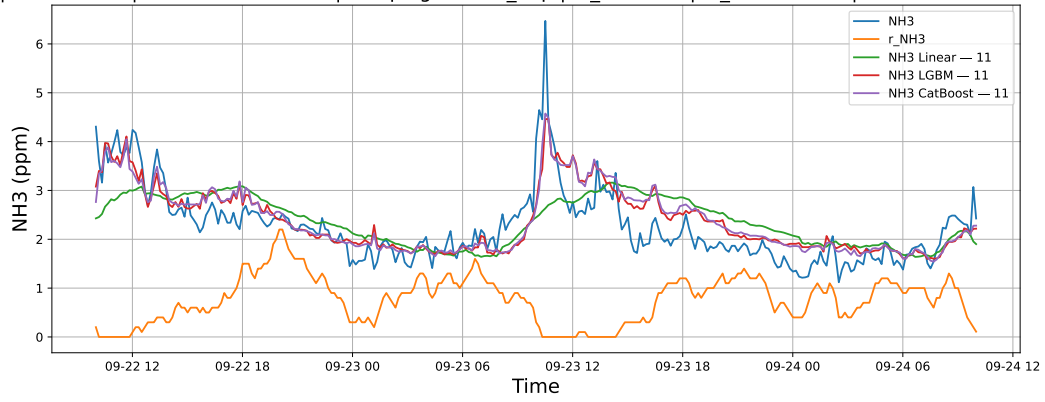
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



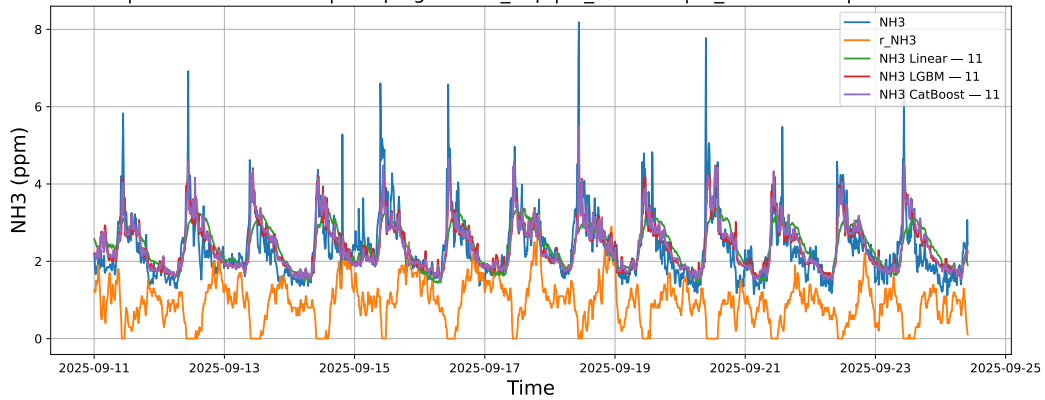
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

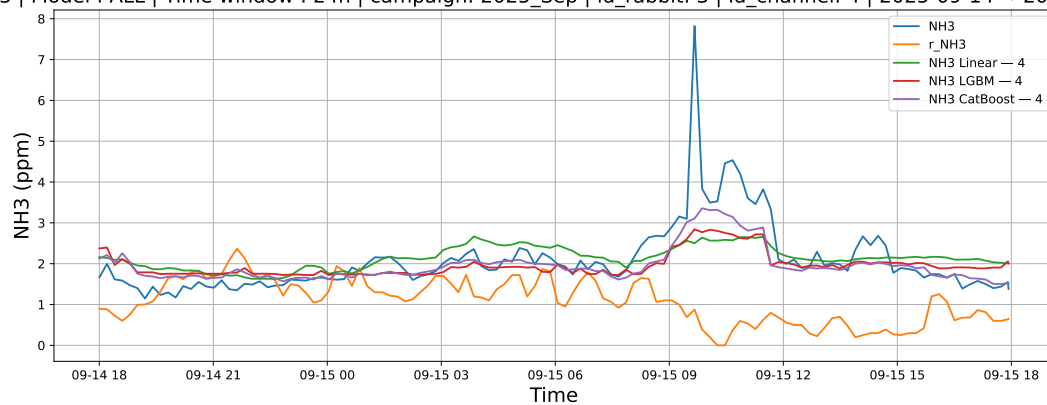
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

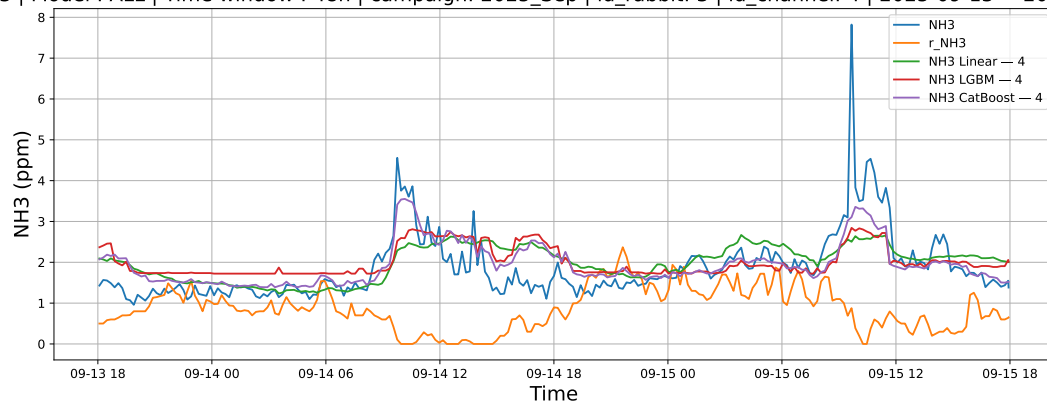
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



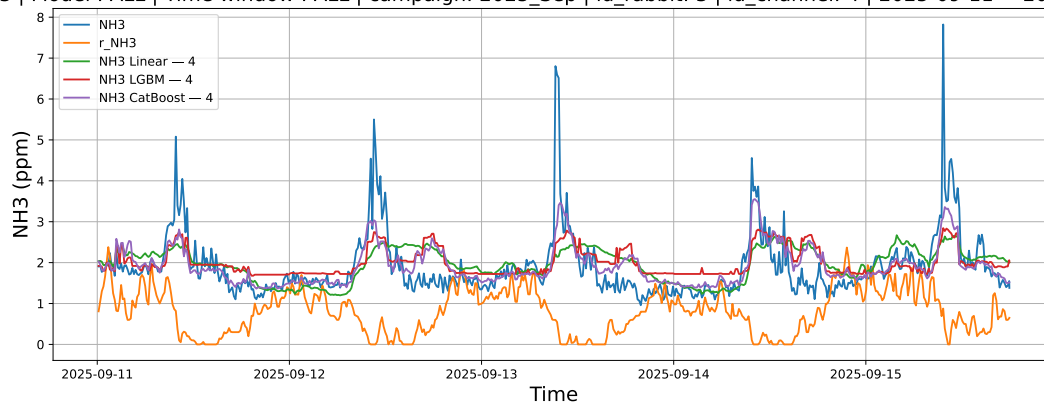
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

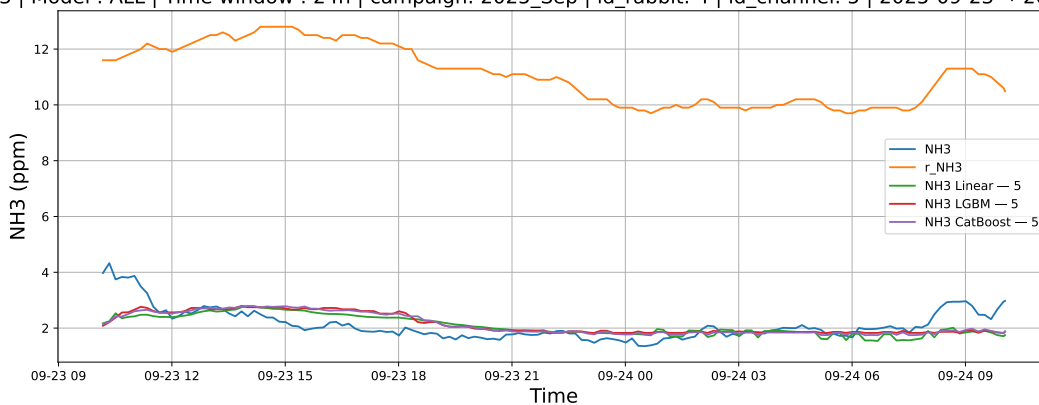
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

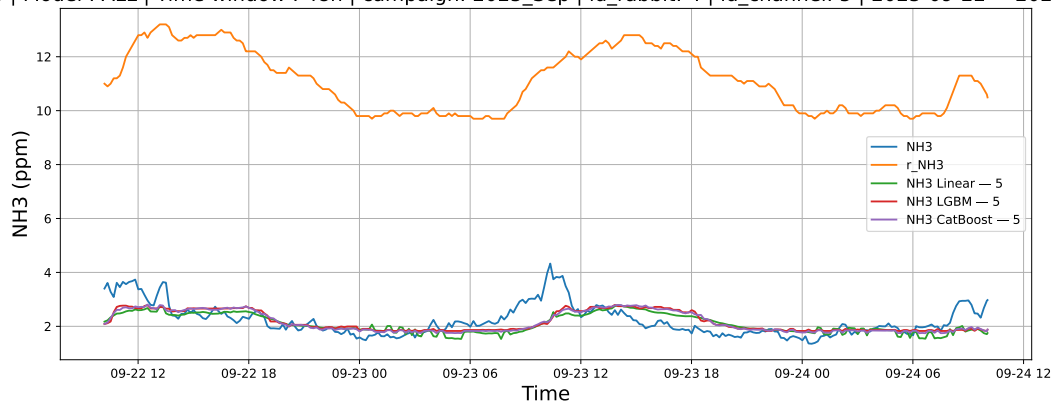
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



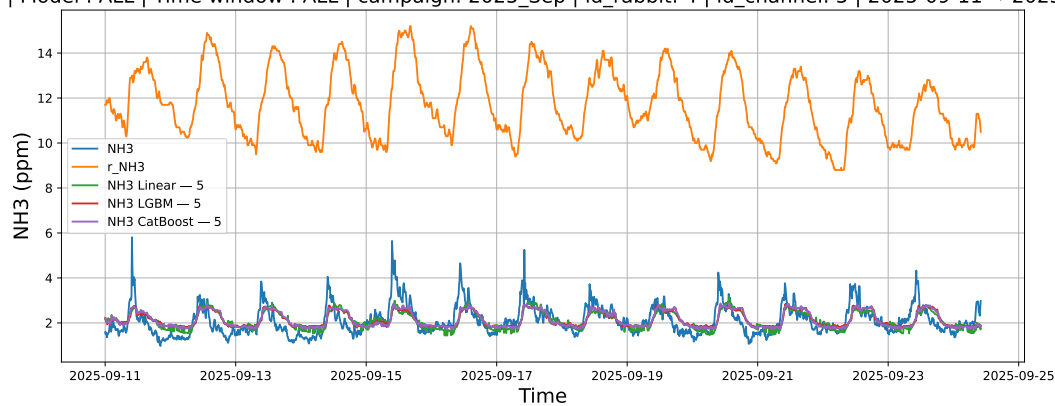
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

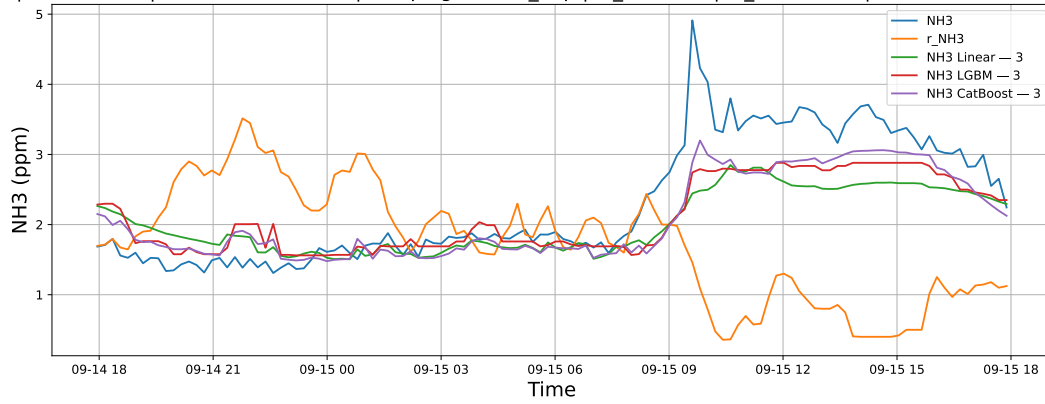
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

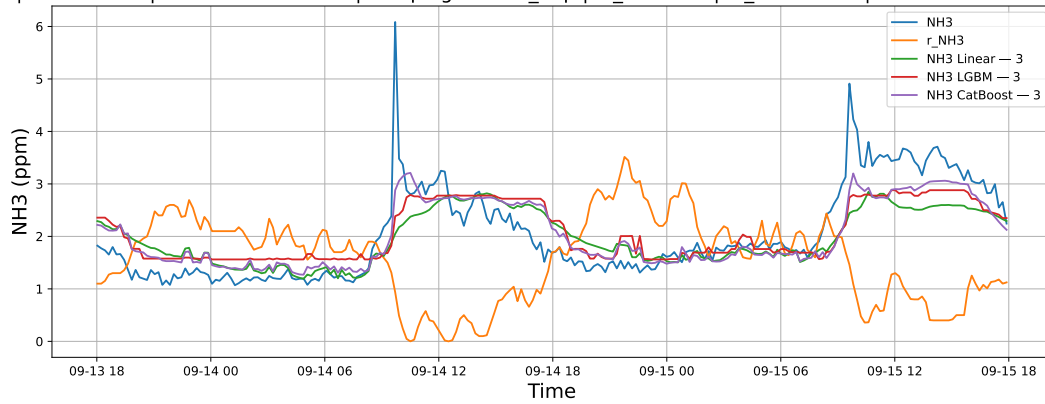
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

